Linear Algebra II

Lecture notes, Chapters 9–16

Based on Lee In-Suk, *Linear Algebra and Groups*

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Chapter 9. Rigid Motions of Euclidean Space

The second half of the course begins here. Euclidean space is the vector space $\mathbb{R}^n$ with its distance; its "rigid" motions, which preserve lengths and distances, motivate the abstract inner product spaces, groups and forms of the following chapters.

9.1 The dot product on $\mathbb{R}^n$

Motivation. The dot product gives $\mathbb{R}^n$ its geometry: length, distance, angle and perpendicularity. Orthonormal bases and orthogonal complements are the basic tools, and the question $\dim W^\perp = n - \dim W$ points ahead to Section 10.3.

Definition 9.1.1 (Dot product)

For $X = (a_1, \ldots, a_n)^t$, $Y = (b_1, \ldots, b_n)^t \in \mathbb{R}^n$,

$$\langle X, Y \rangle = X^t \cdot Y = \sum_{i=1}^{n} a_i b_i$$

is the **dot product** on $\mathbb{R}^n$; $\mathbb{R}^n$ with the dot product is **Euclidean space**.

Exercise 9.1.2

(a) $(X, Y) \mapsto \langle X, Y \rangle$ is a bilinear form on $\mathbb{R}^n$ (Definition 6.1.1).
(b) $\langle X, Y \rangle = \langle Y, X \rangle$.
(c) $\langle X, X \rangle \geq 0$, with equality iff $X = 0$.

Solution.

(a) $X^t(aY + bZ) = aX^tY + bX^tZ$, and similarly in X.

(b) $\sum a_i b_i = \sum b_i a_i$.

(c) $\langle X, X \rangle = \sum a_i^2 \geq 0$, and a sum of squares of reals vanishes iff every $a_i = 0$. □

Exercise 9.1.3

If $\langle Y, X \rangle = 0$ for all $Y \in \mathbb{R}^n$, then $X = 0$.

Solution. Take $Y = X$ and use Exercise 9.1.2(c). □

The dot product gives $\mathbb{R}^n$ a geometric (topological) structure; studying it is **Euclidean geometry**.

Definition 9.1.4 (Norm)

$\|X\| = \sqrt{\langle X, X \rangle}$ is the (**Euclidean**) **norm**, or length, of X.

Exercise 9.1.5

For $X \in \mathbb{R}^n$, $c \in \mathbb{R}$:
(a) $\|X\| \geq 0$, and $\|X\| = 0$ iff $X = 0$.
(b) $\|cX\| = |c|\|X\|$.
(c) If $X \neq 0$, then $\left\| \frac{1}{\|X\|} X \right\| = 1$. A vector of norm 1 is a **unit vector**.

Solution.

(a) Exercise 9.1.2(c).

(b) $\|cX\|^2 = c^2\langle X, X\rangle$.

(c) By (b) with $c = 1/\|X\|$. □

Exercise 9.1.6 (Parallelogram law)

$\|X+Y\|^2 + \|X-Y\|^2 = 2(\|X\|^2 + \|Y\|^2)$.

Solution. Expand: $\|X \pm Y\|^2 = \|X\|^2 \pm 2\langle X, Y\rangle + \|Y\|^2$, and add. □

The angle $\theta \in [0, \pi]$ between X and Y is measured in the plane $\langle X\rangle \oplus \langle Y\rangle \cong \mathbb{R}^2$, where

$$\langle X, Y\rangle = \|X\|\|Y\| \cos\theta. \qquad (\angle)$$

Exercise 9.1.7

(a) (Cauchy–Schwarz) $|\langle X, Y\rangle| \le \|X\|\|Y\|$.
(b) (Triangle inequality) $\|X+Y\| \le \|X\| + \|Y\|$.

Solution.

(a) If $Y = 0$ both sides vanish. Otherwise $0 \le \|X - tY\|^2 = \|X\|^2 - 2t\langle X, Y\rangle + t^2\|Y\|^2$ for all $t \in \mathbb{R}$, so the discriminant is ≤ 0: $\langle X, Y\rangle^2 \le \|X\|^2\|Y\|^2$. Equality holds iff X, Y are linearly dependent.

(b) $\|X+Y\|^2 = \|X\|^2 + 2\langle X, Y\rangle + \|Y\|^2 \le (\|X\| + \|Y\|)^2$ by (a). □

So $\|X - Y\|$ is a distance (metric) on $\mathbb{R}^n$:

Exercise 9.1.8

(a) $\|X - X\| = 0$.
(b) $\|X - Y\| = \|Y - X\|$.
(c) $\|X - Y\| \le \|X - Z\| + \|Z - Y\|$.

Solution.

(a) Clear.

(b) Exercise 9.1.5(b) with $c = -1$.

(c) Exercise 9.1.7(b) applied to $(X - Z) + (Z - Y)$. □

Definition 9.1.9 (Perpendicularity)

(a) $X \perp Y$ (X, Y are **perpendicular**, **orthogonal**) if $\langle X, Y\rangle = 0$.
(b) For $S, T \subseteq \mathbb{R}^n$, $S \perp T$ if $\langle X, Y\rangle = 0$ for all $X \in S$, $Y \in T$.

Exercise 9.1.10

The following are equivalent: (1) $X \perp Y$; (2) $\|X+Y\|^2 = \|X\|^2 + \|Y\|^2$ (Pythagoras); (3) $\|X+Y\| = \|X-Y\|$.

Solution. $\|X \pm Y\|^2 = \|X\|^2 + \|Y\|^2 \pm 2\langle X, Y\rangle$. So (2) and (3) both say $\langle X, Y\rangle = 0$. □

Definition 9.1.11 (Orthogonal and orthonormal sets)

(a) Nonzero $X_1, \ldots, X_m \in \mathbb{R}^n$ with $X_i \perp X_j$ for $i \neq j$ form an **orthogonal subset**; if moreover all are unit vectors, an **orthonormal subset**.
(b) For $W \leq \mathbb{R}^n$, a basis $\mathfrak{B}$ of W is an **orthogonal basis** if it is an orthogonal subset, and an **orthonormal basis** if it is orthonormal. Normalizing (Exercise 9.1.5(c)) turns an orthogonal basis into an orthonormal one.

Exercise 9.1.12

(a) An orthogonal subset $\{X_1, \ldots, X_m\}$ of $\mathbb{R}^n$ is linearly independent.
(b) For $W \leq \mathbb{R}^n$, a basis $\{X_1, \ldots, X_m\}$ of W is orthonormal iff $\langle X_i, X_j \rangle = \delta_{ij}$.

Solution.

(a) If $\sum c_i X_i = 0$, then $0 = \langle \sum c_i X_i, X_j \rangle = c_j \|X_j\|^2$, so $c_j = 0$.

(b) This is the definition. □

Exercise 9.1.13

The following are orthonormal bases:
(a) the standard basis $\mathcal{E}$ of $\mathbb{R}^n$;
(b) $\{(1,0)^t, (0,-1)^t\}$ of $\mathbb{R}^2$;
(c) $\{(\cos\theta, \sin\theta)^t, (-\sin\theta, \cos\theta)^t\}$ of $\mathbb{R}^2$;
(d) $\{(\cos\theta, \sin\theta, 0)^t, (-\sin\theta, \cos\theta, 0)^t, (0,0,1)^t\}$ of $\mathbb{R}^3$;
(e) $\{\frac{1}{\sqrt{5}}(2,1)^t, \frac{1}{\sqrt{5}}(-1,2)^t\}$ of $\mathbb{R}^2$;
(f) $\{\frac{1}{\sqrt{2}}(-1,1,0)^t, \frac{1}{\sqrt{3}}(1,1,1)^t, \frac{1}{\sqrt{6}}(1,1,-2)^t\}$ of $\mathbb{R}^3$;
(g) $\{(\cos\theta, \sin\theta, 0, 0)^t, (-\sin\theta, \cos\theta, 0, 0)^t, (0,0,\cos\eta, \sin\eta)^t, (0,0,-\sin\eta, \cos\eta)^t\}$ of $\mathbb{R}^4$.

Solution. In each case the inner products of distinct vectors vanish and each vector has norm 1 (using $\cos^2 + \sin^2 = 1$). An orthonormal set of n vectors in $\mathbb{R}^n$ is independent (Exercise 9.1.12(a)), hence a basis. □

So orthonormal bases of $\mathbb{R}^n$ exist, and (f) shows how an orthogonal basis is normalized.

Definition 9.1.14 (Orthogonal complement)

For $S \subseteq \mathbb{R}^n$, $S^\perp = \{X \in \mathbb{R}^n : Y \perp X \text{ for all } Y \in S\}$ ("S perp"). For $W \leq \mathbb{R}^n$, $W^\perp$ is the **orthogonal complement** of W.

Observation 9.1.15

Let $S \subseteq \mathbb{R}^n$, $W \leq \mathbb{R}^n$, and $\mathfrak{C}$ a basis of W.
(a) $S^\perp$ is a subspace of $\mathbb{R}^n$.
(b) $S^\perp = \langle S \rangle^\perp$.
(c) $W^\perp = \mathfrak{C}^\perp$.

Proof.

(a) If $S \perp X$ and $S \perp Y$, then $S \perp X + Y$ and $S \perp cX$.

(b) $S \subseteq \langle S \rangle$ gives $\langle S \rangle^\perp \subseteq S^\perp$. Conversely, elements of $\langle S \rangle$ are linear combinations of vectors of S (Proposition 3.1.6), so a vector perpendicular to S is perpendicular to $\langle S \rangle$.

(c) The case $S = \mathfrak{C}$ of (b). ■

By (b) it suffices to consider subspaces, and by (c) $X \perp W$ iff X is perpendicular to a basis of W.

Exercise 9.1.16

Let $\mathfrak{B}$ be a basis of $\mathbb{R}^n$ and $W \leq \mathbb{R}^n$.
(a) $0^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \mathfrak{B}^\perp = 0$.
(b) $W \cap W^\perp = 0$.
(c) $W \leq (W^\perp)^\perp$.

Solution.

(a) Every X is perpendicular to 0; $(\mathbb{R}^n)^\perp = 0$ by Exercise 9.1.3, and $\mathfrak{B}^\perp = (\mathbb{R}^n)^\perp$ by Observation 9.1.15(c).

(b) $X \in W \cap W^\perp$ gives $\langle X, X\rangle = 0$.

(c) Each $X \in W$ is perpendicular to every vector of $W^\perp$. □

Question 9.1.17

If $W \leq \mathbb{R}^n$, is $W = (W^\perp)^\perp$?

There seems to be no direct proof of $(W^\perp)^\perp \leq W$; one is led to compute $\dim W^\perp$.

Example 9.1.18

For $0 \neq Y \in \mathbb{R}^n$, the linear map $\varphi_Y : \mathbb{R}^n \to \mathbb{R}$, $\varphi_Y(X) = \langle Y, X\rangle = Y^t X = L_{Y^t}(X)$, is surjective, since $\varphi_Y(Y) \neq 0$. By the dimension theorem,

$$\dim\langle Y\rangle^\perp = \dim\ker\varphi_Y = n - \dim\operatorname{im}\varphi_Y = n - 1.$$

So $\dim W^\perp = n - \dim W$ holds when $\dim W = 1$.

If $\dim W^\perp = n - \dim W$ held for all $W \leq \mathbb{R}^n$, then $\dim(W^\perp)^\perp = \dim W$, so $W = (W^\perp)^\perp$ by Exercise 9.1.16(c) (a *dimension argument*: $U \leq V$ and $\dim U = \dim V$ force $U = V$). And by Exercise 9.1.16(b) and Observation 7.6.4, $\mathbb{R}^n = W \oplus W^\perp$.

Question 9.1.19

If $W \leq \mathbb{R}^n$, do we always have $\dim W^\perp = n - \dim W$ and $\mathbb{R}^n = W \oplus W^\perp$?

The answer (yes) uses the homogeneous system $AX = 0$ and the rank theorem; it is given in §10.3.

9.2 Rigid motions of $\mathbb{R}^n$

Motivation. Rigid motions preserve distances. Using only the dimension theorem and the basic theorem on linear maps, we show that every rigid motion is a translation composed with a *linear* rigid motion; in particular it is a bijection.

Definition 9.2.1 (Rigid motion)

$M : \mathbb{R}^n \to \mathbb{R}^n$ is a **rigid motion** (**isometry**) of $\mathbb{R}^n$ if $\|M(X) - M(Y)\| = \|X - Y\|$ for all $X, Y \in \mathbb{R}^n$.

Example 9.2.2

For $Y \in \mathbb{R}^n$, the **translation** by Y, $T_Y(X) = X + Y$, is a rigid motion; T_Y is a bijection with $(T_Y)^{-1} = T_{-Y}$, and $T_0 = I$.

Example 9.2.3

For $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, the **rotation** L_{R_θ} by θ about the origin is a linear rigid motion of $\mathbb{R}^2$ (Exercise 4.3.5). Abusing notation, write $L_{R_\theta} = R_\theta$; R_θ is a bijection, $(R_\theta)^{-1} = R_{-\theta}$, $R_0 = I$.

Example 9.2.4

$S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ gives a linear rigid motion $L_S = S$ of $\mathbb{R}^2$ (Exercise 4.3.6), a bijection with $S^{-1} = S$. (It is a reflection; see §9.4.)

The columns of R_θ and S are the orthonormal bases of Exercise 9.1.13; this is explained in Observation 9.3.6.

Observation 9.2.5

A rigid motion M is injective.

Proof. $M(X) = M(Y)$ gives $\|X - Y\| = 0$, so $X = Y$. ■

Question 9.2.6

Is every rigid motion of $\mathbb{R}^n$ a bijection?

The answer (Corollary 9.2.12) needs the dimension theorem and the basic theorem of Chapter 5.

Exercise 9.2.7

(a) A composite of rigid motions is a rigid motion.
(b) The inverse of a bijective rigid motion is a rigid motion.

Solution.

(a) $\|MN(X) - MN(Y)\| = \|N(X) - N(Y)\| = \|X - Y\|$.

(b) Put $X = M(X')$, $Y = M(Y')$: $\|M^{-1}X - M^{-1}Y\| = \|X' - Y'\| = \|MX' - MY'\| = \|X - Y\|$. □

For a rigid motion M, $L = T_{-M(0)} \circ M$ is a rigid motion with $L(0) = 0$, and $M = T_{M(0)} \circ L$. We claim such an L is linear.

Observation 9.2.8

A rigid motion L of $\mathbb{R}^n$ with $L(0) = 0$ satisfies:
(a) $\|L(X)\| = \|X\|$;
(b) $\langle L(X), L(X)\rangle = \langle X, X\rangle$;
(c) $\langle L(X), L(Y)\rangle = \langle X, Y\rangle$;
(d) if $\mathfrak{B}$ is an orthonormal basis of $\mathbb{R}^n$, so is $L(\mathfrak{B})$.

Proof.

(a) Put $Y = 0$ in $\|L(X) - L(Y)\| = \|X - Y\|$.

(b) Square (a).

(c) Square $\|L(X)-L(Y)\| = \|X-Y\|$ and expand: $\langle LX, LX\rangle - 2\langle LX, LY\rangle + \langle LY, LY\rangle = \langle X, X\rangle - 2\langle X, Y\rangle + \langle Y, Y\rangle$; by (b), $\langle LX, LY\rangle = \langle X, Y\rangle$.

(d) If $\mathfrak{B} = \{X_i\}$, then $\langle L(X_i), L(X_j)\rangle = \langle X_i, X_j\rangle = \delta_{ij}$ by (c), so $L(\mathfrak{B})$ is an orthonormal set of n vectors, hence a basis (Exercise 9.1.12). ■

Observation 9.2.9

If the columns $[A]^1, \ldots, [A]^n$ of $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ form an orthonormal basis of $\mathbb{R}^n$, then L_A is a rigid motion.

Proof. For $X = \sum a_i e_i$, $AX = \sum a_i Ae_i = \sum a_i [A]^i$, so $\|AX\|^2 = \sum_{i,j} a_i a_j \delta_{ij} = \sum a_i^2 = \|X\|^2$. Replacing X by $X - Y$ gives $\|AX - AY\| = \|X - Y\|$. ■

Observation 9.2.10

If a rigid motion L with $L(0) = 0$ fixes every vector of a basis $\mathfrak{B} = \{X_i\}$ of $\mathbb{R}^n$, then $L = I$.

Proof. For $X \in \mathbb{R}^n$, $\langle X_i, L(X)\rangle = \langle L(X_i), L(X)\rangle = \langle X_i, X\rangle$ by Observation 9.2.8(c). So $L(X) - X \in \mathfrak{B}^\perp = 0$ (Exercise 9.1.16(a)). ■

Theorem 9.2.11

(a) A rigid motion L of $\mathbb{R}^n$ with $L(0) = 0$ is linear.
(b) Hence every rigid motion M of $\mathbb{R}^n$ is a translation composed with a linear rigid motion.

Proof.

(a) We look for A with $L = L_A$. Put $X_j = L(e_j)$ and $A = (X_1, \ldots, X_n)$, so $Ae_j = X_j$. By Observation 9.2.8(d), $\{X_j\}$ is an orthonormal basis, so A is invertible (basic theorem, Chapter 5) and L_A is invertible. Consider $M' = (L_A)^{-1} \circ L$. Then $M'(0) = 0$ and $M'(e_j) = A^{-1}X_j = e_j$. M' is a rigid motion by Observation 9.2.9 and Exercise 9.2.7, so $M' = I$ by Observation 9.2.10, i.e. $L = L_A$.

(b) With $L = T_{-M(0)} \circ M$, L is linear by (a) and $M = T_{M(0)} \circ L$. ■

Second proof of (a). By Observation 9.2.8(c), for all j,

$$\langle L(e_j), L(X+Y)\rangle = \langle e_j, X+Y\rangle = \langle e_j, X\rangle + \langle e_j, Y\rangle = \langle L(e_j), L(X) + L(Y)\rangle.$$

Since $L(\mathcal{E})$ is a basis (Observation 9.2.8(d)), $L(X+Y) - L(X) - L(Y) \in L(\mathcal{E})^\perp = 0$. Likewise $\langle L(e_j), L(cX)\rangle = c\langle e_j, X\rangle = \langle L(e_j), cL(X)\rangle$ gives $L(cX) = cL(X)$. ■

The second proof is shorter but offers no motivation; the first shows why the basic theorem and Exercise 9.2.7 matter.

Corollary 9.2.12

Every rigid motion of $\mathbb{R}^n$ is a bijection.

Proof. A linear rigid motion is injective (Observation 9.2.5), hence bijective by the dimension theorem. Translations are bijections, so by Theorem 9.2.11(b) every rigid motion is a bijection. ■

Exercise 9.2.13

Let $\mathcal{A}(\mathbb{R}^n)$ be the set of rigid motions of $\mathbb{R}^n$. (a) If $L, M \in \mathcal{A}(\mathbb{R}^n)$, then $L \circ M \in \mathcal{A}(\mathbb{R}^n)$.
(b) If $L \in \mathcal{A}(\mathbb{R}^n)$, then $L^{-1} \in \mathcal{A}(\mathbb{R}^n)$.

Solution. Exercise 9.2.7, with Corollary 9.2.12 for (b). □

A set with these two properties (and containing I) is a **group**; groups are studied from Chapter 11.

Exercise 9.2.14

Let $L_A, L_B \in \mathcal{A}(\mathbb{R}^n)$ and $Y, Z \in \mathbb{R}^n$.
(a) If $T_Y \circ L_A = T_Z \circ L_B$, then $A = B$ and $Y = Z$. (So every element of $\mathcal{A}(\mathbb{R}^n)$ is uniquely $T_Y \circ L_A$.)
(b) $L_A \circ T_Y = T_{AY} \circ L_A$.

Solution.

(a) Evaluate at 0: $Y = Z$. Then $L_A = L_B$, so $A = B$.

(b) $A(X + Y) = AX + AY$. □

Since translations are well understood, from now on we study *linear* rigid motions.

9.3 Orthogonal operators and orthogonal matrices

Motivation. Linear rigid motions are the orthogonal operators. They are characterised by preserving lengths, dot products or orthonormal bases, and their matrices by $A^tA = I$. Those with determinant 1 form the special orthogonal group.

Definition 9.3.1 (Orthogonal operators and matrices)

(a) A *linear* rigid motion of $\mathbb{R}^n$ is a (real) **orthogonal operator**; if $L = L_A$ is one, the real $n \times n$ matrix A is an **orthogonal matrix**.
(b) The **orthogonal group** on $\mathbb{R}^n$ is $\mathbf{O}(\mathbb{R}^n) = \{L \in \mathfrak{L}(\mathbb{R}^n, \mathbb{R}^n) : \|L(X) - L(Y)\| = \|X - Y\| \ \forall X, Y\}$, and

$$\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : \|AX - AY\| = \|X - Y\| \ \forall X, Y \in \mathbb{R}^n\}.$$

Corollary 9.3.2

For every rigid motion M of $\mathbb{R}^n$ there are unique $A \in \mathbf{O}(n)$, $B \in \mathbb{R}^n$ with $M(X) = AX + B$.

Proof. In Theorem 9.2.11, $M = T_{M(0)} \circ L$ with $L = L_A$ linear; put $B = M(0)$. Uniqueness is Exercise 9.2.14(a). ■

Exercise 9.3.3

(a) If $L, M \in \mathbf{O}(\mathbb{R}^n)$, then $L \circ M \in \mathbf{O}(\mathbb{R}^n)$; if $A, B \in \mathbf{O}(n)$, then $AB \in \mathbf{O}(n)$.
(b) If $L \in \mathbf{O}(\mathbb{R}^n)$, then $L^{-1} \in \mathbf{O}(\mathbb{R}^n)$; if $A \in \mathbf{O}(n)$, then $A^{-1} \in \mathbf{O}(n)$.

Solution. Composites and inverses of linear rigid motions are linear rigid motions (Exercise 9.2.13); $L_A L_B = L_{AB}$ and $(L_A)^{-1} = L_{A^{-1}}$. □

Observation 9.3.4

For linear $L : \mathbb{R}^n \to \mathbb{R}^n$ the following are equivalent:
(1) $\|LX - LY\| = \|X - Y\|$ for all X, Y (i.e. $L \in \mathbf{O}(\mathbb{R}^n)$);
(2) $\|LX\| = \|X\|$ for all X;
(3) $\langle LX, LX\rangle = \langle X, X\rangle$ for all X;
(4) $\langle LX, LY\rangle = \langle X, Y\rangle$ for all X, Y;
(5) if $\mathfrak{B}$ is an orthonormal basis of $\mathbb{R}^n$, so is $L(\mathfrak{B})$.

Proof. (1)⇒(2): $Y = 0$. (2)⇒(1): replace X by $X - Y$, using $L(X - Y) = LX - LY$. (2)⇔(3): square. (3)⇒(4): replace X by $X - Y$ (or $X + Y$) and expand, as in Observation 9.2.8(c). (4)⇒(3): clear. (4)⇒(5): $\langle LX_i, LX_j\rangle = \langle X_i, X_j\rangle = \delta_{ij}$. (5)⇒(2): take an orthonormal basis $\mathfrak{B} = \{X_i\}$ (e.g. $\mathcal{E}$) and $X = \sum a_i X_i$. Then $\|X\|^2 = \sum a_i^2$, and since $L\mathfrak{B}$ is orthonormal, $\|LX\|^2 = \|\sum a_i LX_i\|^2 = \sum a_i^2$. ■

So orthogonal operators preserve distance, length, dot product and orthonormal bases.

Exercise 9.3.5

The conditions of Observation 9.3.4 are also equivalent to
(2)′ if $\|X\| = 1$, then $\|LX\| = 1$ (L preserves unit vectors);
(5)′ some orthonormal basis $\mathfrak{B}$ has $L\mathfrak{B}$ orthonormal.

Solution. (2)⇒(2)′ and (5)⇒(5)′ are clear. (2)′ ⇒(2): for $X \neq 0$, $\|LX\| = \|X\| \left\|L(\frac{X}{\|X\|})\right\| = \|X\|$. (5)′ ⇒(2): the last step of the proof of Observation 9.3.4 uses only one orthonormal basis. □

Observation 9.3.6

For $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ the following are equivalent:
(1) $\|AX - AY\| = \|X - Y\|$ for all X, Y (i.e. $A \in \mathbf{O}(n)$);
(2) $\|AX\| = \|X\|$;
(3) $\langle AX, AX\rangle = \langle X, X\rangle$;
(4) $\langle AX, AY\rangle = \langle X, Y\rangle$;
(5) the columns $\{[A]^1, \ldots, [A]^n\}$ form an orthonormal basis of $\mathbb{R}^n$;
(6) $A^tA = I = AA^t$, i.e. $A^{-1} = A^t$;
(7) $A^t \in \mathbf{O}(n)$;
(8) the rows $\{[A]_1, \ldots, [A]_n\}$ form an orthonormal basis of $\mathfrak{M}_{1,n}(\mathbb{R})$.

Proof. (1)⇔(2)⇔(3)⇔(4)⇔(5): Observation 9.3.4 for $L = L_A$ with $\mathfrak{B} = \mathcal{E}$, using $[A]^i = Ae_i$.

(5)⇔(6): the (i, j) entry of A^tA is the dot product of column i and column j of A; so $A^tA = I$ iff $\langle [A]^i, [A]^j\rangle = \delta_{ij}$. For square matrices $A^tA = I$ implies $AA^t = I$.

(6)⇒(7) [Supplement]: $(A^t)^tA^t = AA^t = I$, so A^t satisfies (6). (7)⇒(6): apply (6) to A^t.

(7)⇔(8) [Supplement]: the rows of A are the transposed columns of A^t; apply (5) to A^t. ■

Second proof of (4)⇔(6). (4) says $X^tY = (AX)^tAY = X^t(A^tA)Y = X^tIY$ for all X, Y, which by Exercise 9.3.7 means $A^tA = I$. The converse reads the computation backwards. ■

Exercise 9.3.7

If $B, C \in \mathfrak{M}_{m,n}(F)$ and $X^tBY = X^tCY$ for all $X \in F^m$, $Y \in F^n$, then $B = C$.

Solution. $e_i^tBe_j = B_{ij}$; take $X = e_i$, $Y = e_j$. □

So finding $n \times n$ orthogonal matrices is the same as finding orthonormal bases of $\mathbb{R}^n$, and

$$\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : A^t A = I\}.$$

With $(L_A)^t = L_{A^t}$, $\mathbf{O}(\mathbb{R}^n) = \{L \in \mathfrak{L}(\mathbb{R}^n, \mathbb{R}^n) : L^t \circ L = I\}$; defining the transpose of a general linear map is one of the goals of later chapters.

Exercise 9.3.8

If $A \in \mathbf{O}(n)$, then $\det A = \pm 1$.

Solution. $1 = \det(A^t A) = \det(A^t) \det A = (\det A)^2$. □

Definition 9.3.9 (Special orthogonal group)

$\mathbf{SO}(n) = \{A \in \mathbf{O}(n) : \det A = 1\}$ and $\mathbf{SO}(\mathbb{R}^n) = \{L \in \mathbf{O}(\mathbb{R}^n) : \det L = 1\}$.

In geometry, L_A with $A \in \mathbf{SO}(n)$ is **orientation preserving**, and with $A \in \mathbf{O}(n) - \mathbf{SO}(n)$ **orientation reversing**.

Exercise 9.3.10

(a) $I \in \mathbf{SO}(n)$.
(b) If $A, B \in \mathbf{SO}(n)$, then $AB \in \mathbf{SO}(n)$.
(c) If $A \in \mathbf{SO}(n)$, then $A^{-1} \in \mathbf{SO}(n)$.

Solution. Exercise 9.3.3 together with $\det(AB) = \det A \det B$ and $\det A^{-1} = (\det A)^{-1}$. □

Exercise 9.3.11

Fix $S \in \mathbf{O}(n) - \mathbf{SO}(n)$.
(a) Indeed $\mathbf{O}(n) - \mathbf{SO}(n) \neq \emptyset$.
(b) $\lambda_S : \mathbf{SO}(n) \to S \cdot \mathbf{SO}(n) = \{SA : A \in \mathbf{SO}(n)\}$, $\lambda_S(A) = SA$, is a bijection.
(c) $\mathbf{O}(n) - \mathbf{SO}(n) = S \cdot \mathbf{SO}(n)$.
(d) $\mathbf{O}(n) = \mathbf{SO}(n) \amalg (S \cdot \mathbf{SO}(n))$ (disjoint union).

Solution.

(a) $\operatorname{diag}(-1, 1, \ldots, 1)$.

(b) $SA = SB$ implies $A = B$; surjective by definition.

(c) If $R \in \mathbf{O}(n) - \mathbf{SO}(n)$, then $S^{-1}R \in \mathbf{SO}(n)$ ($\det = (-1)(-1)$), so $R = S(S^{-1}R)$; conversely $\det(SA) = -1$.

(d) Combine (c) with $\mathbf{SO}(n) \cap (\mathbf{O}(n) - \mathbf{SO}(n)) = \emptyset$. □

This is the first example of a coset decomposition (Chapter 12).

9.4 Reflections

Motivation. Reflecting across the hyperplane $\langle Y \rangle^\perp$ is the simplest orthogonal operator besides the identity. In an orthonormal basis starting with $Y/\|Y\|$ its matrix is $\operatorname{diag}(-1, 1, \ldots, 1)$. This "change to a good orthonormal frame" is the way of thinking used for the rest of the course.

Example 9.4.1

For $X, Y \in \mathbb{R}^3$, $Y \neq 0$, write $X = aY + (X - aY)$ with $(X - aY) \perp Y$; then $a\langle Y, Y\rangle = \langle X, Y\rangle$, so $a = \frac{\langle X,Y\rangle}{\langle Y,Y\rangle}$. Reflecting X in the plane $\langle Y\rangle^\perp$ gives $X - \frac{2\langle X,Y\rangle}{\langle Y,Y\rangle}Y$.

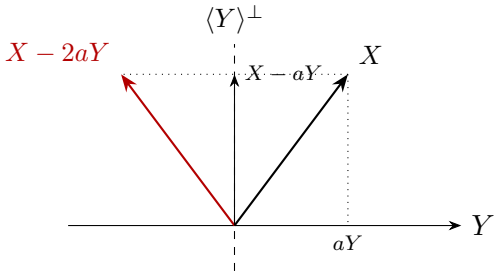


Figure. *Reflection of X in the hyperplane $\langle Y\rangle^\perp$.*

Definition 9.4.2 (Reflection)

For $0 \neq Y \in \mathbb{R}^n$, $S_Y : \mathbb{R}^n \to \mathbb{R}^n$,

$$S_Y(X) = X - \frac{2\langle X, Y\rangle}{\langle Y, Y\rangle}Y,$$

is the **reflection** (**symmetry**) with respect to Y.

Remark 9.4.3

The "reflection in the x-axis" of $\mathbb{R}^2$ is $S_{(0,1)^t}$ in this terminology.

Exercise 9.4.4

For $0 \neq Y \in \mathbb{R}^n$:
(a) $S_Y = S_{cY}$ for $0 \neq c \in \mathbb{R}$;
(b) $S_Y(Y) = -Y$;
(c) if $X \perp Y$, i.e. $X \in \langle Y\rangle^\perp$, then $S_Y(X) = X$.

Solution.
(a) $\frac{\langle X,cY\rangle}{\langle cY,cY\rangle}cY = \frac{c^2\langle X,Y\rangle}{c^2\langle Y,Y\rangle}Y$.
(b) $Y - 2Y = -Y$.
(c) $\langle X, Y\rangle = 0$. □

S_Y is visibly linear, and preserves lengths, so $S_Y \in \mathbf{O}(\mathbb{R}^n)$; one may also check this by computation:

Exercise 9.4.5

$S_Y \in \mathbf{O}(\mathbb{R}^n)$.

Solution. S_Y is linear because $X \mapsto \langle X, Y\rangle$ is. And $\|S_Y X\|^2 = \|X\|^2 - \frac{4\langle X,Y\rangle^2}{\langle Y,Y\rangle} + \frac{4\langle X,Y\rangle^2}{\langle Y,Y\rangle^2}\langle Y, Y\rangle = \|X\|^2$. Apply Observation 9.3.4. □

The way of thinking. By Example 9.1.18, $\langle Y\rangle \cap \langle Y\rangle^\perp = 0$ and $\dim\langle Y\rangle^\perp = n - 1$, so $\mathbb{R}^n = \langle Y\rangle \oplus \langle Y\rangle^\perp$ (Observation 7.6.4). If $\mathfrak{C} = \{X_2, \ldots, X_n\}$ is an ordered basis of $\langle Y\rangle^\perp$, then $\mathfrak{B} =$

$\{Y, X_2, \ldots, X_n\}$ is an ordered basis of $\mathbb{R}^n$. Exercise 9.4.4(b),(c) say $S_Y(Y) = -Y$ and $S_Y|_{\langle Y\rangle^\perp} = I$, which determine S_Y uniquely (linear extension theorem), and

$$[S_Y]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, 1, \ldots, 1).$$

Exercise 9.4.6

For $0 \neq Y \in \mathbb{R}^n$, $(S_Y)^2 = I$ and $(S_Y)^{-1} = S_Y$.

Solution. $[S_Y^2]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, 1, \ldots, 1)^2 = I$. □

Exercise 9.4.7

For $0 \neq Y \in \mathbb{R}^n$, $\det(S_Y) = -1$.

Solution. $\det S_Y = \det[S_Y]_{\mathfrak{B}}^{\mathfrak{B}} = \det \operatorname{diag}(-1, 1, \ldots, 1) = -1$. □

As a matrix, $S_Y = L_A$ for a unique $A \in \mathbf{O}(n)$, and

$$A = [S_Y]_{\mathcal{E}}^{\mathcal{E}} = [I]_{\mathcal{E}}^{\mathfrak{B}} \cdot [S_Y]_{\mathfrak{B}}^{\mathfrak{B}} \cdot [I]_{\mathfrak{B}}^{\mathcal{E}} = [I]_{\mathcal{E}}^{\mathfrak{B}} \cdot \operatorname{diag}(-1, 1, \ldots, 1) \cdot [I]_{\mathfrak{B}}^{\mathcal{E}}.$$

$[I]_{\mathfrak{B}}^{\mathcal{E}}$ (columns the vectors of $\mathfrak{B}$) is in $\mathbf{O}(n)$ iff $\mathfrak{B}$ is an orthonormal basis (Observation 9.3.6(5)). For that we may assume $\|Y\| = 1$ (Exercise 9.4.4(a)); then $\mathfrak{B}$ is orthonormal iff $\mathfrak{C}$ is an orthonormal basis of $\langle Y\rangle^\perp$.

Question 9.4.8

Does every $W \leq \mathbb{R}^n$ have an orthonormal basis?

Yes (Gram–Schmidt, next chapter). With $\mathfrak{C}$ orthonormal, all three factors above lie in $\mathbf{O}(n)$: the identity happens inside the orthogonal group. (Matrices related by $B = U^{-1}AU$ with $U \in \mathbf{O}(n)$ are *orthogonally similar*.) Thinking of orthonormal bases as orthonormal coordinate frames: in the frame $\mathfrak{B}$, the reflection S_Y flips one axis (the Y-axis) and fixes all the others. This is the way of thinking we will use repeatedly.

9.5 $\mathbf{O}(2)$ and $\mathbf{SO}(2)$

Motivation. Every orthogonal operator of the plane is a rotation R_θ or a reflection S_θ, according to its determinant. The composition rules for these, in particular "flip, rotate, flip back = rotate backwards", complete the description of $\mathbf{O}(2)$.

Example 9.5.1

$\mathbf{O}(1) = \{1, -1\}$ and $\mathbf{SO}(1) = \{1\}$.

Theorem 9.5.2

The elements of $\mathbf{SO}(2)$ are exactly the 2-dimensional rotations:

$$\mathbf{SO}(2) = \left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} : 0 \leq \theta < 2\pi \right\} = \left\{ \begin{pmatrix} x & -y \\ y & x \end{pmatrix} : x, y \in \mathbb{R},\ x^2 + y^2 = 1 \right\}.$$

High-school proof. Let $A = \left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right) \in \mathbf{SO}(2)$: $x^2 + y^2 = 1$, $z^2 + w^2 = 1$, $xz + yw = 0$, $xw - yz = 1$. Write $(x, y) = (\cos\theta, \sin\theta)$ and $(z, w) = (\sin\eta, \cos\eta)$. Then $\sin(\theta + \eta) = xz + yw = 0$ and $\cos(\theta + \eta) = xw - yz = 1$, so $\theta + \eta \in 2\pi\mathbb{Z}$. Hence $z = \sin\eta = -\sin\theta$ and $w = \cos\eta = \cos\theta$. ■

Freshman's proof. If $y = 0$, then $x = \pm 1$; $xz = 0$ gives $z = 0$, and $xw = 1$ gives $w = x$: $A = \pm I$, of the stated form. If $y \neq 0$, $w = -\frac{xz}{y}$, and $1 = z^2 + w^2 = \frac{z^2(y^2+x^2)}{y^2} = \frac{z^2}{y^2}$, so $z = \pm y$. If $z = y$, then $w = -x$ and $xw - yz = -(x^2 + y^2) = -1$, impossible; so $z = -y$ and $w = x$. ■

Sophomore's proof. An orthogonal matrix is an orthonormal basis (Observation 9.3.6(5)). Its first vector is $(\cos\theta, \sin\theta)^t$, and the second, a unit vector perpendicular to it, is $(-\sin\theta, \cos\theta)^t$ or $(\sin\theta, -\cos\theta)^t$. The determinants are 1 and -1, so only the first lies in $\mathbf{SO}(2)$. ■

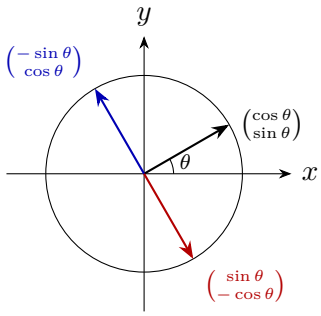


Figure. *The second vector of an orthonormal basis of $\mathbb{R}^2$ is one of two choices: determinant $+1$ (rotation) or -1 (reflection).*

The sophomore's proof describes all of $\mathbf{O}(2)$:

Corollary 9.5.3

$$\mathbf{O}(2) = \mathbf{SO}(2) \amalg \left\{ \begin{pmatrix} x & y \\ y & -x \end{pmatrix} : x, y \in \mathbb{R},\ x^2 + y^2 = 1 \right\}.$$

Proof. By Exercise 9.3.11, $\mathbf{O}(2) - \mathbf{SO}(2) = S \cdot \mathbf{SO}(2)$ for any $S \in \mathbf{O}(2) - \mathbf{SO}(2)$. With $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, the elements are $S\left(\begin{smallmatrix} x & -y \\ y & x \end{smallmatrix}\right) = \left(\begin{smallmatrix} -x & y \\ y & x \end{smallmatrix}\right)$; replace x by $-x$. ■

Write $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $S_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$.

Question 9.5.4

Is S_θ a reflection of $\mathbb{R}^2$?

Question 9.5.5

For reflections S_Y, S_Z of $\mathbb{R}^2$, $\det(S_Y S_Z) = (-1)(-1) = 1$, so $S_Y \circ S_Z = R_\theta$ for some θ (Theorem 9.5.2). How do we find θ?

Observation 9.5.6

For $\theta \in \mathbb{R}$ and $Y_\theta = (-\sin\frac{\theta}{2}, \cos\frac{\theta}{2})^t$, $S_\theta = S_{Y_\theta}$. In particular S_θ is a reflection.

Proof. Both are linear, so compare them on e_1, e_2. With $s = \sin\frac{\theta}{2}$, $c = \cos\frac{\theta}{2}$: $S_{Y_\theta}(e_1) = e_1 - 2\langle e_1, Y_\theta\rangle Y_\theta = e_1 + 2s(-s, c)^t = (1-2s^2, 2sc)^t = (\cos\theta, \sin\theta)^t = S_\theta e_1$, and $S_{Y_\theta}(e_2) = e_2 - 2c(-s, c)^t = (2sc, 1-2c^2)^t = (\sin\theta, -\cos\theta)^t = S_\theta e_2$ (half-angle formulas). ■

Theorem 9.5.7

An orthogonal operator L of $\mathbb{R}^2$ is a rotation or a reflection: a rotation if $\det L = 1$, a reflection if $\det L = -1$.

Proof. $\mathbf{O}(2) = \mathbf{SO}(2) \amalg (\mathbf{O}(2) - \mathbf{SO}(2)) = \{R_\theta\} \amalg \{S_\theta\}$ (Corollary 9.5.3), and each S_θ is a reflection (Observation 9.5.6). ■

Example 9.5.8

(a) For a rotation R and a reflection S of $\mathbb{R}^2$, $R \circ S$ and $S \circ R$ are reflections (determinant -1).
(b) For reflections S, S', $S \circ S'$ is a rotation (determinant 1).

By the addition formulas $R_\theta \circ R_\eta = R_{\theta+\eta}$ and $(R_\theta)^{-1} = R_{-\theta}$; by Exercise 9.4.6, $(S_\theta)^2 = I$ and $(S_\theta)^{-1} = S_\theta$. The last rule:

Observation 9.5.9

For a reflection S of $\mathbb{R}^2$ and $\theta \in \mathbb{R}$: $S^{-1} \circ R_\theta \circ S = (R_\theta)^{-1} = R_{-\theta}$, i.e. $R_\theta \circ S = S \circ R_{-\theta}$, or $(S \circ R_\theta)^2 = I$.

Proof. $S \circ R_\theta$ is a reflection (Example 9.5.8), so $(S \circ R_\theta)^2 = I$ by Exercise 9.4.6. Thus $SR_\theta SR_\theta = I$, i.e. $SR_\theta S = (R_\theta)^{-1} = R_{-\theta}$, and $S = S^{-1}$. ■

"Flip, rotate, flip back = rotate backwards."

Example 9.5.10

Fix a reflection S of $\mathbb{R}^2$. By Exercise 9.3.11, $\mathbf{O}(2) - \mathbf{SO}(2) = S \cdot \mathbf{SO}(2)$, so every reflection is uniquely $S \circ R_\theta$. Products of rotations and reflections reduce to $R_\theta R_\eta$, $(SR_\theta)R_\eta$, $R_\theta(SR_\eta)$ and $(SR_\theta)(SR_\eta)$, and Observation 9.5.9 lets us move S past rotations; e.g. $(S \circ R_\theta) \circ (S \circ R_\eta) = S \circ (S \circ R_{-\theta}) \circ R_\eta = R_{\eta-\theta}$.

Example 9.5.11

Answer to Question 9.5.5: find θ, η with $S_Y = S_\theta$, $S_Z = S_\eta$, i.e. $\frac{1}{\|Y\|}Y = \pm(-\sin\frac{\theta}{2}, \cos\frac{\theta}{2})^t$ and $\frac{1}{\|Z\|}Z = \pm(-\sin\frac{\eta}{2}, \cos\frac{\eta}{2})^t$ (Observation 9.5.6, Exercise 9.4.4(a)). With $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = S_\pi$ we have $S \circ R_{\pi-\theta} = S_\theta$, so by Observation 9.5.9

$$S_\theta \circ S_\eta = (S \circ R_{\pi-\theta}) \circ (S \circ R_{\pi-\eta}) = S^2 \circ R_{\theta-\pi} \circ R_{\pi-\eta} = R_{\theta-\eta}.$$

Exercise 9.5.12

For $Y = (-1, 1)^t$, $Z = (-1, \sqrt{3})^t$, find θ with $S_Y \circ S_Z = R_\theta$.

Solution. $\frac{1}{\|Y\|}Y = (-\sin\frac{\pi}{4}, \cos\frac{\pi}{4})^t$ and $\frac{1}{\|Z\|}Z = (-\frac{1}{2}, \frac{\sqrt{3}}{2})^t = (-\sin\frac{\pi}{6}, \cos\frac{\pi}{6})^t$, so $S_Y = S_{\pi/2}$, $S_Z = S_{\pi/3}$ and $S_Y \circ S_Z = R_{\pi/2-\pi/3} = R_{\pi/6}$. □

Exercise 9.5.13

$R_\theta \circ S_\eta = S_{\theta+\eta}$ and $S_\eta \circ R_\theta = S_{\eta-\theta}$.

Solution. Direct multiplication with the addition formulas: the first column of $R_\theta S_\eta$ is $R_\theta(\cos\eta, \sin\eta)^t = (\cos(\theta+\eta), \sin(\theta+\eta))^t$, and similarly for the second column. For the second identity, $S_\eta R_\theta e_1 = S_\eta(\cos\theta, \sin\theta)^t = (\cos(\eta-\theta), \sin(\eta-\theta))^t$, etc. □

Exercise 9.5.14

For a reflection S of $\mathbb{R}^2$, compute $(S \circ R_{\pi/6} \circ S)^{2000}$.

Solution. $S \circ R_{\pi/6} \circ S = R_{-\pi/6}$ (Observation 9.5.9, $S = S^{-1}$). So the power is $R_{-2000\pi/6} = R_{-1000\pi/3}$, and $-\frac{1000\pi}{3} \equiv \frac{2\pi}{3} \pmod{2\pi}$ ($1000 = 3 \cdot 334 - 2$). The answer is $R_{2\pi/3}$. □

Exercise 9.5.15

Every element of $\mathbf{SO}(2)$ is a product of two reflections, but not uniquely.

Solution. By Example 9.5.11, $S_{\theta+\alpha} \circ S_\alpha = R_\theta$ for every $\alpha \in \mathbb{R}$; different $\alpha \pmod{2\pi}$ give different factorizations. □

The great principle of the plane: flip, rotate, flip back = rotate backwards.

9.6 $\mathbf{SO}(3)$ and $\mathbf{SO}(n)$

Motivation. A rotation of $\mathbb{R}^3$ has an axis and turns the perpendicular plane. Every element of $\mathbf{SO}(3)$ has eigenvalue 1 and is such a rotation, so composites of rotations are rotations. This suggests defining a rotation of $\mathbb{R}^n$ as an element of $\mathbf{SO}(n)$.

Definition 9.6.1 (Rotation of $\mathbb{R}^3$)

Let $Z \in \mathbb{R}^3$ be a unit vector and $\mathfrak{B} = \{Z, X, Y\}$ an orthonormal basis of $\mathbb{R}^3$ (so $\{X, Y\}$ is an orthonormal basis of $\langle Z\rangle^\perp$). Then $(Z, X, Y) \in \mathbf{O}(3)$ has determinant ± 1; order X, Y so that $\det(Z, X, Y) = \det(X, Y, Z) = 1$. The **rotation** $R_{Z,\theta} : \mathbb{R}^3 \to \mathbb{R}^3$ is the linear map with

$$[R_{Z,\theta}]_{\mathfrak{B}}^{\mathfrak{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = \operatorname{diag}(1, R_\theta);$$

Z is the **axis**, $\langle X\rangle \oplus \langle Y\rangle$ the **plane of rotation**, θ the **angle**.

Equivalently (linear extension theorem), $R_{Z,\theta}$ is the unique linear map with $R_{Z,\theta}(Z) = Z$, $R_{Z,\theta}(X) = \cos\theta X + \sin\theta Y$, $R_{Z,\theta}(Y) = -\sin\theta X + \cos\theta Y$. Its independence of the choice of $\{X, Y\}$ is Exercise 9.6.6. As in §9.4,

$$[R_{Z,\theta}]_{\mathcal{E}}^{\mathcal{E}} = [I]_{\mathcal{E}}^{\mathfrak{B}} \cdot \operatorname{diag}(1, R_\theta) \cdot [I]_{\mathfrak{B}}^{\mathcal{E}},$$

where $\operatorname{diag}(1, R_\theta) \in \mathbf{SO}(3)$ and $[I]_{\mathfrak{B}}^{\mathcal{E}} = (Z, X, Y) \in \mathbf{SO}(3)$.

Observation 9.6.2

Rotations $R_{Z,\theta}$ of $\mathbb{R}^3$ belong to $\mathbf{SO}(\mathbb{R}^3)$.

Proof. By the displayed product all three factors lie in $\mathbf{SO}(3)$ (Exercise 9.3.10). ■

Question 9.6.3

Is the composite of two rotations of $\mathbb{R}^3$ a rotation? (The question is not obvious when the axes differ.)

Theorem 9.6.4

Every element of $\mathbf{SO}(\mathbb{R}^3)$ is a rotation of $\mathbb{R}^3$.

Proof. (i) *An axis.* Let $L_A \in \mathbf{SO}(\mathbb{R}^3)$. We need a unit vector Z with $AZ = Z$, i.e. $\det(A - I) = 0$. Since $\det A = \det A^t = 1$ and $A^tA = I$,

$$\begin{aligned}\det(A - I) &= \det(A^t)\det(A - I) = \det(A^t(A - I)) = \det(I - A^t)\\ &= \det((I - A)^t) = \det(I - A) = (-1)^3\det(A - I),\end{aligned}$$

so $\det(A - I) = 0$.

(ii) *The plane.* Choose an orthonormal basis $\mathfrak{B} = \{Z, X, Y\}$ with $\det(Z, X, Y) = 1$. Then $[L_A]^{\mathfrak{B}}_{\mathfrak{B}} = [I]^{\mathcal{E}}_{\mathfrak{B}}\, A\, [I]^{\mathfrak{B}}_{\mathcal{E}} \in \mathbf{SO}(3)$ and has first column e_1: $[L_A]^{\mathfrak{B}}_{\mathfrak{B}} = \left(\begin{smallmatrix}1 & a & b\\ 0 & * & *\\ 0 & * & *\end{smallmatrix}\right)$, with lower right 2×2 block B. For its columns to be orthonormal, $a = b = 0$ and $B \in \mathbf{O}(2)$; $\det B = 1$, so $B \in \mathbf{SO}(2)$ and $B = R_\theta$ (Theorem 9.5.2). ■

Second proof of (i). If $A \in \mathbf{O}(3)$ has a real eigenvalue λ with eigenvector X, then $\langle X, X\rangle = \langle AX, AX\rangle = \lambda^2\langle X, X\rangle$, so $\lambda = \pm 1$. Triangularizing A over $\mathbb{C}$ (§7.3), its eigenvalues satisfy $\lambda_1\lambda_2\lambda_3 = \det A = 1$ (Corollary 7.3.3). A real cubic has a real root, say $\lambda_1 = \pm 1$. If $\lambda_2, \lambda_3 \notin \mathbb{R}$, then $\lambda_3 = \bar{\lambda}_2$ and $\lambda_2\lambda_3 = |\lambda_2|^2 > 0$, so $\lambda_1 = 1$. If all are real, they are ± 1 with product 1, so one of them is 1. ■

Corollary 9.6.5

The composite of two rotations of $\mathbb{R}^3$ is a rotation.

Proof. By Observation 9.6.2 and Exercise 9.3.10 the composite lies in $\mathbf{SO}(\mathbb{R}^3)$; apply Theorem 9.6.4. ■

Finding the axis and angle (Z_3, θ_3) of $R_{Z_1,\theta_1} \circ R_{Z_2,\theta_2}$ means writing $R_{Z_i,\theta_i} = L_{A_i}$ and finding the unit eigenvector of A_1A_2 for the eigenvalue 1.

Exercise 9.6.6

Let $\mathfrak{B}' = \{Z, X', Y'\}$ be another orthonormal basis with $\det(Z, X', Y') = 1$, and $R'_{Z,\theta}$ the rotation defined from it. Show $R_{Z,\theta} = R'_{Z,\theta}$:
(a) since $\langle X\rangle \oplus \langle Y\rangle = \langle X'\rangle \oplus \langle Y'\rangle = \langle Z\rangle^\perp$, $[I]^{\mathfrak{B}}_{\mathfrak{B}'} = \operatorname{diag}(1, U)$ for some $U \in \mathfrak{M}_{2,2}(\mathbb{R})$;
(b) $[I]^{\mathfrak{B}}_{\mathfrak{B}'} = [I]^{\mathfrak{B}}_{\mathcal{E}}[I]^{\mathcal{E}}_{\mathfrak{B}'} \in \mathbf{SO}(3)$, hence $U \in \mathbf{SO}(2)$;
(c) $[R'_{Z,\theta}]^{\mathfrak{B}}_{\mathfrak{B}} = [I]^{\mathfrak{B}}_{\mathfrak{B}'}\operatorname{diag}(1, R_\theta)[I]^{\mathfrak{B}'}_{\mathfrak{B}} = \operatorname{diag}(1, R_\theta) = [R_{Z,\theta}]^{\mathfrak{B}}_{\mathfrak{B}}$.

Solution.

(a) Z is fixed and $X', Y' \in \langle X, Y\rangle$, so the change of coordinates has this block form.

(b) Both factors are in $\mathbf{SO}(3)$ (columns orthonormal, determinant 1), so $\operatorname{diag}(1, U) \in \mathbf{SO}(3)$, whence $U \in \mathbf{O}(2)$ with $\det U = 1$.

(c) $\operatorname{diag}(1, U)\operatorname{diag}(1, R_\theta)\operatorname{diag}(1, U^{-1}) = \operatorname{diag}(1, UR_\theta U^{-1})$, and $UR_\theta U^{-1} = R_\theta$ because $\mathbf{SO}(2)$ is commutative ($R_\alpha R_\theta = R_{\alpha+\theta}$). □

Example 9.6.7

Unlike in the plane (Theorem 9.5.7), an orthogonal operator of $\mathbb{R}^3$ need not be a rotation or a reflection: $\operatorname{diag}(-1, R_\theta) \in \mathbf{O}(3)$ has determinant -1 (not a rotation), and for $R_\theta \neq I$ it does not have 1 as an eigenvalue with a 2-dimensional eigenspace, so it is not a reflection.

Exercise 9.6.8

For $L \in \mathbf{O}(\mathbb{R}^3) - \mathbf{SO}(\mathbb{R}^3)$:
(a) L has eigenvalue -1;
(b) $[L]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, R_\theta)$ for some orthonormal basis $\mathfrak{B}$ and $\theta \in \mathbb{R}$.

Solution.

(a) With $L = L_A$, $\det A = -1$: $\det(A + I) = -\det(A^t)\det(A + I) = -\det(I + A^t) = -\det(A + I)$, so $\det(A + I) = 0$.

(b) Take a unit eigenvector Z for -1 and an orthonormal basis $\{Z, X, Y\}$. As in Theorem 9.6.4(ii), $[L]_{\mathfrak{B}}^{\mathfrak{B}} = \operatorname{diag}(-1, B)$ with $B \in \mathbf{O}(2)$ and $\det B = 1$, so $B = R_\theta$. □

So every $L \in \mathbf{O}(\mathbb{R}^3)$ is of one of three types in a suitable orthonormal frame: (1) $\operatorname{diag}(1, R_\theta)$ (a rotation); (2) $\operatorname{diag}(-1, 1, 1)$ (a reflection); (3) $\operatorname{diag}(-1, R_\eta)$ with $R_\eta \neq I$.

Unlike the plane (Theorem 9.5.2), we cannot solve the quadratic equations $A^t A = I$ directly in higher dimensions; the rest of the book finds ways around them.

Definition 9.6.9 (Rotations of $\mathbb{R}^n$)

For every n, an element of $\mathbf{SO}(\mathbb{R}^n)$ (or of $\mathbf{SO}(n)$) is called a **rotation** of $\mathbb{R}^n$.

Example 9.6.10

The 4-dimensional rotation $\operatorname{diag}(R_\theta, R_\eta)$ has two "planes of rotation" with two angles.

Describing all of $\mathbf{O}(\mathbb{R}^n)$ and $\mathbf{SO}(\mathbb{R}^n)$ is a goal of the second half of the book (Chapter 15).

Exercise 9.6.11

(a) $L \in \mathbf{O}(\mathbb{R}^n)$ iff $[L]_{\mathfrak{B}}^{\mathfrak{B}} \in \mathbf{O}(n)$ for some orthonormal basis $\mathfrak{B}$ of $\mathbb{R}^n$.
(b) The same holds with $\mathbf{SO}$ in place of $\mathbf{O}$.

Solution.

(a) Let $P = [I]_{\mathfrak{B}}^{\mathcal{E}}$, whose columns form the orthonormal basis $\mathfrak{B}$, so $P \in \mathbf{O}(n)$. Then $[L]_{\mathfrak{B}}^{\mathfrak{B}} = P^{-1}[L]_{\mathcal{E}}^{\mathcal{E}}P$, and $\mathbf{O}(n)$ is closed under products and inverses (Exercise 9.3.3). So $[L]_{\mathfrak{B}}^{\mathfrak{B}} \in \mathbf{O}(n)$ iff $[L]_{\mathcal{E}}^{\mathcal{E}} \in \mathbf{O}(n)$ iff $L \in \mathbf{O}(\mathbb{R}^n)$.

(b) Additionally $\det[L]_{\mathfrak{B}}^{\mathfrak{B}} = \det L$. □

The intuition behind this chapter is Euclidean geometry; its algebra (groups, orthonormal frames, change of frame inside $\mathbf{O}(n)$) is what we now develop in general.

Chapter 10. Inner Product Spaces

We now leave Euclidean space for general inner product spaces, real or complex, finite- or infinite-dimensional, carrying along the philosophy and the tools of Chapter 9.

10.1 Inner product spaces

Motivation. Extending the dot product to $\mathbb{C}^n$ requires a complex conjugate in the second slot. Abstracting the four properties that all the arguments of Chapter 9 actually used gives the notion of an inner product on any real or complex vector space.

Definition 10.1.1 (Hermitian dot product)

For $X = (a_1, \dots, a_n)^t$, $Y = (b_1, \dots, b_n)^t \in \mathbb{C}^n$,

$$\langle X, Y \rangle = X^t \cdot \bar{Y} = \sum_{i=1}^{n} a_i \bar{b}_i$$

is the (**Hermitian**) **dot product** of $\mathbb{C}^n$.

Remark 10.1.2

Without conjugation, $(1+i, 1-i)^t$ would have "length" 0 ($(1+i)^2 + (1-i)^2 = 0$) and $(i, 0)^t$ an imaginary "length".

Exercise 10.1.3

For $X, Y, Z \in \mathbb{C}^n$, $c \in \mathbb{C}$: (a) $\langle X+Y, Z\rangle = \langle X, Z\rangle + \langle Y, Z\rangle$, $\langle cX, Y\rangle = c\langle X, Y\rangle$;
(b) $\langle X, Y+Z\rangle = \langle X, Y\rangle + \langle X, Z\rangle$, $\langle X, cY\rangle = \bar{c}\langle X, Y\rangle$;
(c) $\langle X, Y\rangle = \overline{\langle Y, X\rangle}$;
(d) if $X \neq 0$, then $\langle X, X\rangle \in \mathbb{R}$ and $\langle X, X\rangle > 0$.

Solution. (a), (b) from $\overline{b+c} = \bar{b} + \bar{c}$, $\overline{cb} = \bar{c}\bar{b}$.

(c) $\overline{\sum b_i \bar{a}_i} = \sum \bar{b}_i a_i$.

(d) $\langle X, X\rangle = \sum |a_i|^2$. □

Rereading Chapter 9, almost every argument used only Exercises 9.1.2 and 10.1.3; so we start from these properties.

Definition 10.1.4 (Inner product on F^n)

($F = \mathbb{R}$ or $\mathbb{C}$.) A function $(X, Y) \mapsto \langle X, Y\rangle$, $F^n \times F^n \to F$, is an **inner product** on F^n if for all $X, Y, Z \in F^n$, $c \in F$: (1) $\langle X+Y, Z\rangle = \langle X, Z\rangle + \langle Y, Z\rangle$; (2) $\langle cX, Y\rangle = c\langle X, Y\rangle$; (3) $\langle X, Y\rangle = \overline{\langle Y, X\rangle}$; (4) if $X \neq 0$, then $\langle X, X\rangle \in \mathbb{R}$ and $\langle X, X\rangle > 0$. (For $F = \mathbb{C}$, also a **Hermitian inner product**.)

Observation 10.1.5

If $\langle \cdot, \cdot \rangle$ is an inner product on F^n: (a) $\langle X, Y+Z\rangle = \langle X, Y\rangle + \langle X, Z\rangle$;
(b) $\langle X, cY\rangle = \bar{c}\langle X, Y\rangle$.

Proof. By (3), $\langle X, Y+Z\rangle = \overline{\langle Y+Z, X\rangle} = \overline{\langle Y, X\rangle} + \overline{\langle Z, X\rangle} = \langle X, Y\rangle + \langle X, Z\rangle$, and $\langle X, cY\rangle = \overline{c\langle Y, X\rangle} = \bar{c}\,\overline{\langle Y, X\rangle} = \bar{c}\langle X, Y\rangle$. ■

Definition 10.1.6 (Inner product space)

Let V be an F-vector space (possibly infinite-dimensional). $\langle\cdot,\cdot\rangle : V \times V \to F$ is an **inner product** if for all $u, v, w \in V$, $c \in F$: (1) $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$; (2) $\langle cv, w\rangle = c\langle v, w\rangle$; (3) $\langle v, w\rangle = \overline{\langle w, v\rangle}$; (4) if $v \neq 0$, then $\langle v, v\rangle \in \mathbb{R}$ and $\langle v, v\rangle > 0$ (*positive definiteness*). V with an inner product is an **inner product space**; for $F = \mathbb{C}$, $\langle\cdot,\cdot\rangle$ is also called **Hermitian**.

Remark 10.1.7

(a) Observation 10.1.5 still holds.
(b) For $F = \mathbb{R}$, conjugation does nothing, so a real inner product is a *symmetric* bilinear form.
(c) For $F = \mathbb{C}$, an inner product is linear in the first variable and **conjugate linear** (anti-linear) in the second.
(d) Every F-subspace of an inner product space is an inner product space.

Exercise 10.1.8

$\langle X, Y\rangle = \sum_{j=1}^{n} j\, a_j b_j$ is an inner product on $\mathbb{R}^n$.

Solution. (1)–(3) are clear; $\langle X, X\rangle = \sum j a_j^2 > 0$ for $X \neq 0$. □

Exercise 10.1.9

$\left\langle \binom{a}{b}, \binom{c}{d} \right\rangle = 2ac - bc - ad + 2bd$ is an inner product on $\mathbb{R}^2$.

Solution. It is bilinear and symmetric, and $\langle (a,b),(a,b)\rangle = 2a^2 - 2ab + 2b^2 = a^2 + b^2 + (a-b)^2 > 0$ for $(a,b) \neq 0$. □

So $\mathbb{R}^n$ carries infinitely many inner products.

Exercise 10.1.10

If $\langle\cdot,\cdot\rangle_1, \langle\cdot,\cdot\rangle_2$ are inner products on V, so is $\langle v, w\rangle = \langle v, w\rangle_1 + \langle v, w\rangle_2$.

Solution. (1)–(3) are additive; for $v \neq 0$ both summands are positive reals. □

Exercise 10.1.11

$\langle f(t), g(t)\rangle = \int_0^1 f(t)g(t)\,dt$ is an inner product on $\mathbb{R}[t]$.

Solution. Bilinearity and symmetry are clear. If $f \neq 0$, $f^2 \geq 0$ is continuous and positive except at finitely many points, so $\int_0^1 f^2 > 0$. □

For $f = f_1 + i f_2 : \mathbb{R} \to \mathbb{C}$ (real and imaginary parts), $\int_a^b f = \int_a^b f_1 + i \int_a^b f_2$; f is continuous iff f_1, f_2 are.

Exercise 10.1.12

$\langle f(t), g(t)\rangle = \int_0^1 f(t)\overline{g(t)}\,dt$ is an inner product on $\mathbb{C}[t]$.

Solution. As in Exercise 10.1.11, with $\langle f, f\rangle = \int_0^1 |f|^2$ and $\overline{\int f\bar{g}} = \int \bar{f}g$. □

Exercise 10.1.13

$C^0[0, 2\pi] = \{f : [0, 2\pi] \to \mathbb{C} \text{ continuous}\}$ is a $\mathbb{C}$-vector space, and $\langle f, g\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(x)\overline{g(x)}\,dx$ is an inner product on it.

Solution. Sums and multiples of continuous functions are continuous. (1)–(3) as before; if $f \neq 0$, $|f|^2$ is continuous, nonnegative and positive on an interval, so $\langle f, f\rangle > 0$. □

Exercise 10.1.14

$\langle A, B\rangle = \operatorname{tr}(A^t \cdot \bar{B})$ is an inner product on $\mathfrak{M}_{n,n}(F)$ (the **trace form**).

Solution. $\operatorname{tr}(A^t\bar{B}) = \sum_{i,j} A_{ji}\bar{B}_{ji}$ is the dot product of A and B viewed as vectors in F^{n^2}, so Exercise 10.1.3 applies. □

10.2 Properties of inner product spaces

Motivation. Norm, Pythagoras, projection, Cauchy–Schwarz and the triangle inequality hold verbatim in any inner product space. Two questions remain: do orthonormal bases exist, and is $V = W \oplus W^\perp$? Also, over $\mathbb{C}$ distance-preserving maps fixing 0 need not be linear.

Throughout, V is an inner product space over F.

Exercise 10.2.1

If $\langle w, v\rangle = 0$ for all $w \in V$, then $v = 0$.

Solution. Take $w = v$ and use positive definiteness. □

Definition 10.2.2 (Norm)

$\|v\| = \sqrt{\langle v, v\rangle}$.

Exercise 10.2.3

(a) $\|v\| \geq 0$, with equality iff $v = 0$.
(b) $\|cv\| = |c|\|v\|$.
(c) If $v \neq 0$, $\frac{1}{\|v\|}v$ is a **unit vector**.

Solution. (b) $\langle cv, cv\rangle = c\bar{c}\langle v, v\rangle = |c|^2\langle v, v\rangle$. The rest as in Exercise 9.1.5. □

Exercise 10.2.4 (Parallelogram law)

$\|v + w\|^2 + \|v - w\|^2 = 2(\|v\|^2 + \|w\|^2)$.

Solution. $\|v \pm w\|^2 = \|v\|^2 \pm (\langle v, w\rangle + \langle w, v\rangle) + \|w\|^2$; add. □

Definition 10.2.5 (Perpendicularity)

(a) $v \perp w$ if $\langle v, w\rangle = 0$ (equivalently $\langle w, v\rangle = 0$).
(b) For $S, T \subseteq V$, $S \perp T$ if $\langle v, w\rangle = 0$ for all $v \in S$, $w \in T$.

Exercise 10.2.6

If $v \perp w$: (a) $\|v + w\|^2 = \|v\|^2 + \|w\|^2$ (Pythagoras);
(b) $\|v + w\| = \|v - w\|$.

Solution. In the expansion of Exercise 10.2.4 the cross terms $\langle v, w\rangle + \langle w, v\rangle$ vanish. □

Exercise 10.2.7

If $w \neq 0$, then $\left(v - \frac{\langle v,w\rangle}{\langle w,w\rangle} w\right) \perp w$ (compare Example 9.4.1).

Solution. $\langle v, w\rangle - \frac{\langle v,w\rangle}{\langle w,w\rangle}\langle w, w\rangle = 0$. □

Observation 10.2.8

(a) (Cauchy–Schwarz) $|\langle v, w\rangle| \leq \|v\|\|w\|$, with equality iff $\{v, w\}$ is linearly dependent.
(b) (Triangle inequality) $\|v + w\| \leq \|v\| + \|w\|$.

Proof. (a) Trivial if $w = 0$. Otherwise let $a = \frac{\langle v,w\rangle}{\langle w,w\rangle}$. By Exercise 10.2.7 and Pythagoras, $\|v\|^2 = \|v - aw\|^2 + \|aw\|^2 \geq |a|^2\|w\|^2$, i.e. $|a|\|w\| \leq \|v\|$, which is the inequality. Equality holds iff $v = aw$.
(b) Write $\operatorname{Re} z$ for the real part; $z + \bar{z} = 2 \operatorname{Re} z \leq 2|z|$. Then $\|v+w\|^2 = \|v\|^2 + \langle v, w\rangle + \overline{\langle v, w\rangle} + \|w\|^2 = \|v\|^2 + 2\operatorname{Re}\langle v, w\rangle + \|w\|^2 \leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2$. ■

So $\|v - w\|$ is a distance on V: every inner product space carries a geometry.

Exercise 10.2.9

(a) $\|v - v\| = 0$.
(b) $\|v - w\| = \|w - v\|$.
(c) $\|v - w\| \leq \|v - u\| + \|u - w\|$.

Solution. As in Exercise 9.1.8, using Observation 10.2.8(b). □

Definition 10.2.10 (Orthogonal and orthonormal sets)

(a) Nonzero vectors $\{v_i\}_{i \in I}$ with $v_i \perp v_j$ for $i \neq j$ form an **orthogonal subset**; if all are unit vectors, an **orthonormal subset**.
(b) For $W \leq V$, a basis $\mathfrak{B}$ of W is an **orthogonal basis** if it is an orthogonal subset, an **orthonormal basis** if it is orthonormal.

Exercise 10.2.11

An orthogonal subset $\{v_i\}_{i \in I}$ is linearly independent.

Solution. If $\sum_{\text{finite}} c_i v_i = 0$, take the inner product with v_j: $c_j \|v_j\|^2 = 0$. □

Exercise 10.2.12

A basis $\{v_i\}$ of $W \leq V$ is orthonormal iff $\langle v_i, v_j \rangle = \delta_{ij}$.

Solution. This restates the definition. □

Exercise 10.2.13

(a) $\{(0,-i)^t, (i,0)^t\}$ is an orthonormal basis of $\mathbb{C}^2$ (dot product).
(b) $\{\frac{1}{\sqrt{3}}(1+i,i)^t, \frac{1}{\sqrt{6}}(1-i,-2)^t\}$ is an orthonormal basis of $\mathbb{C}^2$.
(c) $\{\frac{1}{\sqrt{2}}(1,0)^t, \frac{1}{\sqrt{6}}(1,2)^t\}$ is an orthonormal basis of $\mathbb{R}^2$ with the inner product of Exercise 10.1.9.
(d) $\{1, \sqrt{12}(t-\frac{1}{2})\}$ is an orthonormal basis of $\mathcal{P}_1(\mathbb{R})$ with the inner product of Exercise 10.1.11.
(e) For $f_n(x) = e^{inx} = \cos nx + i \sin nx$ $(n \in \mathbb{Z})$, $\{f_n\}_{n\in\mathbb{Z}}$ is an orthonormal subset of $C^0[0, 2\pi]$ (Exercise 10.1.13).

Solution. (a) Norms 1, and $0 \cdot \bar{i} + (-i) \cdot 0 = 0$. (b) $|1+i|^2 + |i|^2 = 3$, $|1-i|^2 + 4 = 6$, and $(1+i)\overline{(1-i)} + i \cdot \overline{(-2)} = (1+i)^2 - 2i = 0$. (c) $\langle (1,0),(1,0)\rangle = 2$, $\langle (1,2),(1,2)\rangle = 2-2-2+8 = 6$, $\langle (1,0),(1,2)\rangle = 2-0-2+0 = 0$. (d) $\int_0^1 1 = 1$, $\int_0^1 12(t-\frac{1}{2})^2 dt = 1$, $\int_0^1 (t-\frac{1}{2})dt = 0$. (e) $\frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)x}dx = \delta_{mn}$, since for $m \neq n$ the antiderivative $\frac{e^{i(m-n)x}}{i(m-n)}$ is 2π-periodic. □

Question 10.2.14

If V is a (finite-dimensional) inner product space and $W \leq V$, does W always have an orthonormal basis?

Definition 10.2.15 (Orthogonal complement)

$S^\perp = \{v \in V : w \perp v \text{ for all } w \in S\}$; for $W \leq V$, $W^\perp$ is the **orthogonal complement**.

Exercise 10.2.16

For $S \subseteq V$, $W \leq V$ with basis $\mathfrak{C}$: (a) $S^\perp \leq V$;
(b) $S^\perp = \langle S \rangle^\perp$;
(c) $W^\perp = \mathfrak{C}^\perp$.

Solution. As in Observation 9.1.15; in (a) use conjugate linearity: $\langle w, cv\rangle = \bar{c}\langle w, v\rangle = 0$. □

Exercise 10.2.17

For a basis $\mathfrak{B}$ of V and $W \leq V$: (a) $0^\perp = V$, $V^\perp = \mathfrak{B}^\perp = 0$;
(b) $W \cap W^\perp = 0$;
(c) $W \leq (W^\perp)^\perp$.

Solution. As in Exercise 9.1.16, using Exercise 10.2.1. □

Question 10.2.18

If V is finite-dimensional and $W \leq V$, is $\dim W^\perp = \dim V - \dim W$ and $V = W \oplus W^\perp$?

A **rigid motion** of V is $M : V \to V$ with $\|M(v) - M(w)\| = \|v - w\|$. Do Theorem 9.2.11 and Corollary 9.2.12 still hold? For $F = \mathbb{C}$ the answer is no: Observation 9.2.8(c) fails.

Exercise 10.2.19

(a) On $\mathbb{C} = \mathbb{C}^1$ with the dot product, $M(\alpha) = \bar{\alpha}$ is a non-linear rigid motion fixing 0 (M is conjugate linear).
(b) On $\mathbb{C}^2$, $M((\alpha, \beta)^t) = (\bar{\alpha}, \beta)^t$ is a non-linear rigid motion fixing 0, not even conjugate linear.

Solution. (a) $|\bar{\alpha} - \bar{\beta}| = |\overline{\alpha - \beta}| = |\alpha - \beta|$, but $M(i \cdot 1) = -i \neq iM(1)$. (b) $\|M(v) - M(w)\|^2 = |\overline{\alpha - \alpha'}|^2 + |\beta - \beta'|^2 = \|v - w\|^2$; $M(i(1,0)^t) = (-i, 0)^t \neq iM((1,0)^t)$, and $M(i(0,1)^t) = (0, i)^t \neq iM((0,1)^t)$. □

Remark 10.2.20

Via the $\mathbb{R}$-isomorphism $\gamma : \mathbb{C} \to \mathbb{R}^2$, $\gamma(a + ib) = (a, b)^t$, the norms (metrics) of $\mathbb{C}$ and $\mathbb{R}^2$ agree: $\|(a, b)^t - (c, d)^t\| = |(a + ib) - (c + id)|$. So rigid motions of $\mathbb{C}$ and of $\mathbb{R}^2$ are the same thing. Conjugation is a reflection of $\mathbb{R}^2$, which is $\mathbb{R}$-linear but not $\mathbb{C}$-linear. Conversely, every $\mathbb{C}$-linear rigid motion of $\mathbb{C}$ (multiplication by $e^{i\theta}$) is a rotation of $\mathbb{R}^2$.

In Chapter 9 we leaned on the standard orthonormal basis of $\mathbb{R}^n$; in general we do not yet know that orthonormal bases exist.

10.3 Gram–Schmidt orthogonalization

Motivation. Subtracting from each vector its components along the previous ones (the projection of Example 9.4.1) turns any independent list into an orthogonal one with the same spans. This answers Questions 9.1.19, 9.4.8, 10.2.14 and 10.2.18 at once, and ties together the dimension theorem, the rank theorem and the "perp theorem".

Theorem 10.3.1 (Gram–Schmidt orthogonalization)

Let V be an inner product space and $\{v_1, \dots, v_r\}$ linearly independent. (a) Put $w_1 = v_1$ and, for $2 \le i \le r$,

$$w_i = v_i - \frac{\langle v_i, w_{i-1}\rangle}{\langle w_{i-1}, w_{i-1}\rangle} w_{i-1} - \cdots - \frac{\langle v_i, w_1\rangle}{\langle w_1, w_1\rangle} w_1.$$

Then $\langle v_1, \dots, v_r\rangle = \langle w_1, \dots, w_r\rangle$ and $\{w_1, \dots, w_r\}$ is an orthogonal subset.
(b) Hence $\{\frac{1}{\|w_1\|} w_1, \dots, \frac{1}{\|w_r\|} w_r\}$ is an orthonormal subset.
(c) In particular, if $r = \dim V$, it is an orthonormal basis of V.

Proof. Induction on i: assume $\langle v_1, \dots, v_i\rangle = \langle w_1, \dots, w_i\rangle$ with $\{w_1, \dots, w_i\}$ orthogonal (nonzero vectors). Then $w_{i+1} \in v_{i+1} + \langle w_1, \dots, w_i\rangle$, so $\langle v_1, \dots, v_{i+1}\rangle = \langle w_1, \dots, w_{i+1}\rangle$. Also $w_{i+1} \neq 0$, since otherwise $v_{i+1} \in \langle v_1, \dots, v_i\rangle$. For $j \le i$, by orthogonality, $\langle w_{i+1}, w_j\rangle = \langle v_{i+1}, w_j\rangle - \frac{\langle v_{i+1}, w_j\rangle}{\langle w_j, w_j\rangle}\langle w_j, w_j\rangle = 0$. (b), (c) follow by normalizing (Exercise 10.2.11). ■

So finite-dimensional inner product spaces and their subspaces have orthonormal bases.

Exercise 10.3.2

Apply Gram–Schmidt to the basis $\mathfrak{B}$: (a) $\mathfrak{B} = \{(1,1,0)^t, (-1,0,2)^t, (2,1,1)^t\}$ of $\mathbb{R}^3$ (dot product); (b) $\mathfrak{B} = \{(3,1,0,1)^t, (0,-1,2,0)^t, (1,1,-1,-1)^t\}$ of a subspace W of $\mathbb{R}^4$; (c) $\mathfrak{B} =$

$\{(1,1)^t,(0,1)^t\}$ of $\mathbb{R}^2$ with the inner product of Exercise 10.1.9; (d) $\mathfrak{B}=\{1,t,t^2\}$ of $\mathcal{P}_2(\mathbb{R})$ with the inner product of Exercise 10.1.11.

Solution. (a) $w_1=(1,1,0)$, $w_2=(-\frac{1}{2},\frac{1}{2},2)$, $w_3=(\frac{2}{3},-\frac{2}{3},\frac{1}{3})$; orthonormal basis $\frac{1}{\sqrt{2}}(1,1,0)$, $\frac{1}{3\sqrt{2}}(-1,1,4)$, $\frac{1}{3}(2,-2,1)$.

(b) $w_1=(3,1,0,1)$, $w_2=\frac{1}{11}(3,-10,22,1)$, $w_3=\frac{1}{9}(3,2,1,-11)$; orthonormal basis $\frac{1}{\sqrt{11}}(3,1,0,1)$, $\frac{1}{\sqrt{594}}(3,-10,22,1)$, $\frac{1}{\sqrt{135}}(3,2,1,-11)$.

(c) $\langle(1,1),(1,1)\rangle=2$, $\langle(0,1),(1,1)\rangle=1$, so $w_2=(0,1)-\frac{1}{2}(1,1)=(-\frac{1}{2},\frac{1}{2})$ with $\langle w_2,w_2\rangle=\frac{3}{2}$; orthonormal basis $\frac{1}{\sqrt{2}}(1,1)$, $\frac{1}{\sqrt{6}}(-1,1)$.

(d) 1, $t-\frac{1}{2}$, $t^2-t+\frac{1}{6}$; orthonormal basis 1, $\sqrt{3}(2t-1)$, $\sqrt{5}(6t^2-6t+1)$. □

Exercise 10.3.3

If $\{v_1,\dots,v_r\}$ in Theorem 10.3.1 is already orthogonal, then $w_i=v_i$ for all i.

Solution. All the subtracted terms have $\langle v_i,w_j\rangle=\langle v_i,v_j\rangle=0$, by induction. □

Corollary 10.3.4 (Perp theorem, inner product case)

If V is a finite-dimensional inner product space and $W\le V$, then $\dim V=\dim W+\dim W^\perp$.

Proof. Extend an orthonormal basis $\{w_1,\dots,w_r\}$ of W to a basis of V and apply Gram–Schmidt; by Exercise 10.3.3 we obtain an orthonormal basis $\{w_1,\dots,w_n\}$ of V beginning with the given one. Then $W^\perp=\langle w_{r+1},\dots,w_n\rangle$: these vectors are perpendicular to $w_1,\dots,w_r$, and if $v=\sum a_iw_i\in W^\perp$, then $a_j=\langle v,w_j\rangle=0$ for $j\le r$. ■

Corollary 10.3.5

If V is a finite-dimensional inner product space and $W\le V$, then $V=W\oplus W^\perp$.

Proof. $W\cap W^\perp=0$ (Exercise 10.2.17) and the dimensions add up (Corollary 10.3.4). ■

Exercise 10.3.6

For a finite-dimensional inner product space V and $U,W\le V$: (a) $W=(W^\perp)^\perp$;
(b) $(U+W)^\perp=U^\perp\cap W^\perp$;
(c) $(U\cap W)^\perp=U^\perp+W^\perp$.

Solution. (a) $W\le(W^\perp)^\perp$ and both have dimension $\dim W$ by Corollary 10.3.4 (applied twice). (b) $v\perp U+W$ iff $v\perp U$ and $v\perp W$. (c) Apply (b) to $U^\perp,W^\perp$: $(U^\perp+W^\perp)^\perp=U^{\perp\perp}\cap W^{\perp\perp}=U\cap W$, and take $\perp$ using (a). □

Remark 10.3.7

Cancelling $W^\perp$ from $W\oplus W^\perp=V=W^\perp\oplus(W^\perp)^\perp$ to get (a) is nonsense: complements in a direct sum are not unique (Remark 7.6.10).

With the dot product on $\mathbb{R}^n$, $AX=0$ reads $\langle[A]_i^t,X\rangle=0$ for all rows i, so

$$\ker L_A=\langle[A]_1^t,\dots,[A]_m^t\rangle^\perp=[\text{row space of } A]^\perp.$$

Example 10.3.8

For $F = \mathbb{R}$, the perp theorem for the dot product follows from the dimension and rank theorems. For $W = \langle X_1, \dots, X_m \rangle \leq \mathbb{R}^n$, let A have rows X_i^t. Then $W^\perp = \ker L_A$, and by the rank theorem $\dim W = \text{row rank} = \text{column rank} = \dim \operatorname{im} L_A$. The dimension theorem gives $\dim W^\perp = n - \dim W$.

Proposition 10.3.9 (The three pillars, $F = \mathbb{R}$)

Of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (for the dot product on $\mathbb{R}^n$), any two imply the third.

Proof. All three are relations among n, $\dim \ker L_A = \dim W^\perp$ (with W the row space), $\dim W$ (row rank) and $\dim \operatorname{im} L_A$ (column rank): (1) $\dim \ker L_A + \dim \operatorname{im} L_A = n$; (2) $\dim W = \dim \operatorname{im} L_A$; (3) $\dim W + \dim W^\perp = n$. Any two of these linear relations imply the third. ■

In particular this gives a third proof of the rank theorem, one that needs no row reduction. Over $\mathbb{C}$ the "ordinary product" X^tY is not an inner product, and $X \mapsto A\bar{X}$ is not $\mathbb{C}$-linear, so a trick is needed:

Exercise 10.3.10

For $W \leq \mathbb{C}^n$, $\overline{W} = \{\bar{X} : X \in W\}$ is a subspace of $\mathbb{C}^n$ with $\dim \overline{W} = \dim W$.

Solution. $\overline{W}$ is closed under sums and under $c\bar{X} = \overline{\bar{c}X}$. If $\{X_i\}$ is a basis of W, then $\{\bar{X}_i\}$ is a basis of $\overline{W}$ (conjugate any relation). □

Proposition 10.3.11 (The three pillars)

For $F = \mathbb{R}$ or $\mathbb{C}$, of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (for the dot product on F^n), any two imply the third.

Proof. As in Proposition 10.3.9, using $A\bar{X} = 0 \iff \langle [A]_i^t, X \rangle = 0$: the solution space of $A\bar{X} = 0$ is $W^\perp$, and it equals $\overline{\ker L_A}$, of dimension $\dim \ker L_A$ (Exercise 10.3.10). ■

10.4 Standard basis versus orthonormal basis

Motivation. What made the standard basis of $\mathbb{R}^n$ special is that coordinates are inner products and the inner product is the dot product of coordinates. Every orthonormal basis has these properties, which gives orthogonal projections, closest vectors and Bessel's inequality.

For the dot product on F^n and the standard basis $\mathcal{E}$: (a) $X = \sum \langle X, e_i \rangle e_i$;
(b) $\langle X, Y \rangle = [X]_\mathcal{E}^t \overline{[Y]_\mathcal{E}}$.

Observation 10.4.1

If $\mathfrak{B} = \{v_1, \dots, v_n\}$ is an orthogonal basis of V:
(a)′ $v = \sum_i \frac{\langle v, v_i \rangle}{\langle v_i, v_i \rangle} v_i$. If $\mathfrak{B}$ is an orthonormal basis:
(a) $v = \sum_i \langle v, v_i \rangle v_i$, i.e. the i-th coordinate of $[v]_\mathfrak{B}$ is $\langle v, v_i \rangle$;
(b) $\langle v, w \rangle = [v]_\mathfrak{B}^t \cdot \overline{[w]_\mathfrak{B}}$.

Proof. (a) If $v = \sum a_i v_i$, then $\langle v, v_j \rangle = a_j \langle v_j, v_j \rangle$; this gives (a)′ and (a).

(b) For $w = \sum b_i v_i$, $\langle v, w\rangle = \sum_{i,j} a_i \bar{b}_j \langle v_i, v_j\rangle = \sum a_i \bar{b}_i$. ■

So in an orthonormal basis the inner product is "really" the dot product (Example 10.5.4). For example, the foot of the perpendicular from $X = (1,2,3)^t$ to the xy-plane is $(1,2,0)^t = \langle X, e_1\rangle e_1 + \langle X, e_2\rangle e_2$.

Observation 10.4.2

Let W be a finite-dimensional subspace of an inner product space V with orthonormal basis $\{v_1, \ldots, v_m\}$. (a) $V = W \oplus W^\perp$.
(b) (Closest vector problem) For $v \in V$, $w = \langle v, v_1\rangle v_1 + \cdots + \langle v, v_m\rangle v_m \in W$ satisfies $v - w \in W^\perp$, and w is the unique vector of W closest to v.

Proof. (a) With w as in (b), $\langle v - w, v_i\rangle = \langle v, v_i\rangle - \langle v, v_i\rangle = 0$ for all i, so $v - w \in W^\perp$ and $V = W + W^\perp$; also $W \cap W^\perp = 0$.

(b) For $u \in W$, $w - u \in W$ is perpendicular to $v - w$, so by Pythagoras $\|v-u\|^2 = \|v-w\|^2 + \|w-u\|^2 \geq \|v - w\|^2$, with equality iff $u = w$. ■

Exercise 10.4.3

(a) In Euclidean $\mathbb{R}^3$, find an orthonormal basis of $W = \langle (1,1,2)^t\rangle^\perp$ and the vector of W closest to $(1,0,-1)^t$.
(b) Find the foot of the perpendicular from $(1,0,0,0)^t$ to the subspace W of Exercise 10.3.2(b).

Solution. (a) $\frac{1}{\sqrt{2}}(1,-1,0)$, $\frac{1}{\sqrt{3}}(1,1,-1)$. The closest vector is $X - \frac{\langle X,Y\rangle}{\langle Y,Y\rangle} Y = (1,0,-1) + \frac{1}{6}(1,1,2) = (\frac{7}{6}, \frac{1}{6}, -\frac{2}{3})$ (with $Y = (1,1,2)$).
(b) Summing $\langle v, u_i\rangle u_i$ over the orthonormal basis found there: $(\frac{9}{10}, \frac{4}{15}, \frac{2}{15}, \frac{1}{30})$. □

Exercise 10.4.4

If $\{v_1, \ldots, v_n\}$ is an orthonormal basis and $v = \sum a_i v_i$, then $\|v\|^2 = \sum |a_i|^2 = \sum |\langle v, v_i\rangle|^2$.

Solution. Observation 10.4.1(b) with $w = v$, and (a). □

Observation 10.4.5 (Bessel's inequality)

If $\{v_1, \ldots, v_m\}$ is an orthonormal subset of an inner product space V (possibly infinite-dimensional), then $\sum_{i=1}^m |\langle v, v_i\rangle|^2 \leq \|v\|^2$ for all $v \in V$.

Proof. With $W = \langle v_1, \ldots, v_m\rangle$, Observation 10.4.2 gives $v = w + w'$, $w = \sum \langle v, v_i\rangle v_i \in W$, $w' \in W^\perp$, and by Pythagoras $\|v\|^2 = \|w\|^2 + \|w'\|^2 \geq \|w\|^2 = \sum |\langle v, v_i\rangle|^2$. ■

In infinite dimensions one uses a (large) orthonormal subset $\{v_i\}$ and calls $\langle v, v_i\rangle$ the i-th **Fourier coefficient** of v.

Exercise 10.4.6

With $f_n(x) = e^{inx}$ in $C^0[0, 2\pi]$ (Exercise 10.2.13(e)) and $f(x) = x$: (a) $\langle f, f_0\rangle = \pi$ and $\langle f, f_n\rangle = \frac{i}{n}$ for $n \neq 0$;
(b) $\|f\|^2 = \frac{4}{3}\pi^2$;
(c) Bessel's inequality for $\{f_n\}_{-k \leq n \leq k}$ gives $\sum_{n=1}^k \frac{1}{n^2} \leq \frac{\pi^2}{6}$, hence $\sum_{n=1}^\infty \frac{1}{n^2} \leq \frac{\pi^2}{6}$.

Solution. (a) $\frac{1}{2\pi}\int_0^{2\pi} x\,dx = \pi$; for $n \neq 0$, integrating by parts, $\frac{1}{2\pi}\int_0^{2\pi} xe^{-inx}dx = \frac{1}{2\pi}\left[\frac{xe^{-inx}}{-in}\right]_0^{2\pi} = \frac{1}{-in} = \frac{i}{n}$. (b) $\frac{1}{2\pi}\int_0^{2\pi} x^2 dx = \frac{4\pi^2}{3}$. (c) $\frac{4}{3}\pi^2 \geq \pi^2 + 2\sum_{n=1}^{k}\frac{1}{n^2}$. □

10.5 Isomorphisms of inner product spaces

Motivation. Two inner product spaces are "the same" if a linear isomorphism carries one inner product to the other. Orthonormal bases show that a finite-dimensional inner product space is determined by its dimension: it is F^n with the dot product under another name.

Definition 10.5.1 (Isomorphism of inner product spaces)

Inner product spaces V, W over F are **isomorphic** if there is a vector space isomorphism $\varphi : V \to W$ with $\langle \varphi(u), \varphi(v)\rangle = \langle u, v\rangle$ for all $u, v \in V$; φ is an **inner product space isomorphism**. Equivalently, with $(\varphi \times \varphi)(u, v) = (\varphi(u), \varphi(v))$, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{\langle\,,\rangle} & F \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle I} \\ W \times W & \xrightarrow[\langle\,,\rangle]{} & F \end{array}$$

commutes.

Exercise 10.5.2

Is $\varphi \times \varphi : V \times V \to W \times W$ a vector space isomorphism between the product spaces?

Solution. Yes: it is linear componentwise, with inverse $\varphi^{-1} \times \varphi^{-1}$. □

Isomorphic inner product spaces share every property expressible with sums, scalar multiples and inner products.

Exercise 10.5.3

Isomorphism of inner product spaces is an equivalence relation.

Solution. I_V works; inverses of isomorphisms preserve inner products ($\langle \varphi^{-1}u, \varphi^{-1}v\rangle = \langle \varphi\varphi^{-1}u, \varphi\varphi^{-1}v\rangle$); composites of isomorphisms preserve them. □

Example 10.5.4

Let V be finite-dimensional with orthonormal basis $\mathfrak{B} = \{v_1, \ldots, v_n\}$ (Gram–Schmidt), and let $\alpha_{\mathfrak{B}} : V \to F^n$ be the coordinate isomorphism, $\alpha_{\mathfrak{B}}(v_i) = e_i$. Then $\langle \alpha_{\mathfrak{B}}(v_i), \alpha_{\mathfrak{B}}(v_j)\rangle = \delta_{ij} = \langle v_i, v_j\rangle$, and in general, for $v = \sum a_i v_i$, $w = \sum b_i v_i$, Observation 10.4.1(b) gives $\langle \alpha_{\mathfrak{B}}(v), \alpha_{\mathfrak{B}}(w)\rangle = \sum a_i \bar{b}_i = \langle v, w\rangle$. So $\alpha_{\mathfrak{B}}$ is an inner product space isomorphism onto F^n with the dot product.

Theorem 10.5.5 (Classification of finite-dimensional inner product spaces)

For finite-dimensional inner product spaces V, W over F the following are equivalent: (1) V and W are isomorphic as inner product spaces; (2) $\dim V = \dim W$.

Proof. (1)⇒(2): an isomorphism of vector spaces preserves dimension. (2)⇒(1): if $\dim V = \dim W = n$, both are isomorphic to F^n with the dot product (Example 10.5.4). Explicitly, for orthonormal bases $\mathfrak{B}, \mathfrak{C}$, $(\alpha_{\mathfrak{C}})^{-1} \circ \alpha_{\mathfrak{B}}$ is an inner product space isomorphism mapping $\mathfrak{B}$ to $\mathfrak{C}$. ■

Exercise 10.5.6

Write explicit inner product space isomorphisms (a) from $\mathbb{R}^2$ with the dot product to $\mathbb{R}^2$ with the inner product of Exercise 10.1.9; (b) from F^4 with the dot product to $\mathfrak{M}_{2,2}(F)$ with the trace form of Exercise 10.1.14.

Solution. (a) Send e_1, e_2 to the orthonormal basis of Exercise 10.2.13(c): $\varphi(a,b) = \frac{a}{\sqrt{2}}(1,0) + \frac{b}{\sqrt{6}}(1,2)$. (b) The matrix units $E_{11}, E_{12}, E_{21}, E_{22}$ are orthonormal for $\operatorname{tr}(A^t \bar{B})$; send $e_1, \dots, e_4$ to them, i.e. $(a,b,c,d)^t \mapsto \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. □

Now return to rigid motions (Exercise 10.2.19). For $F = \mathbb{R}$ the results of §9.2 carry over:

Theorem 10.5.7

Let V be a finite-dimensional inner product space over $\mathbb{R}$. (a) A rigid motion L of V with $L(0) = 0$ is linear. (b) Every rigid motion of V is a translation composed with a linear rigid motion. (c) Every rigid motion of V is a bijection.

First proof. V is isomorphic as an inner product space to $\mathbb{R}^n$ with the dot product (Theorem 10.5.5), and the statements are Theorem 9.2.11 and Corollary 9.2.12. ■

Second proof. The second proof of Theorem 9.2.11(a) works verbatim with an orthonormal basis in place of $\mathcal{E}$; (b), (c) follow as before. ■

Exercise 10.5.8

Let $\gamma : \mathbb{C}^n \to \mathbb{R}^{2n}$, $\gamma((a_1 + ib_1, \dots, a_n + ib_n)^t) = (a_1, b_1, \dots, a_n, b_n)^t$, with the dot products on both spaces. (a) γ is an isomorphism of $\mathbb{R}$-vector spaces.
(b) $\|\gamma(X) - \gamma(Y)\| = \|X - Y\|$.
(c) If M is a rigid motion of $\mathbb{C}^n$, then $\gamma \circ M \circ \gamma^{-1}$ is a rigid motion of $\mathbb{R}^{2n}$; hence M is a bijection.

Solution. (a) Clear.

(b) $|a + ib|^2 = a^2 + b^2$ coordinatewise.

(c) Distances are preserved by γ, M, γ^{-1}, so $\gamma M \gamma^{-1}$ is a rigid motion of $\mathbb{R}^{2n}$, a bijection by Corollary 9.2.12; then $M = \gamma^{-1}(\gamma M \gamma^{-1})\gamma$ is a bijection. □

10.6 Orthogonal groups and unitary groups

Motivation. The orthogonal operators of Chapter 9 generalize to $\mathbf{O}(V)$ (real) and $\mathbf{U}(V)$ (complex): linear operators preserving distance, length, inner product and orthonormal bases. Their matrices form $\mathbf{O}(n)$ and $\mathbf{U}(n)$, characterised by $A^*A = I$.

Throughout this section spaces are finite-dimensional.

Definition 10.6.1 (Orthogonal and unitary groups)

(a) For a real inner product space V: $\mathbf{O}(V) = \mathbf{O}(V, \langle\cdot,\cdot\rangle) = \{L \in \mathfrak{L}(V,V) : \|Lv - Lw\| = \|v - w\| \ \forall v, w\}$, the **orthogonal group**; its elements are **orthogonal operators**.
(b) For a complex inner product space V: $\mathbf{U}(V) = \mathbf{U}(V, \langle\cdot,\cdot\rangle)$ is defined the same way, the **unitary group**; its elements are **unitary operators**.

Different inner products on V give different groups. Elements of these groups are injective linear maps, hence invertible.

Exercise 10.6.2

For $G = \mathbf{O}(V)$ or $\mathbf{U}(V)$: (a) $I_V \in G$;
(b) $L, M \in G \Rightarrow L \circ M \in G$;
(c) $L \in G \Rightarrow L^{-1} \in G$.

Solution. As in Exercise 9.2.7. □

Definition 10.6.3 (Unitary matrices)

With the Hermitian dot product on $\mathbb{C}^n$, $\mathbf{U}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{C}) : L_A \in \mathbf{U}(\mathbb{C}^n)\} = \{[L]_{\mathcal{E}}^{\mathcal{E}} : L \in \mathbf{U}(\mathbb{C}^n)\}$, the (complex) **unitary group**; its elements are **unitary matrices**. Likewise $\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : L_A \in \mathbf{O}(\mathbb{R}^n)\}$.

Remark 10.6.4

Matrix versions of $\mathbf{O}(V)$, $\mathbf{U}(V)$ give nothing new: if $\mathfrak{B}$ is an orthonormal basis of n-dimensional V, then $\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{U}(V)\} = \mathbf{U}(n)$ (similarly for $\mathbf{O}$). This is essentially Exercise 10.6.11 (compare Exercise 9.6.11); see also §13.3.

Observation 10.6.5

For an n-dimensional inner product space V and $L \in \mathfrak{L}(V,V)$ the following are equivalent: (1) $\|Lv - Lw\| = \|v - w\|$ (i.e. $L \in \mathbf{O}(V)$ or $\mathbf{U}(V)$); (2) $\|Lv\| = \|v\|$; (3) $\langle Lv, Lv\rangle = \langle v, v\rangle$; (4) $\langle Lv, Lw\rangle = \langle v, w\rangle$; (5) if $\mathfrak{B}$ is an orthonormal basis, so is $L(\mathfrak{B})$.

Proof. As in Observation 9.3.4 (for (5)⇒(2) use Exercise 10.4.4), except (3)⇒(4) when $F = \mathbb{C}$. Expanding $\langle L(v+w), L(v+w)\rangle = \langle v+w, v+w\rangle$ gives $\langle Lv, Lw\rangle + \langle Lw, Lv\rangle = \langle v, w\rangle + \langle w, v\rangle$, i.e. $2\,\mathrm{Re}\langle Lv, Lw\rangle = 2\,\mathrm{Re}\langle v, w\rangle$. Expanding $\langle L(v+iw), L(v+iw)\rangle = \langle v+iw, v+iw\rangle$ gives $-i\langle Lv, Lw\rangle + i\langle Lw, Lv\rangle = -i\langle v, w\rangle + i\langle w, v\rangle$, i.e. $2\,\mathrm{Im}\langle Lv, Lw\rangle = 2\,\mathrm{Im}\langle v, w\rangle$. ■

Exercise 10.6.6

One may add to Observation 10.6.5: (2)′ $\|v\| = 1 \Rightarrow \|Lv\| = 1$; (5)′ some orthonormal basis $\mathfrak{B}$ has $L\mathfrak{B}$ orthonormal.

Solution. As in Exercise 9.3.5. □

Definition 10.6.7 (Adjoint matrix)

For $A \in \mathfrak{M}_{m,n}(F)$, $A^* = \overline{A^t} = \bar{A}^t$ is the **adjoint matrix** (for real A, $A^* = A^t$). (It is unrelated to the classical adjoint of §6.7.)

Exercise 10.6.8

For $A, B \in \mathfrak{M}_{m,n}(F)$, $C \in \mathfrak{M}_{n,r}(F)$, $c \in F$: (a) $(A+B)^* = A^* + B^*$;
(b) $(cA)^* = \bar{c}A^*$;
(c) $(A^*)^* = A$;
(d) $(AC)^* = C^*A^*$.

Solution. Conjugation and transposition are additive, conjugation is multiplicative, $(AC)^t = C^tA^t$, and both operations are involutions. □

Exercise 10.6.9

For $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ (Hermitian dot product on $\mathbb{C}^n$) the following are equivalent: (1) $\|AX - AY\| = \|X - Y\|$ (i.e. $A \in \mathbf{U}(n)$); (2) $\|AX\| = \|X\|$; (3) $\langle AX, AX\rangle = \langle X, X\rangle$; (4) $\langle AX, AY\rangle = \langle X, Y\rangle$; (5) the columns form an orthonormal basis of $\mathbb{C}^n$; (6) $A^t\bar{A} = I = \bar{A}A^t$, i.e. $A^*A = I = AA^*$, $A^{-1} = A^*$; (7) $A^t \in \mathbf{U}(n)$; (7)′ $\bar{A} \in \mathbf{U}(n)$; (7)″ $A^* \in \mathbf{U}(n)$; (8) the rows form an orthonormal basis of $\mathfrak{M}_{1,n}(\mathbb{C})$.

Solution. (1)–(5) is Observation 10.6.5 with $\mathfrak{B} = \mathcal{E}$. The (i,j) entry of A^*A is $\overline{\langle [A]^i, [A]^j\rangle}$, which gives (5)⇔(6). (6) is preserved by $A \mapsto A^t, \bar{A}, A^*$ (take transposes and conjugates of $A^*A = I = AA^*$), giving (7), (7)′, (7)″. (8) is (5) for A^t. □

Exercise 10.6.10

For $G = \mathbf{O}(n)$ or $\mathbf{U}(n)$: (a) $I_n \in G$;
(b) $A, B \in G \Rightarrow AB \in G$;
(c) $A \in G \Rightarrow A^{-1} \in G$.

Solution. $(AB)^*AB = B^*A^*AB = I$; $(A^{-1})^*A^{-1} = (A^*)^*A^* = AA^* = I$. □

So $\mathbf{U}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{C}) : A^*A = I\}$. One would like to write $\mathbf{U}(V) = \{L : L^* \circ L = I\}$, but how should L^* be defined? (See Question 10.7.2.)

Exercise 10.6.11

If $\mathfrak{B}, \mathfrak{C}$ are orthonormal bases of an n-dimensional complex inner product space V, then $[I]_{\mathfrak{C}}^{\mathfrak{B}} \in \mathbf{U}(n)$.

Solution. The j-th column of $[I]_{\mathfrak{C}}^{\mathfrak{B}}$ is $[w_j]_{\mathfrak{B}}$ for $w_j \in \mathfrak{C}$. By Observation 10.4.1(b), $\langle [w_i]_{\mathfrak{B}}, [w_j]_{\mathfrak{B}}\rangle = \langle w_i, w_j\rangle = \delta_{ij}$ (dot product of coordinates), so the columns are orthonormal. □

Exercise 10.6.12

If $A \in \mathbf{U}(n)$, then $|\det A| = 1$.

Solution. $1 = \det(A^*A) = \overline{\det A} \det A = |\det A|^2$. □

Definition 10.6.13 (Special unitary group)

$\mathbf{SU}(n) = \{A \in \mathbf{U}(n) : \det A = 1\}$.

Exercise 10.6.14

(a) $\mathbf{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} : \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$. What is $\mathbf{U}(2)$?

(b) Are the elements of $\mathbf{SU}(2)$ diagonalizable over $\mathbb{C}$? Those of $\mathbf{U}(2)$?

Solution. (a) Let $A = \left(\begin{smallmatrix} \alpha & z \\ \beta & w \end{smallmatrix}\right) \in \mathbf{SU}(2)$. The first column is a unit vector, $|\alpha|^2+|\beta|^2 = 1$. The second column $(z, w)^t$ is a unit vector orthogonal to it; the orthogonal complement of $(\alpha, \beta)^t$ is spanned by $(-\bar{\beta}, \bar{\alpha})^t$, so $(z, w) = \lambda(-\bar{\beta}, \bar{\alpha})$ with $|\lambda| = 1$, and $\det A = \lambda(|\alpha|^2 + |\beta|^2) = \lambda = 1$. Conversely such matrices are unitary with determinant 1. Hence $\mathbf{U}(2) = \left\{ \left(\begin{smallmatrix} \alpha & -\lambda\bar{\beta} \\ \beta & \lambda\bar{\alpha} \end{smallmatrix}\right) : |\alpha|^2 + |\beta|^2 = 1,\ |\lambda| = 1 \right\}$.

(b) Yes in both cases: unitary matrices are normal and hence unitarily diagonalizable (Chapter 15). Directly: a unitary A has a unit eigenvector v_1; complete it to an orthonormal basis $\{v_1, v_2\}$. Then $Av_2 \perp Av_1 = \lambda v_1$, so $Av_2 \in \langle v_2 \rangle$, and $\{v_1, v_2\}$ is an eigenbasis. □

For $n \geq 3$ solving the quadratic equations is hopeless; we will need the statement "an inner product can be represented by a matrix", taken up in Chapter 13.

10.7 The adjoint matrix and its applications

Motivation. The identity $\langle AX, Y \rangle = \langle X, A^*Y \rangle$ is the heart of the adjoint. It yields $F^m = \operatorname{im} A \oplus \ker A^*$, the normal equations $A^*AX = A^*B$ of least squares, and minimal solutions $X_0 = A^*Y_0$.

Observation 10.7.1

With dot products on F^n, F^m and $A \in \mathfrak{M}_{m,n}(F)$: $\langle AX, Y \rangle = \langle X, A^*Y \rangle$ and $\langle Y, AX \rangle = \langle A^*Y, X \rangle$ for $X \in F^n$, $Y \in F^m$.

Proof. $\langle AX, Y \rangle = (AX)^t\bar{Y} = X^tA^t\bar{Y}$, and since $\overline{A^*} = A^t$, $\langle X, A^*Y \rangle = X^t\overline{A^*Y} = X^tA^t\bar{Y}$. The second identity is the conjugate of the first. ■

Question 10.7.2

For finite-dimensional inner product spaces V, W and $L \in \mathfrak{L}(V, W)$, does there exist $L^* \in \mathfrak{L}(W, V)$ with $\langle Lv, w \rangle = \langle v, L^*w \rangle$ for all $v \in V$, $w \in W$? What does it mean?

For the rest of this section F^n carries the dot product.

Least squares approximation (LSA). If $AX = B$ ($A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$) has no solution, we seek X_0 minimizing $\|AX_0 - B\|$, an **LSA solution**. Example: given distinct $a_0, \ldots, a_m$ and arbitrary $b_0, \ldots, b_m$, find $g(t) = c_nt^n + \cdots + c_0 \in \mathcal{P}_n(F)$ $(m > n)$ minimizing $\sum_i |g(a_i) - b_i|^2$. With the Vandermonde-type matrix $A = (a_i^j)$, $X = (c_0, \ldots, c_n)^t$, $B = (b_0, \ldots, b_m)^t$, this is $\|AX - B\|^2$; for $m = n$ it is Lagrange interpolation.

Proposition 10.7.3

For $A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$: (a) an LSA solution X_0 ($\|AX_0 - B\| \leq \|AX - B\|$ for all X) always exists; (b) if L_A is injective ($\operatorname{rk} A = n$), it is unique.

Proof. By the closest vector problem (Observation 10.4.2), $F^m = \operatorname{im} A \oplus (\operatorname{im} A)^\perp$, and $B = AX_0 + C$ with $C \in (\operatorname{im} A)^\perp$ gives the unique vector AX_0 of $\operatorname{im} A$ closest to B. If L_A is injective, X_0 is determined by AX_0. ■

Proposition 10.7.4

The solutions of $(A^*A)X = A^*B$ are exactly the LSA solutions.

Proof. X_0 is an LSA solution iff $AX_0 - B \in (\operatorname{im} A)^\perp$, i.e. $0 = \langle AX, AX_0 - B\rangle = \langle X, A^*(AX_0 - B)\rangle$ for all X (Observation 10.7.1), i.e. $A^*(AX_0 - B) = 0$. ■

Exercise 10.7.5

Find the least squares line for the points $(1,1)$, $(2,3)$, $(3,3)$, $(4,2)$ of $\mathbb{R}^2$.

Solution. For $y = ct + d$: $A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}$, $B = (1,3,3,2)^t$, $A^*A = \left(\begin{smallmatrix} 30 & 10 \\ 10 & 4 \end{smallmatrix}\right)$, $A^*B = (24,9)^t$. Solving, $(c,d) = (\frac{3}{10}, \frac{3}{2})$: $y = 0.3t + 1.5$. □

Exercise 10.7.6

Solve the LSA problem for $x + 3y + z = 3$, $2x - y - 2z = 2$, $5x + y - 3z = 6$.

Solution. The coefficient matrix has rank 2, so the normal equations $(A^*A)X = A^*B$ have a line of solutions: $X = (\frac{47}{42}, \frac{4}{7}, 0)^t + t(5,-4,7)^t$, $t \in \mathbb{R}$. □

Observation 10.7.7

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{rk}(A^*A) = \operatorname{rk} A$;
(b) if $m \geq n$ and $\operatorname{rk} A = n$, then $A^*A \in \mathfrak{M}_{n,n}(F)$ is invertible.

Proof. (a) By the dimension theorem it suffices that $\ker(A^*A) = \ker A$. If $A^*AX = 0$, then $0 = \langle X, A^*AX\rangle = \langle (A^*)^*X, AX\rangle = \langle AX, AX\rangle$ (Observation 10.7.1, Exercise 10.6.8(c)), so $AX = 0$. (b) follows. ■

Exercise 10.7.8

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{rk} A = \operatorname{rk} A^t = \operatorname{rk} \bar{A} = \operatorname{rk} A^*$;
(b) $\operatorname{rk}(AA^*) = \operatorname{rk} A$.

Solution. (a) Rank theorem and Exercise 10.3.10.

(b) Observation 10.7.7(a) for A^* and (a). □

Observation 10.7.9

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{im}(A^*) = (\ker A)^\perp$, i.e. $\ker A = (\operatorname{im} A^*)^\perp$;
(b) $\operatorname{im} A = (\ker A^*)^\perp$, i.e. $\ker A^* = (\operatorname{im} A)^\perp$.

Proof. The four equalities are equivalent by Exercise 10.3.6(a) and by replacing A with A^*. We show $\ker A = (\operatorname{im} A^*)^\perp$. If $AX = 0$, then $\langle X, A^*Y\rangle = \langle AX, Y\rangle = 0$. If $X \in (\operatorname{im} A^*)^\perp$, then $\langle Y, AX\rangle = \langle A^*Y, X\rangle = 0$ for all Y, so $AX = 0$. ■

Minimal solution problem. If $AX = B$ is solvable, find the solution of smallest norm (the foot of the perpendicular from 0 to the solution set).

Proposition 10.7.10

Let $AX = B$ ($A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$) have a solution. (a) There is a unique minimal solution X_0 ($AX_0 = B$ and $\|X_0\| \leq \|X\|$ for every solution X).
(b) X_0 is the unique solution lying in $\mathrm{im}(A^*)$.
(c) If $(AA^*)Y_0 = B$, then $X_0 = A^*Y_0$.

Proof. (i) Let $AX_1 = B$. By Observation 10.7.9, $F^n = \mathrm{im}(A^*) \oplus \ker A$, so $X_1 = A^*Y_0 + Z$ with $Z \in \ker A$. Put $X_0 = A^*Y_0$; then $AX_0 = AX_1 = B$, and X_0 is a solution in $\mathrm{im}\, A^*$.

(ii) Every solution is $X = X_0 + Z'$ with $Z' \in \ker A = (\mathrm{im}\, A^*)^\perp$, so $X_0 \perp Z'$ and $\|X\|^2 = \|X_0\|^2 + \|Z'\|^2 \geq \|X_0\|^2$, with equality iff $Z' = 0$.

(iii) If $X \in \mathrm{im}\, A^*$ is a solution, then $X_0 - X \in \mathrm{im}\, A^* \cap \ker A = 0$. (c) follows: $A^*Y_0 \in \mathrm{im}\, A^*$ is a solution. ■

So: solve $(AA^*)Y = B$ for any Y_0; then $X_0 = A^*Y_0$.

Exercise 10.7.11

Find the minimal solutions of (a) $x + 3y + z = 3$, $2x - y - 2z = 2$;
(b) $x + 3y + z = 3$, $2x - y - 2z = 2$, $5x + y - 3z = 7$.

Solution. (a) $AA^* = \left(\begin{smallmatrix} 11 & -3 \\ -3 & 9 \end{smallmatrix}\right)$, and $Y_0 = (AA^*)^{-1}(3,2)^t$ gives $X_0 = A^*Y_0 = (\frac{19}{18}, \frac{34}{45}, -\frac{29}{90})^t$. (b) The third equation is the first plus twice the second, so the system is consistent with the same solution set as (a), and the minimal solution is the same. □

Chapter 11. Groups

A "new beginning": we descend to the bottom layer of abstraction, binary operations and groups, where the whole story of the book actually starts.

11.1 Binary operations and groups

Motivation. Addition of vectors, multiplication of invertible matrices and composition of rigid motions are all associative operations with an identity and inverses. Isolating exactly these properties defines a group, the first abstract algebraic structure we meet.

$X \times Y = \{(x, y) : x \in X, y \in Y\}$ is the **Cartesian product**; (x, y) is an **ordered pair**.

Definition 11.1.1 (Binary operation)

A function $* : X \times X \to X$ is a **binary operation** on X; we write $x * y$, or $x \cdot y$, xy.

Definition 11.1.2 (Associativity)

A binary operation is **associative** if $(xy)z = x(yz)$ for all $x, y, z \in X$; then xyz is meaningful.

Example 11.1.3

On $\mathfrak{M}_{n,n}(F)$, the bracket $[A, B] = AB - BA$ is not associative: for $n = 2$, $[[E_{12}, E_{21}], E_{21}] = [E_{11} - E_{22}, E_{21}] = -2E_{21}$, while $[E_{12}, [E_{21}, E_{21}]] = 0$.

From now on every binary operation is assumed associative.

Definition 11.1.4 (Group)

A set G with a binary operation is a **group** if (G1) $(xy)z = x(yz)$ for all $x, y, z \in G$; (G2) there is $e \in G$ with $xe = x = ex$ for all $x \in G$; (G3) for each $x \in G$ there is $\tilde{x} \in G$ with $x\tilde{x} = e = \tilde{x}x$.

Example 11.1.5

(a) $(\{-1, 1\}, +)$ is not a group ($1 + 1 \notin \{-1, 1\}$), but $(\{-1, 1\}, \times)$ is.
(b) $\{e\}$ with $ee = e$ is the **trivial group**.
(c) $(\mathbb{Z}, +)$ is a group; $(\mathbb{Z}, \times)$ is not.
(d) $(F, +)$ and $(F - \{0\}, \times)$ are groups; $(F, \times)$ is not.
(e) For a vector space V, $(V, +)$ is a group (axioms (V1)–(V4)).
(f) The symmetric group $(S_n, \circ)$ of Chapter 6 is a group.

Observation 11.1.6

A group is not empty.

Proof. By (G2) it contains e. ∎

Observation 11.1.7

The element e of (G2) is unique; it is the **identity** of G, written e_G.

Proof. If e' is another, $e = ee' = e'$. ■

Observation 11.1.8

For each $x \in G$, the $\tilde{x}$ of (G3) is unique; it is the **inverse** x^{-1}.

Proof. If $xx' = e = x'x$ as well, then $\tilde{x} = \tilde{x}e = \tilde{x}(xx') = (\tilde{x}x)x' = ex' = x'$. ■

Exercise 11.1.9

For $g, h \in G$: (a) $(gh)^{-1} = h^{-1}g^{-1}$;
(b) $(g^{-1})^{-1} = g$.

Solution. (a) $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = e$ and likewise on the other side; use uniqueness of inverses.

(b) g satisfies the defining property of the inverse of g^{-1}. □

Definition 11.1.10 (Order)

G is a **finite group** if it is a finite set; then $|G|$ is its **order**. Otherwise G is an **infinite group**.

Definition 11.1.11 (Commutative group)

G is **commutative** (**abelian**) if $xy = yx$ for all $x, y \in G$.

Exercise 11.1.12

$\mu_n = \{\alpha \in \mathbb{C} : \alpha^n = 1\}$, under multiplication of complex numbers, is a finite commutative group of order n.

Solution. If $\alpha^n = \beta^n = 1$, then $(\alpha\beta)^n = 1$ and $(\alpha^{-1})^n = 1$; $1 \in \mu_n$; associativity and commutativity are inherited from $\mathbb{C}$. The roots are $e^{2\pi ik/n}$, $0 \le k < n$, which are distinct, and a polynomial of degree n has at most n roots, so $|\mu_n| = n$. □

Exercise 11.1.13

For $n \ge 3$, S_n is a finite non-commutative group of order $n!$.

Solution. $|S_n| = n!$. $(1\,2)(1\,3)$ sends $1 \mapsto 3 \mapsto 3$ but $(1\,3)(1\,2)$ sends $1 \mapsto 2$; so they differ. □

Exercise 11.1.14

On F^3 define $(a_1, a_2, a_3)^t \bullet (b_1, b_2, b_3)^t = (a_1 + b_1,\ a_2 + b_2 + a_1b_3,\ a_3 + b_3)^t$. Then $(F^3, \bullet)$ is an infinite non-commutative group.

Solution. Both $(a\bullet b)\bullet c$ and $a\bullet(b\bullet c)$ equal $(a_1+b_1+c_1,\ a_2+b_2+c_2+a_1b_3+a_1c_3+b_1c_3,\ a_3+b_3+c_3)$. The identity is 0, and the inverse of a is $(-a_1, -a_2+a_1a_3, -a_3)$. Non-commutative: $e_1\bullet e_3 = (1, 1, 1)$

but $e_3 \bullet e_1 = (1, 0, 1)$. (This is the Heisenberg group of upper unitriangular 3×3 matrices in disguise.) □

11.2 Elementary properties of groups

Motivation. Powers, exponent laws, cancellation, and multiplication tables. Small groups can be classified by their tables, but whether a table is associative cannot be read off it; one needs an actual group, such as μ_n or a product, to realize it.

Notation 11.2.1

(a) (Multiplicative notation) The identity is 1; $g^1 = g$, $g^2 = gg$, …, $g^0 = 1$, $g^{-2} = g^{-1}g^{-1}$, ….
(b) (Additive notation) For an abelian group A written with $+$: identity 0, inverse $-a$, and $\ldots, (-2)a = (-a) + (-a)$, $0a = 0$, $1a = a$, $2a = a + a, \ldots$; $a - b = a + (-b)$.
(c) Commutative groups are usually written multiplicatively and abelian groups additively.
(d) Unless stated otherwise we use multiplicative notation.

Exercise 11.2.2

(a) $(g^n)^{-1} = (g^{-1})^n = g^{-n}$ for $n \in \mathbb{Z}$.
(b) Additively: $-(na) = n(-a) = (-n)a$.

Solution. $g^n(g^{-1})^n = 1$ by cancelling adjacent gg^{-1}; and $g^{-n} = (g^{-1})^n$ by definition (for $n > 0$; the other cases follow). □

Exercise 11.2.3

$(h^{-1}gh)^n = h^{-1}g^nh$ for $n \in \mathbb{Z}$.

Solution. For $n > 0$, the inner hh^{-1} cancel; $n = 0$ is clear; for $n < 0$ apply this to g^{-1}, using $(h^{-1}gh)^{-1} = h^{-1}g^{-1}h$. □

Exercise 11.2.4

Elements $ghg^{-1}h^{-1}$ are **commutators**. G is commutative iff all commutators are 1.

Solution. $ghg^{-1}h^{-1} = 1 \iff gh = hg$. □

Exercise 11.2.5

If $g^2 = 1$ for all $g \in G$, then G is commutative.

Solution. Then $g = g^{-1}$ for all g, so $ghg^{-1}h^{-1} = ghgh = (gh)^2 = 1$. □

Additively, $g^2 = 1$ reads $2a = 0$: a nonzero element can satisfy $2a = 0$ (or $3a = 0$).

Exercise 11.2.6 (Exponent laws, multiplicative)

For $g, h \in G$, $m, n \in \mathbb{Z}$: (a) $g^m g^n = g^{m+n}$;
(b) $(g^m)^n = g^{mn}$;

(c) if G is commutative, $g^n h^n = (gh)^n$.

Solution. For $m, n \geq 0$ count factors; the cases with negative exponents reduce to these using $g^{-k} = (g^{-1})^k$ and cancellation. (c) uses commutativity to regroup $(gh)(gh)\cdots$. □

Exercise 11.2.7 (Exponent laws, additive)

For $a, b \in A$, $m, n \in \mathbb{Z}$: (a) $ma + na = (m+n)a$;
(b) $m(na) = (mn)a$;
(c) $na + nb = n(a+b)$.

Solution. Exercise 11.2.6 in additive notation. □

(a) and (c) look like distributive laws; they make an abelian group a $\mathbb{Z}$-module.

Observation 11.2.8 (Cancellation law)

If $xy = xz$, then $y = z$; if $xz = yz$, then $x = y$.

Proof. $y = (x^{-1}x)y = x^{-1}(xy) = x^{-1}(xz) = z$; the other side is symmetric. ■

Exercise 11.2.9

State Observation 11.2.8 additively.

Solution. If $a + b = a + c$ then $b = c$; if $a + c = b + c$ then $a = b$. □

Example 11.2.10

$xy = x$ implies $y = 1$, since $xy = x1$.

Exercise 11.2.11

(a) If $y = u^{-1}xu$, then $x = uyu^{-1}$.
(b) If $u^{-1}xu = 1$, then $x = 1$.

Solution. (a) Multiply by u on the left and u^{-1} on the right.

(b) By (a), $x = u1u^{-1} = 1$. □

A finite group $G = \{1, x_2, x_3, \dots\}$ can be described by its **multiplication table**, whose (x_i, x_j) entry is $x_i x_j$.

Example 11.2.12

All groups of order 1 are "the same": $\{e\}$ and $(\{0\}, +)$ differ only in names.

Example 11.2.13

If $G = \{1, x\}$, then $x^2 \neq x$ (cancellation), so $x^2 = 1$: the table is forced, and G is μ_2 (or $\{\text{even}, \text{odd}\}$ under addition) up to renaming. All groups of order 2 are the same.

Observation 11.2.14

In a multiplication table every row contains each element exactly once, and so does every column.

Proof. $xy = xz$ forces $y = z$ (cancellation), so a row has no repetitions; G is finite, so each element appears. Similarly for columns. ■

Observation 11.2.15

For fixed $x \in G$, $\lambda_x : G \to G$, $\lambda_x(y) = xy$, is a bijection (it permutes G).

Proof. Injective by cancellation; surjective since $z = \lambda_x(x^{-1}z)$. ■

Example 11.2.16

If $G = \{1, x, y\}$, then $x^2 \neq x$, and $x^2 = 1$ would force $xy = y$, contradicting cancellation; so $x^2 = y$, and by Observation 11.2.14 the table is

	1	x	y
1	1	x	y
x	x	y	1
y	y	1	x

So there is at most one group of order 3 up to renaming.

Question 11.2.17

How can we check that a given table is associative?

Checking all triples is impractical, and the table alone does not tell us. But if some group is *known* to realise the table, associativity is automatic. For $\zeta = e^{2\pi i/3}$, $\mu_3 = \{1, \zeta, \zeta^2\}$, and the renaming $1 \mapsto 1$, $x \mapsto \zeta$, $y \mapsto \zeta^2$ preserves the table; so the table of Example 11.2.16 is associative because multiplication of complex numbers is. (This shows how much Exercise 11.1.12, which rests on the existence of $\mathbb{C}$, really says.)

Example 11.2.18

For $G = \{1, x, y, z\}$, cancellation and Observation 11.2.14 leave four possible tables (up to symmetry): one in which every element squares to 1, and three in which one non-identity element squares to 1 (the second, third and fourth tables), which are $\mu_4 = \{1, -1, i, -i\}$ up to renaming.

Exercise 11.2.19

The second, third and fourth tables of Example 11.2.18 are associative.

Solution. Each is the table of μ_4 after renaming its elements (for instance $x \mapsto -1$, $y \mapsto i$, $z \mapsto -i$, with appropriate choices); multiplication in μ_4 is associative. □

Example 11.2.20

For groups G_1, G_2, the Cartesian product $G_1 \times G_2$ with $(g_1, g_2)(h_1, h_2) = (g_1h_1, g_2h_2)$ is a group, the **direct product**.

Example 11.2.21

$\mu_2 \times \mu_2$ realises the first table of Example 11.2.18; e.g. $(1,-1)(1,-1) = (1,1)$. It is the **Klein 4-group**. It is not the same as μ_4: in $\mu_2 \times \mu_2$ every element squares to the identity, in μ_4 not.

Exercise 11.2.22

(If you can,) show with tables that a group of order 5 is μ_5 up to renaming.

Solution [Supplement] (via Lagrange's theorem, Observation 12.1.10). Let $x \neq 1$. The powers of x form a subgroup $\langle x \rangle$ (§11.3) whose order divides 5, hence equals 5. So $G = \{1, x, x^2, x^3, x^4\}$ with $x^5 = 1$, and $x^k \mapsto e^{2\pi ik/5}$ is a renaming onto μ_5 preserving the table. □

Exercise 11.2.23

If $|G| \leq 5$, then G is commutative; so S_3 is the smallest non-commutative group.

Solution. By Examples 11.2.12, 11.2.13, 11.2.16, 11.2.18 and Exercise 11.2.22, every such group is μ_n or $\mu_2 \times \mu_2$, whose tables are symmetric. S_3 has order 6 and is non-commutative. □

Remark 11.2.24

Semigroups, monoids, and one-sided identities and inverses appear in other texts; they are not needed here, nor is the classification of finite groups.

11.3 Subgroups

Motivation. Subgroups play the role of subspaces. Closure under products and inverses suffices. Intersections of subgroups are subgroups, which gives the subgroup generated by a set; it consists of all words in the generators and their inverses.

Definition 11.3.1 (Subgroup)

A subset $H \subseteq G$ that is a group under the operation of G is a **subgroup**, $H \leq G$.

Observation 11.3.2

Let $H \leq G$. (a) $e_H = e_G$.
(b) The inverse of $h \in H$ in H equals its inverse h^{-1} in G.

Proof. (a) In G, $e_H e_H = e_H = e_H e_G$; cancel e_H.

(b) In G, $hh' = e_H = e_G = hh^{-1}$; cancel h. ■

Exercise 11.3.3

Why was $0_W = 0_V$ for subspaces $W \leq V$ never an issue?

Solution. Because $0_W = 0 \cdot w$ for any $w \in W$, computed in V, which is 0_V: scalar multiplication pins down the zero vector. □

Observation 11.3.4

$H \subseteq G$ is a subgroup iff (S1) $h_1, h_2 \in H \Rightarrow h_1 h_2 \in H$; (S2) $e \in H$; (S3) $h \in H \Rightarrow h^{-1} \in H$.

Proof. Necessity: the operation must restrict to H (S1), and Observation 11.3.2 gives (S2), (S3). Sufficiency: associativity is inherited from G. ■

Exercise 11.3.5

A nonempty $H \subseteq G$ is a subgroup iff $h_1, h_2 \in H \Rightarrow h_1 h_2^{-1} \in H$.

Solution. Necessity is clear. For sufficiency, take $h \in H$: then $e = hh^{-1} \in H$, $h^{-1} = eh^{-1} \in H$, and $h_1 h_2 = h_1 (h_2^{-1})^{-1} \in H$. □

Example 11.3.6

(a) $\{e\} \leq G \leq G$. (b) Under addition, $\mathbb{Z} < \mathbb{Q} < \mathbb{R} < \mathbb{C}$. (c) Under multiplication, $\mu_2 < \mathbb{Q} - \{0\} < \mathbb{R} - \{0\} < \mathbb{C} - \{0\}$. (d) The alternating group A_n is a subgroup of S_n. (e) If $m \mid n$, then $\mu_m \leq \mu_n$. (f) If W is a subspace of V, then $(W, +) \leq (V, +)$.

Observation 11.3.7

If $K \leq H$ and $H \leq G$, then $K \leq G$.

Proof. K is a group under the operation of H, which is that of G. ■

Exercise 11.3.8

If $I \neq \emptyset$ and $H_i \leq G$ for all $i \in I$, then $\bigcap_i H_i \leq G$.

Solution. Each H_i contains e and is closed under products and inverses; so is the intersection. □

Notation 11.3.9

For $H, K \subseteq G$, $HK = \{hk : h \in H, k \in K\}$.

Exercise 11.3.10

(a) If G is commutative and $H, K \leq G$, then $HK \leq G$.
(b) In $S_3 = \{id, (1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)\}$, $H = \{id, (1\,2)\}$ and $K = \{id, (2\,3)\}$ are subgroups but $HK \not\leq S_3$.

Solution. (a) $(h_1 k_1)(h_2 k_2)^{-1} = h_1 h_2^{-1} k_1 k_2^{-1} \in HK$ by commutativity; apply Exercise 11.3.5. (b) $HK = \{id, (2\,3), (1\,2), (1\,2)(2\,3)\}$ has four elements, while KH contains $(2\,3)(1\,2) \neq (1\,2)(2\,3)$. If HK were a subgroup it would contain $(2\,3)(1\,2) = ((1\,2)(2\,3))^{-1}$, which is not in HK. □

Definition 11.3.11 (Generated subgroup)

For $S \subseteq G$, the smallest subgroup of G containing S is the **subgroup generated by** S, $\langle S \rangle$: $S \subseteq \langle S \rangle \leq G$, and $S \subseteq H \leq G$ implies $\langle S \rangle \subseteq H$.

Observation 11.3.12

$\langle S \rangle = \bigcap_{S \subseteq H \leq G} H$; in particular $\langle S \rangle$ exists and is unique.

Proof. By Exercise 11.3.8 the intersection is a subgroup containing S, and it lies in every subgroup containing S. ■

Example 11.3.13

$\langle \emptyset \rangle = \{e\}$, and for $x \in G$, $\langle x \rangle = \{x^i : i \in \mathbb{Z}\}$ (this set is a subgroup by the exponent laws, and any subgroup containing x contains it). It need not be infinite: $x^i = x^j$ is possible for $i \neq j$.

Definition 11.3.14 (Cyclic group)

(a) $\langle x \rangle$ is the **cyclic subgroup** generated by x.
(b) If $G = \langle x \rangle$ for some x, G is a **cyclic group** with generator x. Cyclic groups are commutative (§11.7).

Definition 11.3.15 (Alphabet and words)

(a) For $S \subseteq G$ let $S^{-1} = \{s^{-1} : s \in S\}$; $S \cup S^{-1}$ is the **alphabet**.
(b) Finite products of letters are **words**; e is the empty word. (Different words may represent the same element.)

Observation 11.3.16

The set of all words in the alphabet $S \cup S^{-1}$ is a subgroup of G.

Proof. It contains the empty word, products of words are words, and $(s_1 \cdots s_k)^{-1} = s_k^{-1} \cdots s_1^{-1}$ is a word (Exercise 11.1.9); apply Observation 11.3.4. ■

Proposition 11.3.17

$\langle S \rangle$ is the set K of all words in the alphabet $S \cup S^{-1}$.

Proof. $\langle S \rangle$ is a subgroup containing S, so it contains all such words: $K \subseteq \langle S \rangle$. Conversely K is a subgroup (Observation 11.3.16) containing S, so $\langle S \rangle \subseteq K$. ■

Exercise 11.3.18

(a) $(\mathbb{Z}, +)$ is cyclic; how many generators does it have?
(b) $3\mathbb{Z} = \{3a : a \in \mathbb{Z}\}$ is a cyclic subgroup of $(\mathbb{Z}, +)$.
(c) $\langle 2, 3 \rangle = \mathbb{Z} = \langle 4, 15 \rangle$, but $\langle 6, 9 \rangle \neq \mathbb{Z}$.

Solution. (a) $\mathbb{Z} = \langle 1 \rangle = \langle -1 \rangle$, and these are the only generators: $\langle n \rangle = n\mathbb{Z} \neq \mathbb{Z}$ for $|n| \neq 1$.

(b) $3\mathbb{Z} = \langle 3 \rangle$.

(c) $1 = 3-2 = 4\cdot 4-15$, so 1 lies in both; but every element of $\langle 6,9\rangle$ is a multiple of 3, so $\langle 6,9\rangle = 3\mathbb{Z}$. □

Exercise 11.3.19

(a) μ_5 is cyclic; how many generators?
(b) μ_6 is cyclic; how many generators?

Solution. $\mu_n = \langle\zeta\rangle$ with $\zeta = e^{2\pi i/n}$, and ζ^k generates iff $\gcd(k,n) = 1$. (a) 4 generators ($k = 1,2,3,4$).

(b) 2 generators ($k = 1,5$). □

Exercise 11.3.20

(a) Find all subgroups of S_3.
(b) Is S_3 cyclic?
(c) $\langle(1\,2),(2\,3)\rangle = S_3 = \langle(1\,2),(1\,2\,3)\rangle$.

Solution. (a) $\{id\}$, $\langle(1\,2)\rangle$, $\langle(1\,3)\rangle$, $\langle(2\,3)\rangle$, $A_3 = \langle(1\,2\,3)\rangle$, S_3. (Any subgroup containing two distinct transpositions, or a transposition and a 3-cycle, contains their products and is all of S_3, by (c).)

(b) No: the elements have orders $1,2,3$, so no single element generates the 6 elements.

(c) $\langle(1\,2),(2\,3)\rangle$ contains $(1\,2)(2\,3)$ and $(2\,3)(1\,2)$, the two 3-cycles, and $(1\,2)(2\,3)(1\,2) = (1\,3)$: all six elements. Likewise $(1\,2)$ and $(1\,2\,3)$ generate $(1\,2\,3)^2$ and the products $(1\,2)(1\,2\,3)$, $(1\,2\,3)(1\,2)$, which are the other transpositions. □

11.4 The heart of undergraduate algebra

Motivation. Every subgroup of $(\mathbb{Z},+)$ is $n\mathbb{Z}$, by the division algorithm applied to its least positive element. The same argument, with polynomials in place of integers, gave the uniqueness of the minimal polynomial; it reappears in many guises.

Example 11.4.1

For $n \in \mathbb{Z}$, $n\mathbb{Z} = \{na : a \in \mathbb{Z}\} = \langle n\rangle$ is a subgroup of $(\mathbb{Z},+)$.

Theorem 11.4.2 (The heart of undergraduate algebra)

The subgroups of $(\mathbb{Z},+)$ are exactly the $n\mathbb{Z}$, $0 \le n \in \mathbb{Z}$.

Proof. Let $H \le \mathbb{Z}$. If $H = \{0\}$, then $H = 0\mathbb{Z}$. Otherwise H contains a positive integer (with $-a$ for a), and we let n be the least positive element of H (well-ordering). For $a \in H$, divide: $a = qn + r$ with $0 \le r < n$. Then $qn \in H$, so $r = a - qn \in H$, and minimality forces $r = 0$. Hence $H \le n\mathbb{Z}$, and $n\mathbb{Z} = \langle n\rangle \le H$. ■

11.5 Group isomorphisms

Motivation. Groups that differ only in the names of their elements, with "rename then multiply = multiply then rename", are isomorphic. Everything a group theorist studies is invariant under isomorphism.

Definition 11.5.1 (Isomorphism)

Groups G, G' are **isomorphic**, $G \approx G'$, if there is a bijection $\varphi : G \to G'$ with $\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2)$; φ is a (**group**) **isomorphism**. Equivalently, with $(\varphi \times \varphi)(g_1, g_2) = (\varphi g_1, \varphi g_2)$, the square

$$\begin{array}{ccc} G \times G & \xrightarrow{*} & G \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ G' \times G' & \xrightarrow[*']{} & G' \end{array}$$

commutes.

Example 11.5.2

Let $\varphi : G \to H$ be an isomorphism, $g \in G$, $S \subseteq G$. Then (a) $\varphi(e_G) = e_H$;
(b) $\varphi(g^{-1}) = \varphi(g)^{-1}$;
(c) if $\langle S \rangle = G$, then $\langle \varphi(S) \rangle = H$;
(d) if $g^2 = e_G$, then $\varphi(g)^2 = e_H$;
(e) if G is commutative, so is H;
(f) if G is cyclic, so is H;
(g) if G has six subgroups, so does H.

Proof [Supplement]. (a) $\varphi(e)\varphi(e) = \varphi(e) = \varphi(e)e_H$; cancel.

(b) $\varphi(g)\varphi(g^{-1}) = \varphi(e) = e_H$.

(c) φ maps words in $S \cup S^{-1}$ onto words in $\varphi S \cup (\varphi S)^{-1}$ (Proposition 11.3.17) and is onto.

(d) Apply φ.

(e) $\varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a)$, and φ is onto.

(f) Case of (c).

(g) $K \mapsto \varphi(K)$ is a bijection between the sets of subgroups (with inverse $K' \mapsto \varphi^{-1}(K')$). ■

Group theory is the study of properties invariant under isomorphism.

Exercise 11.5.3

Is $\mathbb{Z} \approx \mathbb{Z} \times \mathbb{Z}$ (additive groups)?

Solution. No. $\mathbb{Z}$ is cyclic, so its isomorphic image would be cyclic (Example 11.5.2(f)). But $\mathbb{Z} \times \mathbb{Z}$ is not cyclic: $\langle (a, b) \rangle = \{(na, nb)\}$ never contains both $(1, 0)$ and $(0, 1)$. □

Example 11.5.4

Properties *not* preserved by isomorphisms: "the operation is called multiplication", "G has an element named x", "the elements are complex numbers", "$|g| = 7$ as a real number", "g integrates to h".

Observation 11.5.5

$\approx$ is an equivalence relation.

Proof. Exercise 11.5.6. ■

Exercise 11.5.6

(a) id_G is an isomorphism.
(b) If $\varphi : G \to H$ is an isomorphism, so is φ^{-1}.
(c) If $\psi : K \to G$, $\varphi : G \to H$ are isomorphisms, so is $\varphi \circ \psi$.

Solution. (b) With $h_i = \varphi(g_i)$: $\varphi^{-1}(h_1h_2) = \varphi^{-1}(\varphi(g_1g_2)) = g_1g_2 = \varphi^{-1}(h_1)\varphi^{-1}(h_2)$. (a), (c) are direct. □

Example 11.5.7

$\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a group isomorphism, with inverse log: $e^{a+b} = e^a e^b$.

Remark 11.5.8

For vector spaces, isomorphism is decided by the dimension. For groups, deciding whether two given groups are isomorphic (the *isomorphism problem*) is in general hard.

11.6 Group homomorphisms

Motivation. Homomorphisms are the structure-preserving maps of groups, the analogue of linear maps. Kernels, images and injectivity work as for linear maps.

Definition 11.6.1 (Homomorphism)

$\varphi : G \to H$ is a (**group**) **homomorphism** if $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in G$.

Observation 11.6.2

For a homomorphism φ: (a) $\varphi(e) = e$;
(b) $\varphi(x^{-1}) = \varphi(x)^{-1}$.

Proof. As in Example 11.5.2(a), (b). ■

Definition 11.6.3 (Kinds of homomorphisms)

A homomorphism $\varphi : G \to H$ is a **monomorphism** if injective, an **epimorphism** if surjective, an **isomorphism** if bijective, an **endomorphism** if $G = H$, and an **automorphism** if it is a bijective endomorphism.

Observation 11.6.4

A homomorphism $\varphi : G \to H$ is an isomorphism iff there is a homomorphism $\psi : H \to G$ with $\psi\varphi = id_G$ and $\varphi\psi = id_H$.

Proof. If φ is an isomorphism, take $\psi = \varphi^{-1}$ (Exercise 11.5.6). Conversely such a ψ is an inverse function, so φ is bijective. ■

Definition 11.6.5 (Kernel and image)

$\ker\varphi = \varphi^{-1}(e) = \{x \in G : \varphi(x) = e\}$ and $\operatorname{im}\varphi = \varphi(G)$.

Observation 11.6.6

For a homomorphism $\varphi : G \to H$: (a) $\ker\varphi \leq G$ and $\operatorname{im}\varphi \leq H$;
(b) if φ is a monomorphism, $G \approx \operatorname{im}\varphi$;
(c) if $K \leq G$, then $\varphi(K) \leq H$.

Proof [Supplement]. (a) If $\varphi x = \varphi y = e$, then $\varphi(xy^{-1}) = e$, and $e \in \ker\varphi$ (Exercise 11.3.5).

(c) $\varphi(x)\varphi(y)^{-1} = \varphi(xy^{-1}) \in \varphi(K)$; the case $K = G$ gives $\operatorname{im}\varphi \leq H$.

(b) $\varphi : G \to \operatorname{im}\varphi$ is a bijective homomorphism. ■

Observation 11.6.7

For a homomorphism φ the following are equivalent: (1) φ is a monomorphism; (2) $\varphi x = \varphi y \Rightarrow x = y$; (3) $\varphi x = \varphi e = e \Rightarrow x = e$; (4) $\ker\varphi = \{e\}$.

Proof. All but (3)⇒(2) are immediate. If (3) holds and $\varphi x = \varphi y$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = e$, so $xy^{-1} = e$, $x = y$. ■

Observation 11.6.8

For a homomorphism $\varphi : G \to H$ and $S \subseteq G$: (a) $\varphi\langle S\rangle = \langle\varphi S\rangle$;
(b) if $G = \langle S\rangle$, then $\operatorname{im}\varphi = \langle\varphi S\rangle$;
(c) φ is an epimorphism iff it maps a generating set of G to a generating set of H.

Proof. (a) Elements of $\langle S\rangle$ are words in $S \cup S^{-1}$ (Proposition 11.3.17); their images are words in $\varphi S \cup (\varphi S)^{-1}$, so $\varphi\langle S\rangle \leq \langle\varphi S\rangle$. Conversely $\varphi\langle S\rangle$ is a subgroup of H (Observation 11.6.6(c)) containing φS, so $\langle\varphi S\rangle \leq \varphi\langle S\rangle$.

(b) The case $\langle S\rangle = G$.

(c) By (b), $\operatorname{im}\varphi = H$ iff $\langle\varphi S\rangle = H$. ■

Exercise 11.6.9

If $\varphi : G \to H$ is a homomorphism and G is cyclic, then $\operatorname{im}\varphi$ is cyclic.

Solution. If $G = \langle x\rangle$, then $\operatorname{im}\varphi = \langle\varphi x\rangle$ by Observation 11.6.8(b). □

Exercise 11.6.10

Let $\varphi(g) = g^{-1}$ and $\psi(g) = g^2$ on G. (a) φ is a bijection; is it a homomorphism?
(b) Is ψ a homomorphism?

Solution. (a) φ is its own inverse. It is a homomorphism iff $(xy)^{-1} = x^{-1}y^{-1}$, i.e. $y^{-1}x^{-1} = x^{-1}y^{-1}$ for all x, y, iff G is commutative.

(b) $\psi(xy) = \psi(x)\psi(y)$ iff $xyxy = xxyy$, iff $yx = xy$; so again iff G is commutative. □

Exercise 11.6.11

(a) For $n \geq 2$, $\operatorname{sgn} : S_n \to \mu_2$ is an epimorphism.
(b) Describe $\ker(\operatorname{sgn})$.

Solution. (a) $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}\sigma \operatorname{sgn}\tau$, and $\operatorname{sgn}(1\,2) = -1$.

(b) $\ker(\operatorname{sgn}) = A_n$, the even permutations. □

11.7 Cyclic groups

Motivation. The simplest groups: up to isomorphism a cyclic group is $\mathbb{Z}$ or μ_n. The order of an element is read off from the kernel of $a \mapsto x^a$, another avatar of "the heart of undergraduate algebra".

Example 11.7.1

(a) $\mathbb{Z} = \langle 1 \rangle$ is an infinite cyclic group, and so is $n\mathbb{Z} = \langle n \rangle$ for $n > 0$; $\varphi(a) = na$ is an isomorphism $\mathbb{Z} \approx n\mathbb{Z}$, so unlike a finite-dimensional vector space $\mathbb{Z}$ is isomorphic to a proper subgroup.
(b) For $\zeta = e^{2\pi i/n}$, $\mu_n = \{1, \zeta, \ldots, \zeta^{n-1}\} = \langle \zeta \rangle$ is a finite cyclic group of order n (the n-th roots of unity on the unit circle).

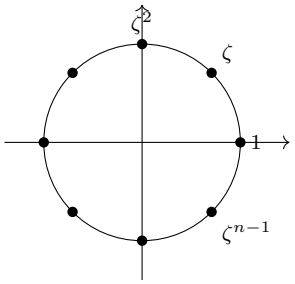


Figure. *μ_n ($n = 8$): the vertices of a regular n-gon on the unit circle.*

Definition 11.7.2 (Order of an element)

The **order** of $x \in G$ is $|x| = |\langle x \rangle|$.

Observation 11.7.3

If x has finite order, then $|x| = n$ iff (1) $x^n = 1$ and (2) $x^m \neq 1$ for $0 < m < n$; i.e. n is the least positive integer with $x^n = 1$.

Proof. $\langle x \rangle$ is finite, so $x^i = x^j$ for some $i < j$, and $x^{j-i} = 1$: the set $\{a \geq 1 : x^a = 1\}$ is nonempty. Let n be its least element. Then $\langle x \rangle = \{1, x, \ldots, x^{n-1}\}$: for $b \in \mathbb{Z}$, $b = qn + r$ with $0 \leq r < n$ and $x^b = (x^n)^q x^r = x^r$. These n powers are distinct, since $x^k = x^\ell$ with $0 \leq k < \ell < n$ would give $x^{\ell-k} = 1$ with $0 < \ell - k < n$. So $|x| = n$. Conversely, if n satisfies (1) and (2), it is the least such integer, and the same argument gives $|\langle x \rangle| = n$. ■

Definition 11.7.4 (Order, second definition)

For $x \in G$, $\gamma_x : \mathbb{Z} \to G$, $\gamma_x(a) = x^a$, is a homomorphism (exponent laws), so $\ker \gamma_x = n\mathbb{Z}$ for a unique $n \geq 0$ (Theorem 11.4.2). Set $|x| = \infty$ if $n = 0$ and $|x| = n$ otherwise.

Exercise 11.7.5

Definitions 11.7.2 and 11.7.4 agree.

Solution. If $\ker \gamma_x = 0$, γ_x is injective and $\langle x \rangle = \operatorname{im} \gamma_x$ is infinite. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, then n is the least positive integer with $x^n = 1$, and $|\langle x \rangle| = n$ by Observation 11.7.3. □

Exercise 11.7.6

(a) $|x| = 1$ iff $x = 1$.
(b) $|x| = 2$ iff $1 \neq x = x^{-1}$.
(c) Every nonzero element of $(\mathbb{R}, +)$ has order ∞.
(d) A k-cycle in S_n has order k.

Solution. (a),

(b) by Observation 11.7.3.

(c) $na \neq 0$ for $n \neq 0$, $a \neq 0$.

(d) A k-cycle $(i_1 \ \ldots \ i_k)$ to the power m sends $i_1 \mapsto i_{1+m}$ (indices mod k), which is i_1 only if $k \mid m$, and the k-th power is the identity. □

Exercise 11.7.7

If $|x| = n$ and $a, b \in \mathbb{Z}$, then $x^a = x^b$ iff $n \mid (a - b)$.

Solution. $x^a = x^b \iff x^{a-b} = 1 \iff a - b \in \ker \gamma_x = n\mathbb{Z}$. □

Exercise 11.7.8

Every subgroup of a cyclic group is cyclic.

Solution. Let $G = \langle x \rangle$, $H \leq G$. Then $\gamma_x : \mathbb{Z} \to G$ is onto, $\gamma_x^{-1}(H) \leq \mathbb{Z}$ equals $m\mathbb{Z}$ (Theorem 11.4.2), and $H = \gamma_x(m\mathbb{Z}) = \langle x^m \rangle$. □

Exercise 11.7.9

Up to isomorphism the cyclic groups are $\mathbb{Z}$ and the μ_n.

Solution. If $G = \langle x \rangle$ and $\ker \gamma_x = 0$, then $\gamma_x : \mathbb{Z} \to G$ is an isomorphism. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, define $\varphi : G \to \mu_n$, $\varphi(x^a) = \zeta^a$. It is well defined and injective because $x^a = x^b \iff n \mid a - b \iff \zeta^a = \zeta^b$ (Exercise 11.7.7), it is onto, and it is a homomorphism by the exponent laws. □

11.8 Examples of groups and homomorphisms

Motivation. A catalogue: symmetric groups and their presentation by generators and relations, Cayley's theorem (every group is a permutation group), conjugacy, automorphisms and products.

Exercise 11.8.1

(a) Composites of homomorphisms are homomorphisms.
(b) If $L \in \mathfrak{L}(V,W)$, then $L : (V,+) \to (W,+)$ is a homomorphism.

Solution. (a) $\varphi\psi(xy) = \varphi(\psi x \psi y) = \varphi\psi x\, \varphi\psi y$.

(b) $L(v+w) = Lv + Lw$. □

Exercise 11.8.2

(a) The constant map $\mathbf{1}(g) = 1$ is a homomorphism, the **trivial map** (written 0 for abelian groups). (b) For $H \leq G$, the inclusion $\iota_H : H \to G$ is a monomorphism, the **natural embedding**. (c) For $H \leq G$ and a homomorphism $\varphi : G \to G'$, the restriction $\varphi|_H$ is a homomorphism.

Solution. All follow directly from the definitions ($1 \cdot 1 = 1$; the operation of H is that of G). □

Example 11.8.3

"Every permutation is a product of transpositions" says S_n is *generated* by transpositions. $\mathrm{sgn} : S_n \to \mu_2$ is a homomorphism with $\ker(\mathrm{sgn}) = A_n$, so $A_n \leq S_n$ automatically (Observation 11.6.6).

Example 11.8.4

(a) In S_3, put $\sigma = (1\,2)$, $\tau = (1\,2\,3)$. Then $\sigma^2 = id = \tau^3$, $\tau^2 = (1\,3\,2)$, $\sigma\tau = (2\,3)$, $\sigma\tau^2 = (1\,3)$, so

$$S_3 = \{id, \tau, \tau^2, \sigma, \sigma\tau, \sigma\tau^2\} = \{\sigma^i\tau^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\} = \langle \sigma, \tau \rangle.$$

(b) Moreover $\sigma^{-1}\tau\sigma = \tau^2 = \tau^{-1}$, i.e. $\tau\sigma = \sigma\tau^2$. This rule moves σ to the left, so the three relations

$$\sigma^2 = 1, \qquad \tau^3 = 1, \qquad \tau\sigma = \sigma\tau^2$$

determine the whole multiplication table of S_3; e.g. $(\sigma\tau)(\sigma\tau^2) = \sigma(\tau\sigma)\tau^2 = \sigma\sigma\tau^2\tau^2 = \tau$.
(c) This is the plane rule "flip, rotate, flip back = rotate backwards" (Observation 9.5.9); see Example 11.9.23.

Definition 11.8.5 (Symmetric group on a set)

For a nonempty set X, $S_X = \{\sigma : X \to X \text{ bijection}\}$ is a group under composition, the **symmetric group** on X.

Exercise 11.8.6

(a) How should "X and Y are the same as sets" be defined?
(b) If X, Y are isomorphic as sets, then $S_X \approx S_Y$.

Solution. (a) There is a bijection $f : X \to Y$ (a renaming).

(b) $\sigma \mapsto f \circ \sigma \circ f^{-1}$ is a bijection $S_X \to S_Y$ with inverse $\tau \mapsto f^{-1}\tau f$, and $f\sigma\sigma' f^{-1} = (f\sigma f^{-1})(f\sigma' f^{-1})$. □

Exercise 11.8.7

(a) The **automorphism group** $\mathrm{Aut}(G) = \{\varphi : G \to G \text{ automorphism}\}$ is a subgroup of S_G.
(b) For $g \in G$, $\mathrm{Int}(g)(x) = gxg^{-1}$ defines $\mathrm{Int}(g) \in \mathrm{Aut}(G)$, an **inner automorphism**; $\mathrm{Int}(G) = \{\mathrm{Int}(g) : g \in G\} \leq \mathrm{Aut}(G)$.
(c) (Remark) Describing $\mathrm{Aut}(G)$ in general is hard; one first asks whether there are **outer** automorphisms (not inner).

Solution. (a) Composites and inverses of automorphisms are automorphisms (Exercise 11.5.6), and $id \in \mathrm{Aut}(G)$. (b) $\mathrm{Int}(g)(xy) = gxyg^{-1} = gxg^{-1}gyg^{-1}$. It is bijective with inverse $\mathrm{Int}(g^{-1})$. And $\mathrm{Int}(g) \circ \mathrm{Int}(h) = \mathrm{Int}(gh)$, so $\mathrm{Int}(G)$ is closed under composition and inverses. □

Theorem 11.8.8 (Cayley)

$\lambda : G \to S_G$, $\lambda(x) = \lambda_x$ with $\lambda_x(g) = xg$, is a monomorphism. Hence every group is isomorphic to a subgroup of a symmetric group.

Proof. $\lambda_x \in S_G$ by Observation 11.2.15. For $x, y, g \in G$, $\lambda_{xy}(g) = xyg = \lambda_x(\lambda_y(g))$, so $\lambda_{xy} = \lambda_x \circ \lambda_y$ and λ is a homomorphism. If $\lambda_x = id$, then $x = \lambda_x(1) = 1$, so $\ker \lambda = \{1\}$ (Observation 11.6.7). By Observation 11.6.6(b), $G \approx \mathrm{im}\,\lambda \leq S_G$. ■

(With right multiplication $\rho_x(g) = gx$ one would get $\rho_{xy} = \rho_y \circ \rho_x$, an anti-homomorphism.) S_G is far bigger than G, so Cayley's theorem is of limited practical use.

Definition 11.8.9 (Conjugacy)

$x \sim y$ iff $y = g^{-1}xg$ for some $g \in G$; this is an equivalence relation, the **conjugacy relation**, and its classes are the **conjugacy classes** (x is **conjugate** to y).

Exercise 11.8.10

(a) The conjugacy class of 1 is $\{1\}$.
(b) If G is commutative, conjugacy is equality.
(c) How many conjugacy classes does S_3 have?

Solution. (a) $g^{-1}1g = 1$.

(b) $g^{-1}xg = x$.

(c) Three: $\{id\}$, the three transpositions, the two 3-cycles (conjugation preserves cycle type, and $\sigma^{-1}\tau\sigma = \tau^2$, $\tau^{-1}\sigma\tau = \sigma\tau^2$, ...). □

Exercise 11.8.11

(a) For $H \leq G$, $g \in G$: $g^{-1}Hg = \{g^{-1}hg : h \in H\} \leq G$.
(b) On the set $\mathcal{S}$ of subgroups of G, $H \sim K$ iff $K = g^{-1}Hg$ for some g is an equivalence relation (K is a **conjugate subgroup** of H).

Solution. (a) $g^{-1}Hg = \mathrm{Int}(g^{-1})(H)$, the image of a subgroup under a homomorphism.

(b) Reflexive with $g = 1$; symmetric since $H = gKg^{-1}$; transitive since $h^{-1}(g^{-1}Hg)h = (gh)^{-1}H(gh)$. □

Exercise 11.8.12

(a) Give the Cartesian product $G = G_1 \times \cdots \times G_n$ of groups a natural group structure. (b) The i-th **projection** $\pi_i : G \to G_i$ is an epimorphism and the j-th **embedding** $\iota_j : G_j \to G$ is a monomorphism.

Solution. (a) Componentwise multiplication $(g_i)(h_i) = (g_i h_i)$; identity $(1, \ldots, 1)$, inverse (g_i^{-1}).

(b) $\pi_i(g) = g_i$ is a surjective homomorphism; $\iota_j(g) = (1, \ldots, g, \ldots, 1)$ (g in position j) is an injective homomorphism. □

Exercise 11.8.13

(a) $G \times H \approx H \times G$.
(b) $(G \times H) \times K \approx G \times (H \times K) \approx G \times H \times K$.

Solution. $(g, h) \mapsto (h, g)$ and $((g, h), k) \mapsto (g, h, k)$ and $(g, (h, k)) \mapsto (g, h, k)$ are bijective homomorphisms. □

Exercise 11.8.14

$G_1 \times \cdots \times G_n$ is commutative iff all G_i are commutative.

Solution. If all G_i commute, products commute componentwise. Conversely each $G_i \approx \iota_i(G_i)$ is a subgroup of a commutative group. □

Exercise 11.8.15

(a) $\mu_3 \times \mu_3 \not\approx \mu_9$.
(b) $\mu_2 \times \mu_3 \approx \mu_6$.

Solution. (a) Every element of $\mu_3 \times \mu_3$ has order dividing 3, while μ_9 has an element of order 9; orders of elements are preserved by isomorphisms.

(b) $(-1, \omega)$ ($\omega = e^{2\pi i/3}$) has order 6: its powers are $(-1)^k$ and ω^k, which are both trivial only for $2 \mid k$ and $3 \mid k$. So $\mu_2 \times \mu_3$ is cyclic of order 6, hence $\approx \mu_6$ (Exercise 11.7.9). □

Exercise 11.8.16

In Exercise 11.8.12(b): (a) find $\ker \pi_i$ and $\operatorname{im} \iota_j$;
(b) $\pi_i \circ \iota_i = id$;
(c) $\pi_i \circ \iota_j = \mathbf{1}$ for $i \neq j$.

Solution. (a) $\ker \pi_i = \{g : g_i = 1\}$ and $\operatorname{im} \iota_j = \{g : g_k = 1 \text{ for } k \neq j\}$. (b), (c) are direct. □

Remark 11.8.17

For homomorphisms $\varphi, \psi : G \to H$, the pointwise product $(\varphi\psi)(g) = \varphi(g)\psi(g)$ need not be a homomorphism: $(\varphi\psi)(xy) = \varphi(x)\varphi(y)\psi(x)\psi(y)$ differs from $\varphi(x)\psi(x)\varphi(y)\psi(y)$ unless H is commutative. So homomorphisms do not form a group the way linear maps form a vector space.

11.9 Linear groups

Motivation. Groups of invertible matrices and operators are where groups and linear algebra meet. A basis identifies $\mathrm{GL}(V)$ with $\mathrm{GL}_n(F)$ and $\mathrm{SL}(V)$ with $\mathrm{SL}_n(F)$, and an orthonormal basis identifies $\mathbf{O}(V)$ with $\mathbf{O}(n)$. By Cayley's theorem, every finite group is a matrix group.

Example 11.9.1

(a) $\mathrm{GL}_n(F) = \{A \in \mathfrak{M}_{n,n}(F) : A \text{ invertible}\}$, the **general linear group**, is a group under matrix multiplication; $F^\times = \mathrm{GL}_1(F) = F - \{0\}$ is the **unit** (multiplicative) **group** of F.
(b) Subgroups of $\mathrm{GL}_n(F)$ are **linear** (**matrix**) **groups**.
(c) The **special linear group** $\mathrm{SL}_n(F) = \{A \in \mathrm{GL}_n(F) : \det A = 1\} = \ker(\det : \mathrm{GL}_n(F) \to F^\times)$ is a subgroup.

Example 11.9.2

For a finite-dimensional F-vector space V: (a) $\mathrm{GL}(V) = \{L \in \mathfrak{L}(V,V) : L \text{ invertible}\}$ is a group under composition; (b) its subgroups are **linear groups**;
(c) $\mathrm{SL}(V) = \{L \in \mathrm{GL}(V) : \det L = 1\} \leq \mathrm{GL}(V)$.

Example 11.9.3

(a) For $\dim V = n$ and a basis $\mathfrak{B}$, the vector space isomorphism $\Psi^{\mathfrak{B}}_{\mathfrak{B}} : \mathfrak{L}(V,V) \to \mathfrak{M}_{n,n}(F)$, $L \mapsto [L]^{\mathfrak{B}}_{\mathfrak{B}}$, restricts to a group isomorphism $\mathrm{GL}(V) \to \mathrm{GL}_n(F)$: it preserves products ($[LM] = [L][M]$) and maps invertible maps exactly onto invertible matrices.

$$\begin{array}{ccc} \mathrm{GL}(V) & \hookrightarrow & \mathfrak{L}(V,V) \\ {\scriptstyle \Psi^{\mathfrak{B}}_{\mathfrak{B}}}\downarrow & & \downarrow{\scriptstyle \Psi^{\mathfrak{B}}_{\mathfrak{B}}} \\ \mathrm{GL}_n(F) & \hookrightarrow & \mathfrak{M}_{n,n}(F) \end{array}$$

(c) Since $\det L = \det[L]^{\mathfrak{B}}_{\mathfrak{B}}$, it also restricts to $\mathrm{SL}(V) \approx \mathrm{SL}_n(F)$.

Example 11.9.4

If $\dim V = \dim W = n$, then $\mathrm{GL}(V) \approx \mathrm{GL}(W)$. (a) Via bases: $(\Psi^{\mathfrak{C}}_{\mathfrak{C}})^{-1} \circ \Psi^{\mathfrak{B}}_{\mathfrak{B}}$. (b) Via a vector space isomorphism $\varphi : V \to W$ (e.g. with $\varphi(\mathfrak{B}) = \mathfrak{C}$): $\Gamma(L) = \varphi \circ L \circ \varphi^{-1}$. Γ is well defined, $\Gamma(LL') = \varphi L \varphi^{-1} \varphi L' \varphi^{-1} = \Gamma(L)\Gamma(L')$, and $\Gamma^{-1}(M) = \varphi^{-1} M \varphi$. (c) If $\varphi(\mathfrak{B}) = \mathfrak{C}$, the two isomorphisms coincide: $[\varphi L \varphi^{-1}]^{\mathfrak{C}}_{\mathfrak{C}} = [\varphi]^{\mathfrak{C}}_{\mathfrak{B}} [L]^{\mathfrak{B}}_{\mathfrak{B}} [\varphi^{-1}]^{\mathfrak{B}}_{\mathfrak{C}} = [L]^{\mathfrak{B}}_{\mathfrak{B}}$, since $[\varphi]^{\mathfrak{C}}_{\mathfrak{B}} = I$. ("There is only one natural thing.")

Example 11.9.5

$\mu_2 < \mathbb{R}^\times < \mathbb{C}^\times$ and $\mu_n < \mathbb{C}^\times$.

Example 11.9.6

Theorem 1.3.6(3) (every invertible matrix is a product of elementary matrices) says: $\mathrm{GL}_n(F)$ is generated by elementary matrices.

Example 11.9.7

$\mathbf{T}_n(F) = \{A \in \mathrm{GL}_n(F) : A \text{ upper triangular}\}$ and $\mathbf{V}_n(F) = \{A \in \mathbf{T}_n(F) : A \text{ unipotent}\}$ (diagonal entries 1) are matrix groups.

Proof [Supplement]. Products of upper triangular matrices are upper triangular with diagonal entries multiplied, and the inverse of an invertible upper triangular matrix is upper triangular (back substitution) with inverted diagonal entries. So both sets are closed under products and inverses and contain I. ■

Exercise 11.9.8

$(F, +)$ is a matrix group: $(F, +) \approx \mathbf{V}_2(F)$.

Solution. $a \mapsto \left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)$ is a bijection, and $\left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & a+b \\ 0 & 1 \end{smallmatrix}\right)$. □

Exercise 11.9.9

The group $(F^3, \bullet)$ of Exercise 11.1.14 is isomorphic to $\mathbf{V}_3(F)$.

Solution. $(a_1, a_2, a_3)^t \mapsto \left(\begin{smallmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{smallmatrix}\right)$; the product of two such matrices has entries $a_1 + b_1$, $a_2 + b_2 + a_1 b_3$, $a_3 + b_3$, which is the formula for $\bullet$. (So associativity there came for free.) □

Example 11.9.10

Conjugacy in $\mathrm{GL}_n(F)$ is the similarity relation of §5.5.

Exercise 11.9.11

If $L \in \mathrm{GL}(V)$ is unipotent and $L \neq I$, then $|L| = \infty$ in $\mathrm{GL}(V)$ (over $F = \mathbb{R}$ or $\mathbb{C}$).

Solution. Write $L = I + N$ with N nilpotent, $N \neq 0$, and let $k \geq 1$ be largest with $N^k \neq 0$. Then $L^m = \sum_{j \leq k} \binom{m}{j} N^j$, and the coefficient of N in $L^m - I = mN + \dots$ is $m \neq 0$. Since $I, N, \dots, N^k$ are linearly independent (if $\sum c_j N^j = 0$ with c_r the first nonzero coefficient, multiplying by N^{k-r} gives $c_r N^k = 0$), $L^m \neq I$ for every $m \geq 1$. □

Exercise 11.9.12

For fixed $C \in \mathrm{GL}_n(\mathbb{R})$, $\varphi(A) = CAC^{-1}$ and $\psi(A) = (A^t)^{-1}$ belong to $\mathrm{Aut}(\mathrm{SL}_n(\mathbb{R}))$. If $C \notin \mathrm{SL}_n(\mathbb{R})$, are φ, ψ outer automorphisms?

Solution. Both preserve $\mathrm{SL}_n(\mathbb{R})$ (det is unchanged) and products ($((AB)^t)^{-1} = (A^t)^{-1}(B^t)^{-1}$), and are bijective ($\psi^2 = id$, $\varphi^{-1} = \mathrm{Int}(C^{-1})$).

[Supplement] φ. An element $C' \in \mathrm{SL}_n(\mathbb{R})$ induces the same automorphism iff $C^{-1}C'$ commutes with all of $\mathrm{SL}_n(\mathbb{R})$, i.e. is a scalar λI (it must commute with all elementary matrices of type 3). So φ is inner iff $\lambda^n \det C = 1$ for some real λ. If n is odd this is solvable, and φ is inner. If n is even and $\det C < 0$, it is not, and φ is outer.

ψ. For $n = 2$, ψ is inner: $(A^t)^{-1} = JAJ^{-1}$ with $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$. For $n \geq 3$, ψ is outer, because inner automorphisms preserve the trace. $A = \mathrm{diag}(2, 3, \frac{1}{6}, 1, \dots, 1)$ has trace $2 + 3 + \frac{1}{6} + (n-3)$, while $\psi(A) = \mathrm{diag}(\frac{1}{2}, \frac{1}{3}, 6, 1, \dots, 1)$ has trace $\frac{1}{2} + \frac{1}{3} + 6 + (n-3)$. □

Exercise 11.9.13

For the invertible diagonal matrices $\mathbf{D}_n(F)$: (a) $\mathbf{D}_n(F) \approx F^\times \times \cdots \times F^\times$ (n factors);
(b) $\mathbf{D}_n(F) \cap \mathrm{SL}_n(F) \approx (F^\times)^{n-1}$ for $n \geq 2$.

Solution. (a) $\mathrm{diag}(a_1, \ldots, a_n) \mapsto (a_1, \ldots, a_n)$.

(b) The last entry is determined by $a_n = (a_1 \cdots a_{n-1})^{-1}$; project to the first $n-1$ entries. □

Exercise 11.9.14

(a) If $m \leq n$, $\mathrm{GL}_m(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.
(b) If $n_1 + \cdots + n_r = n$, $\mathrm{GL}_{n_1}(F) \times \cdots \times \mathrm{GL}_{n_r}(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.

Solution. (b) $(A_1, \ldots, A_r) \mapsto A_1 \oplus \cdots \oplus A_r$ (block diagonal) is a monomorphism. (a) is the case $A \mapsto A \oplus I_{n-m}$. □

Exercise 11.9.15

(a) $\mathbf{SO}(n) < \mathbf{O}(n) < \mathrm{GL}_n(\mathbb{R})$;
(b) $\mathbf{SU}(n) < \mathbf{U}(n) < \mathrm{GL}_n(\mathbb{C})$;
(c) $\mathbf{SO}(n) = \mathrm{SL}_n(\mathbb{R}) \cap \mathbf{O}(n)$;
(d) $\mathbf{SU}(n) = \mathrm{SL}_n(\mathbb{C}) \cap \mathbf{U}(n)$.

Solution. By Exercises 9.3.3, 9.3.10, 10.6.10 and the definitions. □

Example 11.9.16

With $\mathbf{O}(\mathbb{R}^n) = \mathbf{O}(\mathbb{R}^n, \text{dot product})$, $\Psi_{\mathcal{E}}^{\mathcal{E}}$ restricts to $\mathbf{O}(\mathbb{R}^n) \approx \mathbf{O}(n)$ (by the definition of $\mathbf{O}(n)$). For any orthonormal basis $\mathfrak{B}$ of $\mathbb{R}^n$, $\Psi_{\mathfrak{B}}^{\mathfrak{B}}$ also restricts to $\mathbf{O}(\mathbb{R}^n) \approx \mathbf{O}(n)$, by Remark 10.6.4: $\mathbf{O}(n) = \{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{O}(\mathbb{R}^n)\}$ (see Exercise 9.6.11).

Exercise 11.9.17

(a) $\mathbf{U}(\mathbb{C}^n) \approx \mathbf{U}(n)$.
(b) $\mathbf{SO}(\mathbb{R}^n) \approx \mathbf{SO}(n)$.
(c) Define $\mathbf{SU}(\mathbb{C}^n)$ and show $\mathbf{SU}(\mathbb{C}^n) \approx \mathbf{SU}(n)$.

Solution. As in Example 11.9.16, restricting $\Psi_{\mathcal{E}}^{\mathcal{E}}$; for (b), (c) also $\det L = \det [L]_{\mathcal{E}}^{\mathcal{E}}$, with $\mathbf{SU}(\mathbb{C}^n) = \{L \in \mathbf{U}(\mathbb{C}^n) : \det L = 1\}$. □

Example 11.9.18

If $\varphi : V \to W$ is an inner product space isomorphism, $\Gamma(L) = \varphi L \varphi^{-1}$ (Example 11.9.4) restricts to $\mathbf{O}(V) \approx \mathbf{O}(W)$ (resp. $\mathbf{U}$), since L preserves inner products iff $\varphi L \varphi^{-1}$ does (orthogonal operators are the inner product space automorphisms).

Example 11.9.19

The **circle group** $S^1 = \{X \in \mathbb{R}^2 : \|X\| = 1\} = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times$, and $\varphi : \mathbf{SO}(2) \to S^1$, $\varphi(R_\theta) = e^{i\theta}$, is a group isomorphism.

Exercise 11.9.20

$\mathbf{SO}(2) \approx S^1 \approx \mathbf{U}(1)$.

Solution. $\varphi(R_\theta R_\eta) = \varphi(R_{\theta+\eta}) = e^{i(\theta+\eta)} = e^{i\theta}e^{i\eta}$, and φ is bijective (Theorem 9.5.2). $\mathbf{U}(1) = \{z \in \mathbb{C} : \bar{z}z = 1\} = S^1$. □

Exercise 11.9.21

$S_n \approx \{I_\sigma \in \mathfrak{M}_{n,n}(F) : \sigma \in S_n\} \approx \{P_\sigma \in \mathfrak{L}(V,V) : \sigma \in S_n\}$ ($\dim V = n$), via permutation matrices and permutation operators.

Solution. The permutation matrix I_σ (with $I_\sigma e_j = e_{\sigma(j)}$) satisfies $I_\sigma I_\tau = I_{\sigma\tau}$, and $\sigma \mapsto I_\sigma$ is injective; similarly for P_σ on a basis. □

Corollary 11.9.22

Every finite group is isomorphic to a linear (matrix) group.

Proof. By Cayley's theorem G embeds in $S_G \approx S_n$ ($n = |G|$), and S_n embeds in $\mathrm{GL}_n(F)$ by Exercise 11.9.21. ■

(Representing a group by matrices is a *linear representation*; by permutations, a *permutation representation*.)

Example 11.9.23

In $\mathbf{O}(2)$ let $R = R_{2\pi/3}$, $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and $D_3 = \langle R, S\rangle \leq \mathbf{O}(2)$.
(a) By the plane rule $S^{-1}RS = R^{-1} = R^2$, i.e. $RS = SR^2$. Since $\{R,S\} \cup \{R,S\}^{-1} = \{R, R^2, S\}$, D_3 consists of words in R, R^2, S; moving every S to the left with $RS = SR^2$ gives $D_3 = \{S^iR^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\}$, and $|D_3| = 6$ (the six matrices are distinct: compare determinants and first columns). The relations $S^2 = I$, $R^3 = I$, $RS = SR^2$ determine the table; e.g. $(SR)(SR^2) = S(RS)R^2 = S^2R^4 = R$.
(b) These are exactly the relations of $S_3 = \langle \sigma, \tau\rangle$ with $\sigma^2 = 1$, $\tau^3 = 1$, $\tau\sigma = \sigma\tau^2$ (Example 11.8.4).
(c) So $\varphi(\sigma^i\tau^j) = S^iR^j$ is a bijection $S_3 \to D_3$ preserving the multiplication tables (which both relations completely determine): $S_3 \approx D_3$.

Exercise 11.9.24

For $n \geq 3$, the **dihedral group** $D_n = \langle R_{2\pi/n}, S\rangle \leq \mathbf{O}(2)$ has $|D_n| = 2n$.

Solution. With $R = R_{2\pi/n}$: $R^n = I$, $S^2 = I$, $RS = SR^{-1}$, so as in Example 11.9.23 $D_n = \{S^iR^j : 0 \leq i \leq 1, 0 \leq j < n\}$. These $2n$ elements are distinct: $\det(S^iR^j) = (-1)^i$ separates i, and R^j, $0 \leq j < n$, are distinct rotations. □

Exercise 11.9.25

(a) For $\sigma = (25)(34)$, $\tau = (12345) \in S_5$, $G_5 = \langle \sigma, \tau\rangle \approx D_5$. (b) For $\sigma = (26)(35)$, $\tau = (123456) \in S_6$, $G_6 = \langle \sigma, \tau\rangle \approx D_6$. (c) Interpret (a),
(b) as permutations of the vertices of a regular pentagon/hexagon, and generalize to the regular n-gon.

Solution. In both cases $\sigma^2 = 1$, $\tau^n = 1$, and $\sigma\tau\sigma = \tau^{-1}$ (check on 1: $\sigma\tau\sigma(1) = \sigma\tau(1) = \sigma(2) = n = \tau^{-1}(1)$, and similarly on the other points). So $\sigma^i\tau^j \mapsto S^iR^j$ is an isomorphism onto D_n, as in Example 11.9.23; both groups have $2n$ elements. Geometrically, numbering the vertices of a regular n-gon $1, \ldots, n$ counterclockwise, τ is the rotation by $2\pi/n$ and σ the reflection in the axis through vertex 1; D_n is the symmetry group of the regular n-gon. □

Chapter 12. Quotients

Quotient groups and quotient spaces raise our language one level and supply powerful abstract tools; in particular, induction on dimension via quotient spaces becomes a basic technique of linear algebra.

12.1 Cosets

Motivation. The left cosets xH of a subgroup partition the group into pieces of equal size. Counting them gives Lagrange's theorem: the order of a subgroup divides the order of the group, which already classifies groups of prime order.

Definition 12.1.1 (Cosets)

For $H \leq G$, $x \in G$: $xH = \{xh : h \in H\}$ (or $x + H$ additively) is the **left coset** of H with **representative** x, and Hx the **right coset**. "Coset" means left coset from now on.

Observation 12.1.2

For $H \leq G$, $x, y \in G$: (a) $x \in xH$, so $G = \bigcup_{x \in G} xH$;
(b) $xH = yH$ iff $y^{-1}x \in H$;
(c) $xH \cap yH = \emptyset$ if $xH \neq yH$.

Proof. (a) $x = x1$. (b) If $xH = yH$, then $x \in yH$ by (a), $x = yh'$, so $y^{-1}x = h' \in H$. Conversely, if $y^{-1}x = h_0 \in H$, then $xh = yh_0h \in yH$, so $xH \subseteq yH$; likewise $yH \subseteq xH$ (as $x^{-1}y = h_0^{-1} \in H$). (c) If $xh_1 = yh_2$, then $y^{-1}x = h_2h_1^{-1} \in H$, so $xH = yH$ by (b). ■

So the distinct cosets partition G, which is the equivalence relation

$$x \equiv y \pmod{H} \iff xH = yH \iff y^{-1}x \in H,$$

with classes $\bar{x} = xH$. The set of cosets $G/H = \frac{G}{H} = \{xH : x \in G\} = \{\bar{x} : x \in G\}$ is the **coset space** of G modulo H. For a **complete set of representatives** $\{x_i\}_{i \in I}$, $G/H = \{\bar{x}_i\}$ and $G = \coprod_i x_i H$ (**coset decomposition**).

Example 12.1.3

$1H = H$ is always a coset.

Exercise 12.1.4

$|G/G| = 1$ and $|G/\{1\}| = |G|$.

Solution. $xG = G$ for all x; $x\{1\} = \{x\}$ are all distinct. □

Example 12.1.5

The solution set $X_0 + \ker L_A$ of $AX = B$ is a coset of the subgroup $\ker L_A$ of the abelian group $(F^n, +)$.

Example 12.1.6

For $n\mathbb{Z} \leq \mathbb{Z}$ ($n \neq 0$): $\bar{a} = a + n\mathbb{Z} = \bar{b} \iff a - b \in n\mathbb{Z} \iff n \mid (a-b)$, i.e. a and b have the same remainder mod n. So $\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \ldots, \overline{n-1}\}$, the residues mod n. In number theory $\bar{a}$ is the **residue class** of a, $a \equiv b \pmod{n}$, and a complete set of representatives is a **complete residue system**.

Example 12.1.7

In S_3 (notation of Example 11.8.4) with $H = \{1, \sigma\}$: $\tau H = \{\tau, \sigma\tau^2\}$, $\tau^2 H = \{\tau^2, \sigma\tau\}$, so $S_3/H = \{\bar{1}, \bar{\tau}, \overline{\tau^2}\}$. Since $\sigma\tau \neq \tau\sigma$, $\tau H \neq H\tau$: left and right cosets need not coincide.

Exercise 12.1.8

For $K = \{1, \tau, \tau^2\} \leq S_3$: $|S_3/K| = 2$, and left and right cosets coincide (e.g. $\sigma K = K\sigma$).

Solution. $S_3 = K \amalg \sigma K$ with $\sigma K = \{\sigma, \sigma\tau, \sigma\tau^2\}$, the three transpositions. Since $K\sigma$ is a coset of size 3 disjoint from K, it equals σK. □

Observation 12.1.9

For $H \leq G$, $x, y \in G$: $|xH| = |yH| = |H|$.

Proof. $\lambda_x : H \to xH$, $h \mapsto xh$, is a bijection (injective by cancellation, onto by definition). ■

Observation 12.1.10 (Lagrange's theorem)

If G is finite and $H \leq G$, then $|H|$ divides $|G|$ and $|G/H| = \frac{|G|}{|H|}$.

Proof. The distinct cosets partition G and each has $|H|$ elements (Observation 12.1.9). ■

Notation 12.1.11

If $|G/H| < \infty$ (G may be infinite), $[G : H] = |G/H|$ is the **index** of H in G.

Observation 12.1.12

If $K \leq H \leq G$ with $|G/H|, |H/K| < \infty$, then $[G : K] = [G : H][H : K]$.

Proof. Let $\{g_i\}$ be a complete set of representatives for G/H and $\{h_j\}$ for H/K. Then $\{g_i h_j\}$ is one for G/K: $G = \coprod_i g_i H = \coprod_i g_i \coprod_j h_j K = \bigcup_{i,j} g_i h_j K$, and the union is disjoint, since $g_i h_j K = g_{i'} h_{j'} K$ forces $g_i H = g_{i'} H$ (both cosets lie in it), so $i = i'$, and then $h_j K = h_{j'} K$, so $j = j'$. ■

Example 12.1.13

By Example 11.3.6(e) and Lagrange, $\mu_m \leq \mu_n$ iff $m \mid n$.

Observation 12.1.14

If G is finite and $x \in G$, then $|x|$ divides $|G|$.

Proof. $|x| = |\langle x \rangle|$; apply Lagrange. ■

Proposition 12.1.15

If $|G| = p$ is prime, then G is cyclic, $G \approx \mu_p$.

Proof. Take $x \neq 1$. Then $|x| \neq 1$ divides p, so $|x| = p$, $\langle x \rangle = G$, and $\langle x \rangle \approx \mu_p$ (Exercise 11.7.9). ■

Example 12.1.16

Subgroups of S_3: besides $\{id\}$ and S_3, Lagrange allows orders $2, 3$, and such subgroups are cyclic (Proposition 12.1.15). Elements of order 2 give $\{id, (1\,2)\}$, $\{id, (2\,3)\}$, $\{id, (1\,3)\}$; elements of order 3, $(1\,2\,3)$ and $(1\,3\,2) = (1\,2\,3)^2$, give the unique subgroup $\{id, (1\,2\,3), (1\,3\,2)\}$ of order 3.

Exercise 12.1.17

If $|G|$ is even, G has an element of order 2.

Solution. Pair each x with x^{-1}. The pairs with $x \neq x^{-1}$ have 2 elements, so the set $\{x : x = x^{-1}\}$ has even size; it contains 1, hence some $x \neq 1$ with $x = x^{-1}$, i.e. $|x| = 2$. □

Remark 12.1.18

But if $d \mid |G|$, G need not have an element, or even a subgroup, of order d.

Exercise 12.1.19

(a) S_4 and A_4 have no elements of order 6, 8 or 12. (b) S_4 has exactly four subgroups of order 6. (c) A_4 has no subgroup of order 6. (d) How many subgroups of order 8 does S_4 have? (e) Of order 12?

Solution. (a) Elements of S_4 have cycle types $1, (2), (3), (4), (2,2)$, of orders $1, 2, 3, 4, 2$.

(b) A subgroup of order 6 fixing a point i is the copy of S_3 on the other three points; there are four of these. Conversely, a subgroup H of order 6 has an element of order 3 (Cauchy, Theorem 12.1.20), a 3-cycle fixing some i, and the order-2 elements of H normalize $\langle 3\text{-cycle} \rangle$, which forces them to fix i too, so H is the stabilizer of i.

(c) A_4 (order 12) has eight 3-cycles and three elements of type $(2,2)$. A subgroup H of order 6 would have index 2, hence be normal (Observation 12.2.9) and contain all squares, in particular all eight 3-cycles ($c = (c^2)^2$), which is too many.

(d) Three (the dihedral groups D_4, Sylow 2-subgroups).

(e) One, namely A_4 (a subgroup of index 2 contains all squares, which generate A_4). □

Theorem 12.1.20 (Cauchy)

If a prime p divides $|G|$, then G has an element of order p.

Proof [Supplement] (McKay; the book defers it to abstract algebra). Let $X = \{(x_1, \ldots, x_p) \in G^p : x_1 x_2 \cdots x_p = 1\}$. Choosing $x_1, \ldots, x_{p-1}$ freely determines x_p, so $|X| = |G|^{p-1}$, divisible by p. If $(x_1, \ldots, x_p) \in X$, then $x_2 \cdots x_p x_1 = x_1^{-1}(x_1 \cdots x_p)x_1 = 1$, so cyclic

shifts preserve X. The orbits of the cyclic shift have size 1 or p (p prime), and orbits of size 1 are the constant tuples $(x, \dots, x)$ with $x^p = 1$. Since $|X| \equiv 0 \pmod p$ and the size-p orbits contribute multiples of p, the number of x with $x^p = 1$ is divisible by p. It is at least 1 ($x = 1$), hence at least $p \geq 2$. So some $x \neq 1$ has $x^p = 1$, i.e. $|x| = p$. ■

12.2 Normal subgroups and quotient groups

Motivation. Multiplying cosets by representatives, $\bar{x}\,\bar{y} = \overline{xy}$, is well defined exactly when the subgroup is normal. Then G/N is a group, the natural projection is a homomorphism with kernel N, and residue arithmetic mod n becomes the group $\mathbb{Z}_n$.

Residue arithmetic ($3^3 - 20 \cdot 3 + 9 \equiv 3^3 + 0 + 4 \equiv 1 \pmod 5$), like adding fractions with different representatives, raises the question of *well-definedness*. To make G/H a group, the natural operation is $(xH)(yH) = (xy)H$, i.e. $\bar{x}\,\bar{y} = \overline{xy}$; we must check that if $\bar{x} = \bar{x}'$ and $\bar{y} = \bar{y}'$, then $\overline{xy} = \overline{x'y'}$, i.e. $x'^{-1}x, y'^{-1}y \in H \Rightarrow (x'y')^{-1}xy = y'^{-1}x'^{-1}xy \in H$.

Example 12.2.1

For $H = \{1, \sigma\} \leq S_3$ (Example 12.1.7): $\bar{1} = \bar{\sigma}$ and $\bar{\tau} = \overline{\sigma\tau^2}$, but $1 \cdot \tau = \tau$ while $\sigma \cdot \sigma\tau^2 = \tau^2$, and $\bar{\tau} \neq \overline{\tau^2}$. So the operation is not well defined. (For $K = \{1, \tau, \tau^2\}$ it is.)

Definition 12.2.2 (Normal subgroup)

$H \leq G$ is **normal**, $H \trianglelefteq G$, if $g^{-1}hg \in H$ for all $g \in G$, $h \in H$.

Observation 12.2.3

For $H \leq G$, $\bar{x}\,\bar{y} = \overline{xy}$ is well defined on G/H iff $H \trianglelefteq G$.

Proof. If $H \trianglelefteq G$ and $x'^{-1}x, y'^{-1}y \in H$, then $y'^{-1}x'^{-1}xy = (y'^{-1}y)(y^{-1}(x'^{-1}x)y) \in H$. Conversely, if the operation is well defined, take $x = h$, $x' = 1$, $y = y' = g$: then $\bar{h} = \bar{1}$, so $\overline{hg} = \bar{g}$, i.e. $g^{-1}hg \in H$. ■

Proposition 12.2.4

If $N \trianglelefteq G$, then G/N with $\bar{x}\,\bar{y} = \overline{xy}$ is a group, the **quotient group** of G modulo N.

Proof. $(\bar{x}\bar{y})\bar{z} = \overline{xyz} = \bar{x}(\bar{y}\bar{z})$; $\bar{1}$ is the identity; $\bar{x}\,\overline{x^{-1}} = \bar{1} = \overline{x^{-1}}\,\bar{x}$, so $\bar{x}^{-1} = \overline{x^{-1}}$. ■

For $N \trianglelefteq G$, $n \in N$, $g \in G$: $ng = gn'$ with $n' = g^{-1}ng \in N$. So elements of N can be moved past elements of G, at the cost of changing n to n'.

Observation 12.2.5

For $H \leq G$ the following are equivalent: (1) $H \trianglelefteq G$, i.e. $g^{-1}Hg \leq H$ for all g; (2) $gHg^{-1} = H$ for all g; (3) $g^{-1}Hg = H$ for all g.

Proof. (3)⇔(2): replace g by g^{-1}. (3)⇒(1) is clear. (1)⇒(3): (1) for g^{-1} gives $gHg^{-1} \leq H$, i.e. $H \leq g^{-1}Hg$. ■

Exercise 12.2.6

For $H \leq G$ the following are equivalent: (1) $H \trianglelefteq G$; (2) $\varphi(H) = H$ for every inner automorphism φ; (3) $gH = Hg$ for all g.

Solution. (1)⇔(2) is Observation 12.2.5 with $\varphi = \mathrm{Int}(g)$. (3)⇔ $gHg^{-1} = H$ (multiply by g^{-1} on the right). □

Example 12.2.7

$\{1\} \trianglelefteq G$ and $G \trianglelefteq G$; $G/\{1\} \approx G$ and $G/G \approx \{1\}$.

Example 12.2.8

(a) In a commutative group every subgroup is normal, and G/N is commutative.
(b) If $G = \langle x \rangle$ is cyclic, so is $G/N = \langle \bar{x} \rangle$.

Observation 12.2.9

If $[G : H] = 2$, then $H \trianglelefteq G$.

Proof. $G = H \amalg xH$ for any $x \notin H$. Let $g \in G$, $h \in H$; we show $ghg^{-1} \in H$. If $g \in H$, this is clear. Otherwise $g = xh_1$, and $ghg^{-1} = xh_2x^{-1}$ with $h_2 = h_1hh_1^{-1} \in H$. If $xh_2x^{-1} \notin H$, then $xh_2x^{-1} = xh_3$ for some $h_3 \in H$, so $h_2x^{-1} = h_3$ and $x = h_3^{-1}h_2 \in H$, a contradiction. ■

(For index $3, 4, \ldots$ this fails, e.g. Example 12.2.1.)

Example 12.2.10

Groups of order 6. By Cauchy there are x, y with $|x| = 2$, $|y| = 3$; $N = \langle y \rangle$ has index 2, so $N \trianglelefteq G$ and $G = N \amalg xN = \{1, y, y^2, x, xy, xy^2\}$. Then $x^{-1}yx \in N$ has order 3, so it is y or y^2. If $x^{-1}yx = y$, then $xy = yx$ and $|xy| = 6$, so $G \approx \mu_6$. If $x^{-1}yx = y^2$, then $G \approx S_3$ (Examples 11.8.4, 11.9.23). So up to isomorphism the groups of order 6 are μ_6 and S_3.

Observation 12.2.11

If $\varphi : G \to H$ is a homomorphism, then $\ker \varphi \trianglelefteq G$.

Proof [Supplement]. For $k \in \ker \varphi$, $\varphi(g^{-1}kg) = \varphi(g)^{-1}\varphi(g) = e$. ■

(But $\mathrm{im}\, \varphi$ need not be normal in H.)

Definition 12.2.12 (Natural projection)

For $N \trianglelefteq G$, $\pi : G \to G/N$, $\pi(x) = \bar{x}$, is the **natural projection**.

Exercise 12.2.13

(a) π is an epimorphism.
(b) $\ker \pi = N$.
(c) For the natural embedding $\iota : N \to G$, $\pi \circ \iota = \mathbf{1}$.

Solution. (a) $\pi(xy) = \overline{xy} = \bar{x}\bar{y}$, onto by definition.

(b) $\bar{x} = \bar{1} \iff x \in N$.

(c) is (b). □

The quotient operation is the unique one making π a homomorphism, so π carries all information about G/N. Residue calculus is the operation of $\mathbb{Z}/n\mathbb{Z}$.

Notation 12.2.14

For $1 \leq n \in \mathbb{Z}$, $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, with $\bar{a} + \bar{b} = \overline{a+b}$.

Observation 12.2.15

$\mathbb{Z}_n$ is cyclic, so $\mathbb{Z}_n \approx \mu_n$.

Proof. $\mathbb{Z}_n = \langle \bar{1} \rangle$ has order n; apply Exercise 11.7.9. ■

Exercise 12.2.16

(a) In A_4, $N = \{id, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\}$ satisfies $N \approx \mathbb{Z}_2 \times \mathbb{Z}_2$ and $N \trianglelefteq A_4$.
(b) $H = \{id, (1\,2)(3\,4)\}$ satisfies $H \trianglelefteq N$ and $N \trianglelefteq A_4$, but H is not normal in A_4.

Solution. (a) Each non-identity element of N has order 2 and the product of two of them is the third, so N is a subgroup in which every element squares to 1: the Klein 4-group. Conjugation preserves cycle type, and N consists of the identity and all elements of type $(2,2)$, so N is normal (even in S_4).

(b) H has index 2 in N, so it is normal in N. But $(1\,2\,3)^{-1}(1\,2)(3\,4)(1\,2\,3) = (1\,4)(2\,3) \notin H$ (a normal subgroup of a normal subgroup need not be normal). □

Remark 12.2.17

The normal subgroup N of A_4 of index 3 is behind the solution of the quartic by radicals, while the absence of such normal subgroups in A_5 explains why there is no formula for the quintic (Galois theory).

Definition 12.2.18 (Center)

$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$ is the **center** of G.

Exercise 12.2.19

(a) $Z(G) \trianglelefteq G$.
(b) If $\varphi \in \mathrm{Aut}(G)$, then $\varphi(Z(G)) = Z(G)$.

Solution. (a) $Z(G)$ is a subgroup, and $g^{-1}zg = z$ for $z \in Z(G)$.

(b) If z commutes with all g, then $\varphi(z)$ commutes with all $\varphi(g)$, i.e. with all of G; apply the same to φ^{-1}. □

Exercise 12.2.20

(a) $Z(\mathrm{GL}_n(F)) = \{cI : c \in F^\times\}$.
(b) $Z(\mathrm{SL}_n(F)) = \{cI : c \in F,\ c^n = 1\}$.

Solution. (a) Scalars are central. If A commutes with $I + E_{ij}$ $(i \neq j)$, it commutes with E_{ij}; $AE_{ij} = E_{ij}A$ forces $A_{ki} = 0$ for $k \neq i$, $A_{jk} = 0$ for $k \neq j$, and $A_{ii} = A_{jj}$. So A is scalar.

(b) The same argument works, since $I + E_{ij} \in \mathrm{SL}_n(F)$; and $cI \in \mathrm{SL}_n(F)$ iff $c^n = 1$. □

Definition 12.2.21 (Projective linear groups)

$\mathbf{PGL}_n(F)$ $= \mathrm{GL}_n(F)/Z(\mathrm{GL}_n(F))$ and **$\mathbf{PSL}_n(F)$** $= \mathrm{SL}_n(F)/Z(\mathrm{SL}_n(F))$; similarly **PSO**, **PSU**, ...

H. Weyl called the groups GL, **PGL**, SL, **PSL**, **O**, **SO**, **PSO**, **U**, **SU**, **PSU** (and **Sp**, **PSp** of Chapter 13) the **classical groups**; introducing them is one aim of this book.

12.3 Quotient spaces

Motivation. For a subspace $W \leq V$ the coset group V/W always admits scalar multiplication, so it is a vector space of dimension $\dim V - \dim W$. This is the basis of induction on dimension.

For $W \leq V$ (any dimension), $(W, +)$ is a normal subgroup of the abelian group $(V, +)$, so V/W is an abelian group with $\bar{u} + \bar{v} = \overline{u+v}$; the cosets are $\bar{v} = v + W$. Define $c\bar{v} = \overline{cv}$. This is well defined: if $\bar{v} = \bar{u}$, then $v - u \in W$, so $cv - cu = c(v-u) \in W$.

Proposition 12.3.1

For $W \leq V$, V/W with $\bar{u} + \bar{v} = \overline{u+v}$ and $c\bar{v} = \overline{cv}$ is a vector space, the **quotient space** of V modulo W.

Proof. The operations are well defined, and each axiom for V/W follows from the same axiom in V by putting bars on; the zero vector is $\bar{0}$ and $-\bar{v} = \overline{-v}$. ■

Example 12.3.2

$V/0 \approx V$ and $V/V \approx 0$.

The natural projection $\pi : V \to V/W$, $\pi(v) = \bar{v}$, is now F-linear.

Exercise 12.3.3

For $W \leq V$: (a) $\pi : V \to V/W$ is a linear epimorphism;
(b) $\ker \pi = W$;
(c) for the inclusion $\iota : W \to V$, $\pi \circ \iota = 0$.

Solution. $\pi(cv + u) = \overline{cv+u} = c\bar{v} + \bar{u}$; onto by definition; $\bar{v} = \bar{0} \iff v \in W$. □

Remark 12.3.4

A heuristic: if $|F| = q < \infty$, $\dim V = n$, $\dim W = m$, then $|V| = q^n$, $|W| = q^m$, and by Lagrange $|V/W| = q^{n-m}$, suggesting $\dim V/W = n - m$.

Theorem 12.3.5

If V is finite-dimensional and $W \leq V$, then $\dim V = \dim W + \dim(V/W)$.

Proof. Apply the dimension theorem to $\pi : V \to V/W$ ($\ker \pi = W$, $\operatorname{im} \pi = V/W$). ■

Exercise 12.3.6

If a basis $\{v_1, \dots, v_m\}$ of W is extended to a basis $\{v_1, \dots, v_n\}$ of V, then $\{\overline{v_{m+1}}, \dots, \overline{v_n}\}$ is a basis of V/W.

Solution. They span: $\bar{v} = \overline{\sum a_i v_i} = \sum_{i>m} a_i \bar{v}_i$ since $\bar{v}_i = \bar{0}$ for $i \le m$. They are independent: if $\sum_{i>m} a_i \bar{v}_i = \bar{0}$, then $\sum_{i>m} a_i v_i \in W = \langle v_1, \dots, v_m \rangle$, so all $a_i = 0$. □

Observation 12.3.7

Every subspace of F^n is the solution space of $AX = 0$ for some $A \in \mathfrak{M}_{m,n}(F)$.

Proof. For $W \le F^n$ it suffices to find a linear $L : F^n \to U$ (finite-dimensional U) with $\ker L = W$; take $L = \pi : F^n \to F^n/W$, and A its matrix in some bases. ■

12.4 Isomorphism theorems

Motivation. A homomorphism identifies elements with the same image. Dividing out its kernel turns it into an isomorphism onto its image (First Isomorphism Theorem), and $K/(H \cap K) \approx HK/H$ (Second). Together they classify cyclic groups and give many quotient computations.

Observation 12.4.1

If $\varphi : G \to H$ is a homomorphism and $\varphi(g) = h$, then $\varphi^{-1}(h) = g \ker \varphi$.

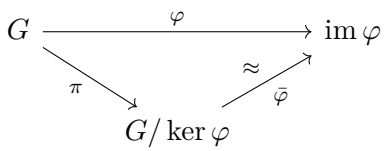
Proof. $x \in \varphi^{-1}(h) \iff \varphi(x) = \varphi(g) \iff g^{-1}x \in \ker \varphi \iff x \in g \ker \varphi$. ■

So the fibres of φ are exactly the cosets of $\ker \varphi$, which is normal; renaming each coset $\bar{g}$ as $\varphi(g)$ is a bijection $G/\ker \varphi \to \operatorname{im} \varphi$.

Theorem 12.4.2 (First isomorphism theorem for groups)

For a homomorphism $\varphi : G \to H$, $\bar{\varphi} : G/\ker \varphi \to \operatorname{im} \varphi$, $\bar{\varphi}(\bar{g}) = \varphi(g)$, is a well-defined isomorphism: $G/\ker \varphi \approx \operatorname{im} \varphi$.

Proof. Well defined: if $\bar{g} = \bar{g}'$, then $g'^{-1}g \in \ker \varphi$, so $\varphi(g) = \varphi(g')$. Homomorphism: $\bar{\varphi}(\bar{g}\bar{g}') = \varphi(gg') = \varphi(g)\varphi(g')$. Onto: clear. Injective: if $\bar{\varphi}(\bar{g}) = 1$, then $g \in \ker \varphi$, so $\bar{g} = \bar{1}$ (Observation 11.6.7). ■



Remark 12.4.3

(a) Writing $\bar{G} = G/\ker \varphi$, $\bar{\varphi}\bar{g} = \varphi g$. (b) Equivalently, $\psi : \operatorname{im} \varphi \to G/\ker \varphi$, $\psi(\varphi(g)) = \bar{g}$, is an isomorphism; its well-definedness ($\varphi(g) = \varphi(g') \Rightarrow \bar{g} = \bar{g}'$) is the injectivity of $\bar{\varphi}$.

Example 12.4.4

$\operatorname{sgn} : S_n \to \mu_2$ $(n \geq 2)$ is onto with kernel A_n; so $A_n \trianglelefteq S_n$, $S_n/A_n \approx \mu_2$, $[S_n : A_n] = 2$, $S_n = A_n \amalg \tau A_n$ $(\tau \notin A_n)$, and $|A_n| = n!/2$.

Example 12.4.5

$\det : \mathrm{GL}_n(F) \to F^\times$ is onto with kernel $\mathrm{SL}_n(F)$; so $\mathrm{SL}_n(F) \trianglelefteq \mathrm{GL}_n(F)$ and $\mathrm{GL}_n(F)/\mathrm{SL}_n(F) \approx F^\times$.

Example 12.4.6

(a) $\det : \mathbf{O}(n) \to \mu_2$ is onto with kernel $\mathbf{SO}(n)$; so $\mathbf{SO}(n) \trianglelefteq \mathbf{O}(n)$, $\mathbf{O}(n)/\mathbf{SO}(n) \approx \mu_2$, and $\mathbf{O}(n) = \mathbf{SO}(n) \amalg S \cdot \mathbf{SO}(n)$ $(S \notin \mathbf{SO}(n)$; Exercise 9.3.11).
(b) $\det : \mathbf{U}(n) \to S^1$ is onto (e.g. $\operatorname{diag}(z, 1, \ldots, 1)$) with kernel $\mathbf{SU}(n)$; so $\mathbf{U}(n)/\mathbf{SU}(n) \approx S^1 \approx \mathbf{SO}(2)$.

Exercise 12.4.7

(a) For $A, B \in \mathrm{GL}_n(F)$, $\bar{A} = \bar{B}$ in $\mathrm{GL}_n(F)/\mathrm{SL}_n(F)$ iff $\det A = \det B$. (b) Describe the coset decomposition of $\mathrm{GL}_n(F)$ modulo $\mathrm{SL}_n(F)$. (c) Describe that of $\mathbf{U}(n)$ modulo $\mathbf{SU}(n)$.

Solution. (a) $\bar{A} = \bar{B} \iff B^{-1}A \in \mathrm{SL}_n(F) \iff \det A = \det B$. (b) $\mathrm{GL}_n(F) = \coprod_{c \in F^\times} \operatorname{diag}(c, 1, \ldots, 1)\,\mathrm{SL}_n(F)$: one coset for each value of the determinant. (c) $\mathbf{U}(n) = \coprod_{z \in S^1} \operatorname{diag}(z, 1, \ldots, 1)\mathbf{SU}(n)$. □

Exercise 12.4.8

$\rho : (\mathbb{R}, +) \to \mathbf{SO}(2)$, $\rho(\theta) = R_\theta$, is a homomorphism, and $\mathbb{R}/2\pi\mathbb{Z} \approx \mathbf{SO}(2) \approx S^1$.

Solution. $R_\theta R_\eta = R_{\theta+\eta}$; ρ is onto (Theorem 9.5.2) with kernel $2\pi\mathbb{Z}$; apply Theorem 12.4.2 and Example 11.9.19. □

Example 12.4.9

For $x \in G$, $\gamma_x : \mathbb{Z} \to G$, $\gamma_x(a) = x^a$, has $\ker \gamma_x = n\mathbb{Z}$ for some $n \geq 0$ (Theorem 11.4.2) and image $\langle x \rangle$, so $\langle x \rangle \approx \mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ (with $\mathbb{Z}_0 = \mathbb{Z}$, $\mathbb{Z}_1 = 0$).

Theorem 12.4.10

Up to isomorphism the cyclic groups are exactly the $\mathbb{Z}_n$, $n \geq 0$.

Proof. Example 12.4.9 with $G = \langle x \rangle$; and $\mathbb{Z}_n$ is cyclic. ■

Exercise 12.4.11

For $\mathrm{Int} : G \to \mathrm{Aut}(G)$, $g \mapsto \mathrm{Int}(g)$: (a) Int is a homomorphism;
(b) $\mathrm{Int}(G) \trianglelefteq \mathrm{Aut}(G)$;
(c) $\ker(\mathrm{Int}) = Z(G)$, so $\mathrm{Int}(G) \approx G/Z(G)$.

Solution. (a) $\mathrm{Int}(gh)(x) = ghxh^{-1}g^{-1} = \mathrm{Int}(g)\,\mathrm{Int}(h)(x)$ (this is why $\mathrm{Int}(g)(x) = gxg^{-1}$ rather than $g^{-1}xg$). (b) For $\varphi \in \mathrm{Aut}(G)$: $\varphi \circ \mathrm{Int}(g) \circ \varphi^{-1}(x) = \varphi(g\varphi^{-1}(x)g^{-1}) = \varphi(g)x\varphi(g)^{-1}$, i.e. $= \mathrm{Int}(\varphi(g))$. (c) $\mathrm{Int}(g) = id \iff gx = xg$ for all x; apply Theorem 12.4.2. □

Exercise 12.4.12

Define $\psi : \mathrm{SL}_2(F) \to \mathrm{GL}_3(F)$ by

$$\psi\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a^2 & -2ac & -c^2 \\ -ab & ad+bc & cd \\ -b^2 & 2bd & d^2 \end{pmatrix}.$$

(a) $\mathrm{im}\,\psi \leq \mathrm{SL}_3(F)$.
(b) ψ is a homomorphism with $\ker\psi = \{\pm I\} = Z(\mathrm{SL}_2(F))$.
(c) Hence $\mathbf{PSL}_2(F) \approx \mathrm{im}\,\psi$ is a matrix group.

Solution. [Supplement] $\psi(A)$ records how A acts on degree-2 polynomials in two variables, which is why it is multiplicative. Concretely, multiplying out (we checked it symbolically) gives $\psi(AB) = \psi(A)\psi(B)$ for all 2×2 matrices A, B, and $\det\psi(A) = (ad-bc)^3$. (a) For $A \in \mathrm{SL}_2(F)$, $\det\psi(A) = 1^3 = 1$. (b) $\psi(A) = I$ forces $c = b = 0$ (off-diagonal entries $-c^2 = -b^2 = 0$) and $a^2 = d^2 = ad = 1$, so $a = d = \pm 1$; conversely $\psi(\pm I) = I$. And $Z(\mathrm{SL}_2(F)) = \{\pm I\}$ (Exercise 12.2.20, $c^2 = 1$). (c) First isomorphism theorem. □

Exercise 12.4.13

Is there an epimorphism $S_3 \to \mathbb{Z}_3$?

Solution. No. Its kernel would be a normal subgroup of order 2 (First Isomorphism Theorem), but the subgroups of order 2 of S_3 are not normal (Example 12.2.1). □

Theorem 12.4.14 (First isomorphism theorem for vector spaces)

For a linear map $L : V \to W$, $\bar{L} : V/\ker L \to \mathrm{im}\,L$, $\bar{L}\bar{v} = Lv$, is a well-defined vector space isomorphism.

Proof. By Theorem 12.4.2 for $L : (V,+) \to (W,+)$, $\bar{L}$ is a well-defined group isomorphism; and $\bar{L}(c\bar{v}) = \bar{L}(\overline{cv}) = L(cv) = cL(v) = c\bar{L}(\bar{v})$. ■

Remark 12.4.15

By Theorem 12.4.14, the dimension theorem and Theorem 12.3.5 are equivalent statements.

Exercise 12.4.16

For $\mathrm{tr} : \mathfrak{M}_{n,n}(F) \to F$: (a) $\ker\mathrm{tr} = \mathfrak{sl}_n(F)$;
(b) $\mathfrak{M}_{n,n}(F)/\mathfrak{sl}_n(F) \approx F$, so $\dim\mathfrak{sl}_n(F) = n^2 - 1$.

Solution. tr is linear and onto ($\mathrm{tr}(cE_{11}) = c$); $\mathfrak{sl}_n(F)$ is by definition the trace-zero matrices; apply Theorem 12.4.14. □

Exercise 12.4.17

For $T \in \mathfrak{L}(V,V)$ and a T-cyclic space $V = F[t]v$, $\varphi : F[t] \to V$, $\varphi(f) = f(T)v$: (a) φ is linear with $\ker\varphi = \mathcal{I}_T$ (the annihilator ideal of v);
(b) $V \approx F[t]/\mathcal{I}_T$.

Solution. φ is linear and onto by definition of $F[t]v$; its kernel is $\{f : f(T)v = 0\}$; apply Theorem 12.4.14. □

Observation 12.4.18

If $H \trianglelefteq G$ and $K \leq G$, then $HK \leq G$.

Proof. $1 \in HK$. For $h,h' \in H$, $k,k' \in K$: $(hk)(h'k') = h(kh'k^{-1})kk' \in HK$ and $(hk)^{-1} = k^{-1}h^{-1} = (k^{-1}h^{-1}k)k^{-1} \in HK$. ■

Theorem 12.4.19 (Second isomorphism theorem for groups)

If $H \trianglelefteq G$ and $K \leq G$, then $\bar{\jmath} : K/(H \cap K) \to HK/H$, $\bar{\jmath}(k(H \cap K)) = kH$, is a well-defined isomorphism.

Proof. By Observation 12.4.18 $HK \leq G$, and clearly $H \trianglelefteq HK$, $H \cap K \trianglelefteq K$.

(i) Well defined: if $k'^{-1}k \in H \cap K$, then $k'^{-1}k \in H$.

(ii) Injective: if $k'^{-1}k \in H$, then (as $k,k' \in K$) $k'^{-1}k \in H \cap K$.

(iii) Homomorphism: $\bar{\jmath}(\bar{k}\bar{k}') = \overline{kk'} = \bar{k}\,\bar{k}'$.

(iv) Onto: for $h \in H$, $k \in K$, $\overline{hk} = \bar{h}\,\bar{k} = \bar{1}\bar{k} = \bar{k}$ in HK/H (H normal). ■

Second proof. $j = \pi' \circ \iota : K \to HK \to HK/H$ is an epimorphism by (iv), with $\ker j = H \cap K$; apply Theorem 12.4.2. ■

Corollary 12.4.20

If G is finite, $H \trianglelefteq G$, $K \leq G$, then $|HK| \cdot |H \cap K| = |H| \cdot |K|$.

Proof. Lagrange and Theorem 12.4.19: $\frac{|K|}{|H \cap K|} = \frac{|HK|}{|H|}$. ■

Theorem 12.4.21 (Second isomorphism theorem for vector spaces)

For $U,W \leq V$, $\bar{\jmath} : W/(U \cap W) \to (U+W)/U$, $\bar{\jmath}(\bar{w}) = \bar{w}$, is a well-defined vector space isomorphism.

Proof. Theorem 12.4.19 for $(V,+)$ gives a group isomorphism; it is F-linear since $\bar{\jmath}(c\bar{w}) = \overline{cw} = c\bar{w}$. ■

Corollary 12.4.22

For finite-dimensional V and $U,W \leq V$: $\dim(U+W) + \dim(U \cap W) = \dim U + \dim W$.

Proof. By Theorems 12.4.21 and 12.3.5, $\dim W - \dim(U \cap W) = \dim(U+W) - \dim U$. ■

12.5 Triangularization II

Motivation. An invariant subspace W splits an operator into $T|_W$ and the induced operator $\bar{T}$ on V/W, with block-triangular matrix. Induction on $\dim V$ via quotients then proves triangularization over $\mathbb{C}$ and simultaneous triangularization of commuting operators.

Throughout, spaces are finite-dimensional. If W is T-invariant with basis $\mathfrak{C} = \{v_1, \dots, v_m\}$ extended to a basis $\mathfrak{B} = \{v_1, \dots, v_n\}$ of V, then $[T]_{\mathfrak{B}}^{\mathfrak{B}} = \left(\begin{smallmatrix} [T|_W]_{\mathfrak{C}}^{\mathfrak{C}} & * \\ 0 & ? \end{smallmatrix}\right)$. The block "?" is explained by $\bar{V} = V/W$ with basis $\bar{\mathfrak{B}} = \{\bar{v}_{m+1}, \dots, \bar{v}_n\}$ (Exercise 12.3.6) and $\bar{T}\bar{v} = \overline{Tv}$.

Observation 12.5.1

If $W \leq V$, $W' \leq V'$ and $L : V \to V'$ is linear with $LW \leq W'$, then $\bar{L} : V/W \to V'/W'$, $\bar{L}\bar{v} = \overline{Lv}$, is a well-defined linear map (L **induces** $\bar{L}$).

Proof. If $v - u \in W$, then $Lv - Lu = L(v - u) \in W'$; linearity is inherited. (Conversely, well-definedness forces $LW \leq W'$: take $u = 0$.) ■

$$\begin{array}{ccc} V & \xrightarrow{L} & V' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ V/W & \xrightarrow[\bar{L}]{} & V'/W' \end{array}$$

Exercise 12.5.2

State and prove Observation 12.5.1 for groups.

Solution. If $N \trianglelefteq G$, $N' \trianglelefteq G'$ and $\varphi : G \to G'$ is a homomorphism with $\varphi(N) \leq N'$, then $\bar{\varphi}(\bar{g}) = \overline{\varphi g}$ is a well-defined homomorphism $G/N \to G'/N'$: if $g'^{-1}g \in N$, then $\varphi(g')^{-1}\varphi(g) = \varphi(g'^{-1}g) \in N'$. □

Exercise 12.5.3

The well-definedness of $\bar{L}$ in Theorem 12.4.14 and of $\bar{\jmath}$ in Theorem 12.4.21 are special cases of Observation 12.5.1.

Solution. $\bar{L}$ is induced by $L : V \to \operatorname{im} L$ with $L(\ker L) \leq 0$; $\bar{\jmath}$ is induced by the inclusion $W \to U+W$ with $U \cap W \leq U$. □

Lemma 12.5.4

Let W be T-invariant, $\bar{T} : \bar{V} \to \bar{V}$ as above, $\mathfrak{C} = \{v_1, \dots, v_m\}$ a basis of W, $\mathfrak{B} = \{v_1, \dots, v_n\}$ a basis of V, $\bar{\mathfrak{B}} = \{\bar{v}_{m+1}, \dots, \bar{v}_n\}$. Then

$$[T]_{\mathfrak{B}}^{\mathfrak{B}} = \begin{pmatrix} [T|_W]_{\mathfrak{C}}^{\mathfrak{C}} & * \\ 0 & [\bar{T}]_{\bar{\mathfrak{B}}}^{\bar{\mathfrak{B}}} \end{pmatrix}.$$

Proof. Let column $m + j$ of $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ be $(a_{1,m+j}, \dots, a_{n,m+j})^t$: $Tv_{m+j} = \sum_i a_{i,m+j}v_i$. Applying bars and using $\bar{v}_1 = \cdots = \bar{v}_m = \bar{0}$, $\bar{T}\bar{v}_{m+j} = \sum_{i>m} a_{i,m+j}\bar{v}_i$. So the j-th column of $[\bar{T}]$ is the lower part of column $m + j$. The first m columns are as stated because $TW \leq W$. ■

Exercise 12.5.5

In Lemma 12.5.4: (a) $\operatorname{tr}T = \operatorname{tr}(T|_W) + \operatorname{tr}\bar{T}$;
(b) $\det T = \det(T|_W)\det\bar{T}$;
(c) $\phi_T(t) = \phi_{T|_W}(t)\phi_{\bar{T}}(t)$.

Solution. Read them off the block-triangular matrix (the determinant of a block-triangular matrix is the product of the determinants of the diagonal blocks; apply this to $tI - [T]$ for (c)). □

Theorem 12.5.6 (Triangularization)

If $F = \mathbb{C}$ and $T \in \mathfrak{L}(V, V)$, then V has a basis $\mathfrak{B}$ with $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ upper triangular.

Second proof. Induction on $n = \dim V$; $n = 1$ is trivial. Over $\mathbb{C}$, T has an eigenvector v, $Tv = \lambda v$; $W = \langle v\rangle$ is T-invariant. With $\bar{V} = V/W$, Lemma 12.5.4 gives $[T]_{\mathfrak{B}}^{\mathfrak{B}} = \left(\begin{smallmatrix}\lambda & * \\ 0 & [\bar{T}]\end{smallmatrix}\right)$, and $\dim\bar{V} = n-1$. By induction $\bar{V}$ has a basis $\bar{\mathfrak{D}} = \{\bar{u}_2, \ldots, \bar{u}_n\}$ with $[\bar{T}]_{\bar{\mathfrak{D}}}^{\bar{\mathfrak{D}}}$ upper triangular. Then $\mathfrak{D} = \{v, u_2, \ldots, u_n\}$ is a basis of V: if $cv + \sum c_i u_i = 0$, applying π gives $\sum c_i\bar{u}_i = 0$, so all $c_i = 0$, then $c = 0$. By Lemma 12.5.4, $[T]_{\mathfrak{D}}^{\mathfrak{D}} = \left(\begin{smallmatrix}\lambda & * \\ 0 & [\bar{T}]_{\bar{\mathfrak{D}}}^{\bar{\mathfrak{D}}}\end{smallmatrix}\right)$ is upper triangular. ■

Example 12.5.7

Let G be finite and $\varphi : G \to \mathrm{GL}_n(\mathbb{C})$ a homomorphism, $\chi = \operatorname{tr}\circ\varphi$ (the **character** of φ). For $g \in G$, triangularize $\varphi(g)$ with diagonal $\lambda_1, \ldots, \lambda_n$. Then $I = \varphi(g^{|g|}) = \varphi(g)^{|g|}$ has diagonal $\lambda_i^{|g|}$, so $\lambda_i^{|g|} = 1$ and $|\lambda_i| = 1$. Since $\varphi(g^{-1}) = \varphi(g)^{-1}$ is triangular with diagonal $\lambda_i^{-1} = \bar{\lambda}_i$,

$$\chi(g^{-1}) = \sum \lambda_i^{-1} = \sum \bar{\lambda}_i = \overline{\chi(g)}.$$

Exercise 12.5.8

In Example 12.5.7, if g is conjugate to h, then $\chi(g) = \chi(h)$.

Solution. $h = x^{-1}gx$ gives $\varphi(h) = \varphi(x)^{-1}\varphi(g)\varphi(x)$, and similar matrices have equal traces. □

Definition 12.5.9 (Simultaneous triangularization)

$T, S \in \mathfrak{L}(V, V)$ are **simultaneously triangularizable** if $[T]_{\mathfrak{B}}^{\mathfrak{B}}$ and $[S]_{\mathfrak{B}}^{\mathfrak{B}}$ are upper triangular for one basis $\mathfrak{B}$ (likewise for any family). Then the first vector of $\mathfrak{B}$ is a common eigenvector.

Proposition 12.5.10

If $F = \mathbb{C}$ and $TS = ST$, then T and S have a common eigenvector.

Proof. Take an eigenvalue λ of T and its eigenspace E_λ^T. It is S-invariant: if $Tv = \lambda v$, then $T(Sv) = S(Tv) = \lambda Sv$. An eigenvector of $S|_{E_\lambda^T}$ (which exists over $\mathbb{C}$) is a common eigenvector. ■

Proposition 12.5.11

If $F = \mathbb{C}$ and $TS = ST$, then T, S are simultaneously triangularizable.

Proof. Take a common eigenvector v (Proposition 12.5.10) and $\bar{V} = V/\langle v\rangle$. $\langle v\rangle$ is invariant under both, so $\bar{T}, \bar{S}$ are defined, and $\bar{T}\bar{S} = \overline{TS} = \overline{ST} = \bar{S}\bar{T}$. By induction on dimension they are

simultaneously triangularizable on $\bar{V}$, and the lifting argument of Theorem 12.5.6 applies to both at once. ■

Exercise 12.5.12

If $\{T_i\}_{i\in I} \subseteq \mathfrak{L}(V,V)$ $(F=\mathbb{C})$ pairwise commute, they are simultaneously triangularizable.

Solution.

Step 1 (a common eigenvector). Choose a nonzero subspace U invariant under all T_i of minimal dimension (e.g. V at first). If some $T_j|_U$ is not a scalar, pick an eigenvalue λ of $T_j|_U$; then $\{u \in U : T_j u = \lambda u\}$ is a smaller nonzero subspace invariant under all T_i (commutativity), contradicting minimality. So every T_i acts on U as a scalar, and any $0 \neq v \in U$ is a common eigenvector.

Step 2 (induction). Pass to $V/\langle v\rangle$, where the induced operators still commute, and argue as in Proposition 12.5.11. □

Chapter 13. Bilinear Forms

Armed with groups, quotient groups and quotient spaces, we return to inner product spaces, aiming at $\mathbf{O}(n)$ and $\mathbf{U}(n)$. To study inner product spaces properly one needs bilinear and Hermitian forms, and for these the dual space is indispensable. From now on all vector spaces are finite-dimensional.

13.1 Bilinear forms

Motivation. Dropping positive definiteness from a real inner product leaves a bilinear form. Such a form is completely determined by its values on pairs of basis vectors, i.e. by a matrix: "bilinear forms are matrices", just as linear maps are.

For a bilinear form $B : V \times V \to F$, a basis $\mathfrak{B} = \{v_i\}$ of V, $v = \sum a_i v_i$, $w = \sum b_i v_i$:

$$B(v,w) = B\Big(\sum_i a_i v_i, \sum_j b_j v_j\Big) = \sum_{i,j} a_i b_j B(v_i, v_j). \tag{$*$}$$

Observation 13.1.1

With $[B]_{\mathfrak{B}} = (B(v_i,v_j))_{n\times n}$, $B(v,w) = [v]_{\mathfrak{B}}^t\,[B]_{\mathfrak{B}}\,[w]_{\mathfrak{B}}$ for all $v, w \in V$. $[B]_{\mathfrak{B}}$ is the **matrix of** B with respect to $\mathfrak{B}$.

Proof. $(*)$ reads $\sum_{i,j} a_i B(v_i,v_j) b_j = [v]_{\mathfrak{B}}^t [B]_{\mathfrak{B}} [w]_{\mathfrak{B}}$. ■

Notation 13.1.2

For $J \in \mathfrak{M}_{n,n}(F)$, $B_{\mathcal{E}}^J(X,Y) = X^t J Y$ is a bilinear form on F^n with $[B_{\mathcal{E}}^J]_{\mathcal{E}} = J$.

Example 13.1.3

(a) $B_{\mathcal{E}}^I$ is the **ordinary inner product** X^tY of F^n (the dot product on $\mathbb{R}^n$; on $\mathbb{C}^n$ it is not Hermitian, but it is one of our objects of study). (b) The inner product $\sum j a_j b_j$ of Exercise 10.1.8 is $B_{\mathcal{E}}^J$ with $J = \operatorname{diag}(1,2,\dots,n)$. (c) The inner product of Exercise 10.1.9 is $B_{\mathcal{E}}^J$ with $J = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$.

Notation 13.1.4

For a basis $\mathfrak{B}$ of V and $J \in \mathfrak{M}_{n,n}(F)$, $B_{\mathfrak{B}}^J(v,w) = [v]_{\mathfrak{B}}^t J [w]_{\mathfrak{B}}$ is a bilinear form on V with $[B_{\mathfrak{B}}^J]_{\mathfrak{B}} = J$.

So $B \leftrightarrow [B]_{\mathfrak{B}}$ is a bijection between bilinear forms on V and $\mathfrak{M}_{n,n}(F)$ (for fixed $\mathfrak{B}$).

Exercise 13.1.5

$\mathfrak{Bil}(V) = \{B : V\times V \to F \text{ bilinear}\}$, with $(B+C)(v,w) = B(v,w)+C(v,w)$ and $(aB)(v,w) = aB(v,w)$, is an F-vector space.

Solution. $B + C$ and aB are bilinear; the axioms hold pointwise. □

Theorem 13.1.6 (Fundamental theorem for bilinear forms)

For a fixed basis $\mathfrak{B}$ of n-dimensional V, $\Omega_{\mathfrak{B}} : \mathfrak{Bil}(V) \to \mathfrak{M}_{n,n}(F)$, $\Omega_{\mathfrak{B}}(B) = [B]_{\mathfrak{B}}$, is a vector space isomorphism.

Proof. Injective by Observation 13.1.1, surjective by Notation 13.1.4, and linear since $[B+C]_{\mathfrak{B}} = [B]_{\mathfrak{B}} + [C]_{\mathfrak{B}}$ and $[aB]_{\mathfrak{B}} = a[B]_{\mathfrak{B}}$ (entrywise). ■

So, given $\mathfrak{B}$, a bilinear form "is" a matrix.

Remark 13.1.7

With the coordinate isomorphism $\alpha_{\mathfrak{B}} : V \to F^n$, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{B} & F \\ {\scriptstyle \alpha_{\mathfrak{B}} \times \alpha_{\mathfrak{B}}}\downarrow & & \downarrow{\scriptstyle I} \\ F^n \times F^n & \xrightarrow[B^J_{\mathcal{E}}]{} & F \end{array}$$

($J = [B]_{\mathfrak{B}}$) commutes, which is Observation 13.1.1. So (V, B) and $(F^n, B^J_{\mathcal{E}})$ are essentially the same.

Observation 13.1.8

If J is invertible, then $B^J_{\mathcal{E}}(AX, Y) = B^J_{\mathcal{E}}(X, (J^{-1}A^tJ)Y)$ for $A \in \mathfrak{M}_{n,n}(F)$, $X, Y \in F^n$.

Proof. $(AX)^tJY = X^tA^tJY$ and $X^tJ(J^{-1}A^tJ)Y = X^tA^tJY$. ■

(This generalizes Observation 10.7.1; compare Question 10.7.2.)

Definition 13.1.9 (Symmetric, alternating)

A bilinear form B is (a) **symmetric** if $B(v, w) = B(w, v)$;
(b) **alternating** if $B(v, v) = 0$ for all v.

Observation 13.1.10

B is symmetric iff $[B]_{\mathfrak{B}}$ is a symmetric matrix.

Proof. If B is symmetric, $B(v_i, v_j) = B(v_j, v_i)$. Conversely, $(*)$ shows $B(v, w) = B(w, v)$ when $B(v_i, v_j) = B(v_j, v_i)$. ■

Exercise 13.1.11

B is alternating iff $B(v, w) = -B(w, v)$ for all v, w (when $1 + 1 \neq 0$ in F).

Solution. If alternating, $0 = B(v + w, v + w) = B(v, w) + B(w, v)$. Conversely, $B(v, v) = -B(v, v)$ gives $2B(v, v) = 0$. (In characteristic 2 only "alternating ⇒ skew" holds.) □

Exercise 13.1.12

Define an **alternating matrix** ($J^t = -J$ with zero diagonal) and show: B is alternating iff $[B]_{\mathfrak{B}}$ is alternating. (For $1 + 1 \neq 0$, alternating = skew-symmetric.)

Solution. If B is alternating, $B(v_i, v_i) = 0$ and $B(v_i, v_j) = -B(v_j, v_i)$. Conversely, by $(*)$, $B(v, v) = \sum_i a_i^2 J_{ii} + \sum_{i<j} a_i a_j (J_{ij} + J_{ji}) = 0$. □

Exercise 13.1.13

Let $\mathfrak{Sym}(V)$ and $\mathfrak{Alt}(V)$ be the symmetric and the alternating bilinear forms on V ($1+1 \neq 0$). (a) They are subspaces of $\mathfrak{Bil}(V)$. (b) $\dim \mathfrak{Bil}(V) = n^2$, $\dim \mathfrak{Sym}(V) = \frac{n(n+1)}{2}$, $\dim \mathfrak{Alt}(V) = \frac{n(n-1)}{2}$. (c) $\mathfrak{Bil}(V) = \mathfrak{Sym}(V) \oplus \mathfrak{Alt}(V)$.

Solution. (a) The defining conditions are linear.

(b) Via $\Omega_{\mathfrak{B}}$: symmetric matrices are determined by the entries on and above the diagonal, alternating ones by those strictly above.

(c) $B = \frac{1}{2}(B+B^t) + \frac{1}{2}(B-B^t)$ with $B^t(v,w) = B(w,v)$, and a form both symmetric and alternating is 0. □

Question 13.1.14

For $F = \mathbb{R}$, how can we tell from $[B]_{\mathfrak{B}}$ whether a symmetric bilinear form is an inner product (positive definite)? (Answered in §15.3.)

13.2 Quadratic forms

Motivation. A symmetric bilinear form is the same thing as a quadratic form $Q(v) = \frac{1}{2}B(v,v)$, i.e. a homogeneous quadratic polynomial in the coordinates. Chapters 9 and 10 were really about quadratic expressions.

Throughout, $1 + 1 \neq 0$ in F.

Example 13.2.1

For $J = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$, $B^J_{\mathcal{E}}((x,y)^t, (x,y)^t) = 2(x^2 - xy + y^2)$, and $Q(x,y) = x^2 - xy + y^2$ is the familiar quadratic form.

Definition 13.2.2 (Quadratic form)

$Q : V \to F$ is a **quadratic form** if (Q1) $B_Q(v,w) = Q(v+w) - Q(v) - Q(w)$ is a symmetric bilinear form, and (Q2) $Q(cv) = c^2 Q(v)$.

Exercise 13.2.3

For a symmetric bilinear form B, $Q(v) = \frac{1}{2}B(v,v)$ is a quadratic form with $B_Q = B$.

Solution. $Q(v+w) - Q(v) - Q(w) = \frac{1}{2}(B(v,w) + B(w,v)) = B(v,w)$, and $Q(cv) = c^2 Q(v)$. □

Observation 13.2.4

$Q \mapsto B_Q$ is a bijection between quadratic forms and symmetric bilinear forms on V.

Proof. $B_Q(v,v) = Q(2v) - 2Q(v) = 4Q(v) - 2Q(v) = 2Q(v)$, so Q is recovered from B_Q: injective. Surjective by Exercise 13.2.3. ■

Exercise 13.2.5

The quadratic forms on V form a vector space, and $Q \mapsto B_Q$ is a vector space isomorphism.

Solution. Pointwise operations preserve (Q1), (Q2), and $B_{aQ+Q'} = aB_Q + B_{Q'}$. □

Why "quadratic". With $[B_Q]_{\mathfrak{B}} = J = (a_{ij})$ (symmetric) and $[v]_{\mathfrak{B}} = (x_1, \dots, x_n)^t$,

$$2Q(v) = B_Q(v,v) = \sum_{i,j} a_{ij}x_ix_j = \sum_i a_{ii}x_i^2 + 2\sum_{i<j} a_{ij}x_ix_j,$$

a homogeneous quadratic polynomial in the coordinates. Conversely every homogeneous quadratic $2Q = \sum_i a_{ii}x_i^2 + 2\sum_{i<j} a_{ij}x_ix_j$ gives, with $a_{ji} = a_{ij}$, $B_Q(v,w) = \sum_{i,j} x_i a_{ij} y_j = [v]^t J [w]$. So symmetric bilinear forms, quadratic forms and homogeneous quadratic polynomials are the same thing.

Definition 13.2.6 (Quadratic space)

A vector space with a symmetric bilinear form B (hence a quadratic form Q) is a **quadratic space** $(V,B) = (V,B,Q)$.

13.3 Orthogonal groups and symplectic groups

Motivation. The orthogonal group of a quadratic space consists of the invertible operators preserving B; in a basis it is $\{A : A^tJA = J\}$, which settles Remark 10.6.4. Replacing symmetric by alternating forms gives the symplectic groups.

Definition 13.3.1 (Orthogonal group of a quadratic space)

$\mathbf{O}(V,B) = \{L \in \mathrm{GL}(V) : B(Lv,Lw) = B(v,w)\ \forall v,w\}$, the **orthogonal group**; its elements are **orthogonal operators**. Studying (V,B) and $\mathbf{O}(V,B)$ is **orthogonal geometry**.

(We consider only elements of $\mathrm{GL}(V)$ from the start; injectivity is not even needed, but we skip that.)

Exercise 13.3.2

(a) $\mathbf{O}(V,B) \leq \mathrm{GL}(V)$.
(b) $\mathbf{SO}(V,B) = \{L \in \mathbf{O}(V,B) : \det L = 1\} \trianglelefteq \mathbf{O}(V,B)$.

Solution. (a) I preserves B; if L, M do, so do LM and L^{-1} ($B(L^{-1}v, L^{-1}w) = B(LL^{-1}v, LL^{-1}w)$).
(b) It is the kernel of $\det : \mathbf{O}(V,B) \to F^\times$. □

Exercise 13.3.3

(a) $\mathbf{O}(V,B) = \{L \in \mathrm{GL}(V) : Q(Lv) = Q(v)\ \forall v\}$.
(b) If $\mathfrak{B} = \{v_i\}$ is a basis, $\mathbf{O}(V,B) = \{L \in \mathrm{GL}(V) : B(Lv_i, Lv_j) = B(v_i,v_j)\ \forall i,j\}$.

Solution. (a) As in Observation 9.3.4: $B(v,w) = Q(v+w) - Q(v) - Q(w)$ is determined by Q.

(b) By bilinearity $(*)$. □

Exercise 13.3.4

Can $\mathbf{O}(V,B) = \mathrm{GL}(V)$?

Solution. Yes, iff $B = 0$ (or $V = 0$). If $B \neq 0$, choose v with $Q(v) \neq 0$ (Lemma: possible since $1+1 \neq 0$, cf. Exercise 13.2.3); then $L = 2I$ satisfies $Q(Lv) = 4Q(v) \neq Q(v)$ (as long as $3 \neq 0$; in characteristic 3 use $L = cI$ with $c^2 \neq 1$ if available, e.g. fields with more than 3 elements). For $B = 0$ every L preserves B. □

Proposition 13.3.5

Let $\mathfrak{B}$ be a basis of n-dimensional (V,B) and $J = [B]_{\mathfrak{B}}$. Then

$$\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{O}(V,B)\} = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}.$$

Proof. By the fundamental theorem and Observation 13.1.1, $B(Lv,Lw) = B(v,w)$ for all v,w means $[Lv]_{\mathfrak{B}}^t J [Lw]_{\mathfrak{B}} = [v]_{\mathfrak{B}}^t J [w]_{\mathfrak{B}}$. Since $[Lv]_{\mathfrak{B}} = [L]_{\mathfrak{B}}^{\mathfrak{B}}[v]_{\mathfrak{B}}$, this is $[v]^t([L]^t J[L])[w] = [v]^t J[w]$ for all coordinate vectors, i.e. $[L]^t J[L] = J$ by Exercise 9.3.7. The argument is reversible. (Symmetry of J is not used.) ■

This settles Remark 10.6.4 ($J = I$ for an orthonormal basis).

Definition 13.3.6 (Matrix orthogonal groups)

For a symmetric $J \in \mathfrak{M}_{n,n}(F)$, $\mathbf{O}_n^J(F) = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}$, the **orthogonal group** of J. (Thus $\mathbf{O}(n) = \mathbf{O}_n^I(\mathbb{R})$.)

$\mathbf{O}_n^J(F) \leq \mathrm{GL}_n(F)$, $\mathbf{SO}_n^J(F) \trianglelefteq \mathbf{O}_n^J(F)$, and with $J = [B]_{\mathfrak{B}}$, $\Psi_{\mathfrak{B}}^{\mathfrak{B}}$ restricts to $\mathbf{O}(V,B) \approx \mathbf{O}_n^J(F)$ (compare §11.9).

Exercise 13.3.7

If J is invertible symmetric and $A \in \mathbf{O}_n^J(F)$, then $\det A = \pm 1$.

Solution. $\det(A)^2 \det J = \det J \neq 0$. □

Definition 13.3.8 (Symplectic group)

For an alternating form B on V, $\mathbf{Sp}(V,B) = \{L \in \mathrm{GL}(V) : B(Lv,Lw) = B(v,w)\ \forall v,w\}$ is the **symplectic group**; this is **symplectic geometry**.

Proposition 13.3.9

If B is alternating, $\mathfrak{B}$ a basis and $J = [B]_{\mathfrak{B}}$ (skew-symmetric), then $\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{Sp}(V,B)\} = \{A \in \mathrm{GL}_n(F) : A^t J A = J\}$.

Proof. Word for word the proof of Proposition 13.3.5, which did not use symmetry of J. ■

Definition 13.3.10 (Matrix symplectic groups)

For skew-symmetric J, $\mathbf{Sp}_n^J(F) = \{A \in \mathrm{GL}_n(F) : A^tJA = J\}$.

Exercise 13.3.11

For $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $\mathbf{Sp}_2^J(F) = \mathrm{SL}_2(F)$.

Solution. For any 2×2 A, $A^tJA = (\det A)J$ (direct computation: A^tJA is alternating, hence a multiple of J, and the $(1,2)$ entry is $ad - bc$). So $A^tJA = J$ iff $\det A = 1$. □

In general, for invertible skew-symmetric J (n even), $\mathbf{Sp}_n^J(F) \leq \mathrm{SL}_n(F)$; the proof is postponed (so there is no "**SSp**").

Question 13.3.12

For invertible symmetric J, is there always $A \in \mathbf{O}_n^J(F)$ with $\det A = -1$? How should "reflections" of a quadratic space be defined? (See §13.5.)

13.4 $\mathbf{O}(1,1)$ and $\mathbf{O}(3,1)$

Motivation. For the form $x^2 - y^2$ on $\mathbb{R}^2$, "rotations" are hyperbolic: cosh and sinh replace cos and sin, and the orthogonal group splits into four copies of a line instead of two circles. Adding three space coordinates gives the Lorentz group of special relativity.

For $J = \mathrm{diag}(1,-1)$, $B_{\mathcal{E}}^J((a,b)^t,(c,d)^t) = ac - bd$ (quadratic form $2Q = x^2 - y^2$), and

$$\mathbf{O}(1,1) = \mathbf{O}_2^J(\mathbb{R}) = \{A \in \mathrm{GL}_2(\mathbb{R}) : A^tJA = J\}.$$

$A = \left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right) \in \mathbf{O}(1,1)$ iff $x^2 - y^2 = 1 = w^2 - z^2$ and $xz - yw = 0$: the columns lie on the hyperbolas $x^2 - y^2 = 1$, $w^2 - z^2 = 1$. Recall $\cosh t = \frac{e^t + e^{-t}}{2}$, $\sinh t = \frac{e^t - e^{-t}}{2}$.

Exercise 13.4.1

For $t, s \in \mathbb{R}$: (a) $\cosh^2 t - \sinh^2 t = 1$;
(b) $\cosh(-t) = \cosh t$, $\sinh(-t) = -\sinh t$;
(c) $\cosh(t+s) = \cosh t \cosh s + \sinh t \sinh s$;
(d) $\sinh(t+s) = \cosh t \sinh s + \sinh t \cosh s$.

Solution. Expand the exponentials; e.g. (a): $\frac{(e^t+e^{-t})^2 - (e^t-e^{-t})^2}{4} = 1$. □

Since $\cosh t \geq 1$, $t \mapsto (\cosh t, \sinh t)^t$ parametrizes the right branch of $x^2 - y^2 = 1$, and $(-\cosh t, \sinh t)^t$ the left one. If $x, w > 0$, write $(x,y) = (\cosh t, \sinh t)$, $(z,w) = (\sinh s, \cosh s)$; then $0 = xz - yw = \sinh(s-t)$, so $s = t$ and

$$A = \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix}.$$

Similarly: if $x > 0$, $w < 0$: $A = \left(\begin{smallmatrix} \cosh u & \sinh u \\ -\sinh u & -\cosh u \end{smallmatrix}\right)$; if $x < 0$, $w > 0$: $A = \left(\begin{smallmatrix} -\cosh t & -\sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)$; if $x, w < 0$: $A = \left(\begin{smallmatrix} -\cosh u & -\sinh u \\ -\sinh u & -\cosh u \end{smallmatrix}\right)$.

Exercise 13.4.2

With $\mathbf{SO}(1,1) = \mathrm{SL}_2(\mathbb{R}) \cap \mathbf{O}(1,1)$: $\mathbf{O}(1,1)/\mathbf{SO}(1,1) \approx \mu_2$, $[\mathbf{O}(1,1) : \mathbf{SO}(1,1)] = 2$, and $\mathbf{O}(1,1) = \mathbf{SO}(1,1) \amalg \mathrm{diag}(-1,1)\mathbf{SO}(1,1)$.

Solution. $\det : \mathbf{O}(1,1) \to \mu_2$ is onto ($\det \mathrm{diag}(-1,1) = -1$) with kernel $\mathbf{SO}(1,1)$; apply the First Isomorphism Theorem. □

Exercise 13.4.3

Let $\mathbf{SO}^\circ(1,1) = \left\{ \left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right) \right\} = \left\{ \left(\begin{smallmatrix} x & y \\ y & x \end{smallmatrix}\right) : x \geq 1,\ x^2 - y^2 = 1 \right\}$. Then $\mathbf{SO}^\circ(1,1) < \mathbf{SO}(1,1)$, $\mathbf{SO}(1,1) = \mathbf{SO}^\circ(1,1) \amalg (-I \cdot \mathbf{SO}^\circ(1,1))$, $[\mathbf{SO}(1,1) : \mathbf{SO}^\circ(1,1)] = 2$, and $\mathbf{SO}^\circ(1,1) \trianglelefteq \mathbf{SO}(1,1)$.

Solution. The product of two such matrices is the matrix for $t+s$ (Exercise 13.4.1), and the inverse is the one for $-t$. From the four cases, $\det A = 1$ exactly in the cases $x, w > 0$ (which is $\mathbf{SO}^\circ$) and $x, w < 0$ (which is $-\mathbf{SO}^\circ$). Index 2 gives normality (Observation 12.2.9). □

Hence $[\mathbf{O}(1,1) : \mathbf{SO}^\circ] = 4$ (Observation 12.1.12) and, writing $\mathbf{SO}^\circ = \mathbf{SO}^\circ(1,1)$,

$$\mathbf{O}(1,1) = \mathbf{SO}^\circ \amalg (-I \cdot \mathbf{SO}^\circ) \amalg (\mathrm{diag}(-1,1)\mathbf{SO}^\circ) \amalg (\mathrm{diag}(1,-1)\mathbf{SO}^\circ),$$

matching the four cases.

Exercise 13.4.4

Is $\mathbf{SO}^\circ(1,1)$ normal in $\mathbf{O}(1,1)$?

Solution. Yes: $S = \mathrm{diag}(1,-1)$ gives $S\left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)S^{-1} = \left(\begin{smallmatrix} \cosh t & -\sinh t \\ -\sinh t & \cosh t \end{smallmatrix}\right) \in \mathbf{SO}^\circ$, and $\mathbf{O}(1,1)$ is generated by $\mathbf{SO}(1,1)$ (in which $\mathbf{SO}^\circ$ is normal) and S. □

Exercise 13.4.5

$\varphi : (\mathbb{R}, +) \to \mathbf{SO}^\circ(1,1)$, $\varphi(t) = \left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)$, is a group isomorphism.

Solution. A homomorphism by Exercise 13.4.1(c), (d); onto by the parametrization; injective since $\sinh t = 0$ only for $t = 0$. □

So $\mathbf{SO}^\circ(1,1)$ looks like $\mathbb{R}$, and $\mathbf{O}(1,1)$ is a disjoint union of four "real lines", whereas $\mathbf{O}(2)$ is a union of two circles.

Exercise 13.4.6

$\mathbf{O}(2) \not\approx \mathbf{O}(1,1)$.

Solution. [Supplement] Elements of finite order: $\mathbf{O}(2)$ contains $R_{2\pi/3}$, of order 3. In $\mathbf{O}(1,1)$ an element of finite order lies in $\mathbf{SO}^\circ(1,1)$ times one of $\pm I, \mathrm{diag}(\pm 1, \mp 1)$. $\mathbf{SO}^\circ \approx (\mathbb{R}, +)$ is torsion-free, so a power A^2 of $A \in \mathbf{O}(1,1)$ lies in $\mathbf{SO}^\circ$ (index 4, quotient $\mu_2 \times \mu_2$), and $A^{2k} = 1$ forces $A^2 = 1$. Hence all elements of finite order in $\mathbf{O}(1,1)$ have order 1 or 2, and isomorphisms preserve orders. □

Definition 13.4.7 (Lorentz group)

For $J = \mathrm{diag}(1,1,1,-1)$, $\mathbf{O}(3,1) = \mathbf{O}_4^J(\mathbb{R})$ is the **Lorentz group**, the symmetry group of space-time $\mathbb{R}^4$.

Exercise 13.4.8

$\mathbf{O}(3)$ and $\mathbf{O}(1,1)$ are isomorphic to subgroups of $\mathbf{O}(3,1)$.

Solution. $A \mapsto \operatorname{diag}(A,1)$ embeds $\mathbf{O}(3)$; $\left(\begin{smallmatrix} x & z \\ y & w \end{smallmatrix}\right) \mapsto \operatorname{diag}(1,1,\cdot)$ with the 2×2 block in positions $3,4$ embeds $\mathbf{O}(1,1)$. Both preserve $A^tJA = J$ blockwise. □

13.5 Non-degenerate bilinear forms

Motivation. Without positive definiteness a vector can be perpendicular to itself, and $W \oplus W^\perp = V$ may fail. What survives is $\dim W^\perp = \dim V - \dim W$ for non-degenerate forms, together with orthogonal bases of quadratic spaces and reflections.

Throughout, V is n-dimensional with a bilinear form B, and $1+1 \neq 0$.

Definition 13.5.1 (Orthogonality)

$v \perp w$ if $B(v,w) = 0$; $S \perp T$ if $B(v,w) = 0$ for all $v \in S$, $w \in T$.

Remark 13.5.2

If B is symmetric or alternating, $v \perp w \iff w \perp v$. Conversely one can show that $\perp$ is symmetric only for such B; this is why we study orthogonal and symplectic geometry.

Definition 13.5.3 (Orthogonal basis)

A basis $\mathfrak{B} = \{v_i\}$ of mutually perpendicular vectors is an **orthogonal basis**; if moreover $B(v_i, v_i) = 1$ for all i, an **orthonormal basis**.

Exercise 13.5.4

(a) $\mathfrak{B}$ is orthogonal iff $[B]_{\mathfrak{B}}$ is diagonal.
(b) $\mathfrak{B}$ is orthonormal iff $[B]_{\mathfrak{B}} = I$.

Solution. The entries of $[B]_{\mathfrak{B}}$ are $B(v_i, v_j)$. □

Now $B(v,v) = 0$ is possible for $v \neq 0$, so orthogonal sets need not be independent, orthogonal bases cannot always be normalized, and Gram–Schmidt is unavailable.

Notation 13.5.5

For $S \subseteq V$, $S^\perp = \{v \in V : B(w,v) = 0 \text{ for all } w \in S\}$.

Exercise 13.5.6

(a) $S^\perp = \langle S \rangle^\perp \leq V$.
(b) If B is symmetric or alternating, $W \leq (W^\perp)^\perp$.

Solution. (a) Linearity in the second variable, and Exercise 10.2.16.

(b) For $w \in W$, $v \in W^\perp$: $B(v,w) = \pm B(w,v) = 0$. □

If $B = 0$, then $W^\perp = V$, so the perp theorem fails in general.

Exercise 13.5.7

If $B(w,w) \neq 0$, then $\dim\langle w\rangle^\perp = \dim V - 1$.

Solution. $v \mapsto B(w,v)$ is a nonzero linear functional (it is $\neq 0$ at w); its kernel $\langle w\rangle^\perp$ has dimension $n-1$. □

Question 13.5.8

When is $\dim W^\perp = \dim V - \dim W$ for $W \leq V$?

Definition 13.5.9 (Non-degenerate)

B is **non-degenerate** if $V^\perp = 0$.

(Exercises 9.1.3, 10.2.1 said inner products are non-degenerate; the ordinary inner product $B_{\mathcal{E}}^I$ is non-degenerate.)

Proposition 13.5.10

For a basis $\mathfrak{B}$ of V: B is non-degenerate iff $\det[B]_{\mathfrak{B}} \neq 0$.

First proof (elementary). With $J = [B]_{\mathfrak{B}}$: B is degenerate iff some $0 \neq Y$ has $X^tJY = 0$ for all $X \in F^n$ (Observation 13.1.1), iff (taking $X = e_i$) some $0 \neq Y$ has $JY = 0$, iff $\det J = 0$. ■

(A second proof via the dual space is given after Observation 13.8.3.)

Theorem 13.5.11 (Perp theorem, non-degenerate case)

If B is non-degenerate and $W \leq V$, then $\dim W^\perp = \dim V - \dim W$. (This does *not* say $V = W \oplus W^\perp$.)

Proof. Extend a basis $\{v_1,\dots,v_m\}$ of W to a basis $\mathfrak{B}$ of V, $J = [B]_{\mathfrak{B}}$. Then $v \in W^\perp$ iff $0 = [v_i]^tJ[v] = e_i^tJ[v] = [J]_i[v]$ for $i \leq m$. With K the $m \times n$ matrix of the first m rows of J, $v \in W^\perp$ iff $[v]_{\mathfrak{B}} \in \ker L_K$, so $\dim W^\perp = \dim\ker L_K$. J is invertible, so its rows are independent and $\operatorname{rk} K = m$. By the rank theorem $\dim\operatorname{im} L_K = m$, and by the dimension theorem $\dim\ker L_K = n - m$. ■

Proposition 13.5.12 (The three pillars)

Of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (non-degenerate case), any two imply the third.

Proof. As in Proposition 10.3.11, using the ordinary inner product $B_{\mathcal{E}}^I$ on F^n (non-degenerate): the row space W of A has $W^\perp = \ker L_A$, and the three statements are the three linear relations among n, $\dim\ker L_A$, $\dim W$ and $\dim\operatorname{im} L_A$. ■

These proofs do not depend on F (the rank theorem's proof in §10.3 for $F = \mathbb{C}$ needed a trick because the Hermitian dot product was used). A proof via the dual space follows in §13.8.

For the rest of this section (V,B) is a quadratic space (B symmetric).

Exercise 13.5.13

If (V, B) is a non-degenerate quadratic space and $U, W \leq V$: (a) $(W^\perp)^\perp = W$;
(b) $(U + W)^\perp = U^\perp \cap W^\perp$;
(c) $U^\perp + W^\perp = (U \cap W)^\perp$.

Solution. As in Exercise 10.3.6, using Theorem 13.5.11 and Exercise 13.5.6(b). □

Lemma 13.5.14

If B is symmetric, $w \in V$ and $B(w, w) \neq 0$, then $V = \langle w \rangle \oplus \langle w \rangle^\perp$.

Proof. For $v \in V$, $v = \frac{B(v,w)}{B(w,w)} w + \left(v - \frac{B(v,w)}{B(w,w)} w \right)$ with the second term in $\langle w \rangle^\perp$. And $cw \in \langle w \rangle^\perp$ means $cB(w, w) = 0$, so $c = 0$. ■

Observation 13.5.15

If B is symmetric and $B(v, v) = 0$ for all v, then $B = 0$.

Proof. $2B(v, w) = B(v + w, v + w) - B(v, v) - B(w, w) = 0$. ■

Theorem 13.5.16

Every quadratic space (V, B) $(V \neq 0)$ has an orthogonal basis (B need not be non-degenerate).

Proof. If $B = 0$, every basis is orthogonal. Otherwise there is w with $B(w, w) \neq 0$ (Observation 13.5.15), and $V = \langle w \rangle \oplus \langle w \rangle^\perp$ (Lemma 13.5.14). By induction on dimension, $\langle w \rangle^\perp$ (with B restricted) has an orthogonal basis; add w. ■

Definition 13.5.17 (Reflection)

For symmetric B and w with $B(w, w) \neq 0$, $S_w(v) = v - \frac{2B(v,w)}{B(w,w)} w$ is the **reflection** (**symmetry**) with respect to w.

Exercise 13.5.18

For symmetric B and $B(w, w) \neq 0$: (a) S_w is linear and $S_w \in \mathbf{O}(V, B)$;
(b) $S_w = S_{cw}$ for $c \neq 0$;
(c) $S_w(w) = -w$;
(d) $S_w(v) = v$ for $v \in \langle w \rangle^\perp$.

Solution. (a) Linearity is clear. $B(S_w v, S_w u) = B(v, u) - \frac{2B(v,w)B(w,u)}{B(w,w)} - \frac{2B(u,w)B(v,w)}{B(w,w)} + \frac{4B(v,w)B(u,w)}{B(w,w)^2} B(w, w) = B(v, u)$, using symmetry ($B(w, u) = B(u, w)$). (b)–(d) as in Exercise 9.4.4. □

By Lemma 13.5.14, S_w is the unique linear operator satisfying (c), (d). With an orthogonal basis $\mathfrak{C} = \{v_2, \ldots, v_n\}$ of $\langle w \rangle^\perp$, $\mathfrak{B} = \{w, v_2, \ldots, v_n\}$ is an orthogonal basis, $[S_w]_\mathfrak{B}^\mathfrak{B} = \operatorname{diag}(-1, 1, \ldots, 1) \in \mathbf{O}_n^J(F)$ (Proposition 13.3.5, $J = [B]_\mathfrak{B}$ diagonal), $\det S_w = -1$ and $S_w^2 = I$.

Exercise 13.5.19

For a quadratic space (V, B): (a) some $L \in \mathbf{O}(V, B)$ has $\det L = -1$;
(b) if B is non-degenerate, $\mathbf{O}(V, B)/\mathbf{SO}(V, B) \approx \mu_2$.

Solution. (a) If $B \neq 0$, a reflection S_w. If $B = 0$, $\mathbf{O}(V, B) = \mathrm{GL}(V)$ contains $\mathrm{diag}(-1, 1, \dots)$.

(b) $\det : \mathbf{O}(V, B) \to F^\times$ takes values ± 1 (Exercise 13.3.7) and is onto μ_2 by (a); apply the First Isomorphism Theorem. □

Exercise 13.5.20

(a) $-1+2\cosh^2\frac{t}{2} = \cosh t = 1+2\sinh^2\frac{t}{2}$ and $2\cosh\frac{t}{2}\sinh\frac{t}{2} = \sinh t$. (b) For $J = \mathrm{diag}(1, -1)$ and $Y = (\cosh\frac{t}{2}, -\sinh\frac{t}{2})^t$, $B^J_{\mathcal{E}}(Y, Y) = 1$ and $[S_Y]_{\mathcal{E}} = \begin{pmatrix} -\cosh t & -\sinh t \\ \sinh t & \cosh t \end{pmatrix}$, so this matrix is a reflection of $\mathbf{O}(1, 1)$.

Solution. (a) Exercise 13.4.1(c),

(d) with $s = t$, replaced by $t/2$. (b) $B(Y, Y) = \cosh^2\frac{t}{2} - \sinh^2\frac{t}{2} = 1$. $S_Y e_1 = e_1 - 2\cosh\frac{t}{2}\,Y = (1-2\cosh^2\frac{t}{2},\ 2\cosh\frac{t}{2}\sinh\frac{t}{2})^t = (-\cosh t, \sinh t)^t$, and $S_Y e_2 = e_2 - 2B(e_2, Y)Y = e_2 - 2\sinh\frac{t}{2}\,Y = (-2\sinh\frac{t}{2}\cosh\frac{t}{2},\ 1 + 2\sinh^2\frac{t}{2})^t = (-\sinh t, \cosh t)^t$. □

The orthogonal-geometry story continues with "every orthogonal operator is a product of finitely many reflections" (compare Exercise 9.5.15).

13.6 Dual spaces and dual maps

Motivation. The dual space V^* of linear functionals, its dual basis, the natural isomorphism $V \cong V^{**}$ and the dual map $L^*(g) = g \circ L$ are the tools for a basis-free treatment of forms. The matrix of a dual map is the transpose: transposition is born here.

$V^* = \mathfrak{L}(V, F)$ is the **dual space**, its elements **linear functionals**; $V^{**} = (V^*)^*$ is the **double dual**. For a basis $\mathfrak{B} = \{v_1, \dots, v_n\}$, $v_i^*(v_j) = \delta_{ij}$ defines the **dual basis** $\mathfrak{B}^* = \{v_1^*, \dots, v_n^*\}$ of V^* (§5.1), so $\dim V = \dim V^* = \dim V^{**}$.

Proposition 13.6.1

$\psi^V : V \to V^{**}$, $\psi^V(v)(f) = f(v)$, is a vector space isomorphism, independent of any choice of basis (a **natural isomorphism**); we use it to identify V and V^{**}.

Proof. $\psi^V(v)$ is linear in f, so $\psi^V(v) \in V^{**}$; ψ^V is linear in v. If $\psi^V(v) = 0$, then $f(v) = 0$ for all $f \in V^*$, so $v = 0$ (Observation 13.6.2). As $\dim V = \dim V^{**}$, the dimension theorem gives surjectivity. ■

Observation 13.6.2

If $f(v) = 0$ for all $f \in V^*$, then $v = 0$.

Proof. With $v = \sum a_i v_i$, $0 = v_i^*(v) = a_i$ for all i. ■

Observation 13.6.3

For a basis $\{v_i\}$ of V, $\psi^V(v_i) = v_i^{**}$ (the dual basis of $\{v_i^*\}$).

Proof. $\psi^V(v_i)(v_j^*) = v_j^*(v_i) = \delta_{ij} = v_i^{**}(v_j^*)$; two linear maps agreeing on a basis are equal. ■

Notation 13.6.4

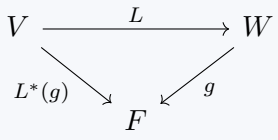
When there is a natural isomorphism, we identify the two spaces and abuse notation: $V = V^{**}$, $v_i = v_i^{**}$.

Remark 13.6.5

The isomorphisms of the first and second isomorphism theorems (Theorems 12.4.14, 12.4.21) are also natural.

Definition 13.6.6 (Dual map)

For linear $L : V \to W$, $L^* : W^* \to V^*$, $L^*(g) = g \circ L$, is the **dual map** of L.



Exercise 13.6.7

(a) For $g \in W^*$, $g \circ L \in V^*$.
(b) L^* is linear.

Solution. (a) A composite of linear maps.

(b) $(ag + h) \circ L = a(g \circ L) + h \circ L$. □

Observation 13.6.8

For bases $\mathfrak{B} = \{v_j\}$ of V, $\mathfrak{C} = \{w_i\}$ of W and $L \in \mathfrak{L}(V, W)$: $[L^*]_{\mathfrak{C}^*}^{\mathfrak{B}^*} = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^t$.

Proof. Let $[L]_{\mathfrak{B}}^{\mathfrak{C}} = (a_{ij})$, i.e. $Lv_j = \sum_i a_{ij} w_i$. We must show $L^* w_i^* = \sum_j a_{ij} v_j^*$, i.e. $(L^* w_i^*)(v_\ell) = a_{i\ell}$. Indeed $(L^* w_i^*)(v_\ell) = w_i^*(Lv_\ell) = w_i^*\left(\sum_k a_{k\ell} w_k\right) = a_{i\ell}$. ■

So the transpose of a matrix corresponds to the dual of a linear map; this answers Question 5.4.22. (The meaning of $L^t \in \mathfrak{L}(W, V)$ itself is still open.)

Exercise 13.6.9

For $L \in \mathfrak{L}(V, W)$, $M \in \mathfrak{L}(W, U)$: (a) $(I_V)^* = I_{V^*}$;
(b) $(M \circ L)^* = L^* \circ M^*$;
(c) if L is invertible, so is L^*, with $(L^*)^{-1} = (L^{-1})^*$.

Solution. (a) $g \circ I = g$.

(b) $(ML)^*(g) = g \circ M \circ L = L^*(M^*(g))$.

(c) Apply (a), (b) to $LL^{-1} = I$ and $L^{-1}L = I$. □

Exercise 13.6.10

(a) $(L_1 + L_2)^* = L_1^* + L_2^*$;
(b) $(cL)^* = cL^*$;
(c) $* : \mathfrak{L}(V, W) \to \mathfrak{L}(W^*, V^*)$, $L \mapsto L^*$, is an isomorphism.

Solution. (a), (b): $g \circ (L_1 + L_2) = g \circ L_1 + g \circ L_2$, $g \circ (cL) = c(g \circ L)$.

(c) $*$ is linear by (a), (b), and via Observation 13.6.8 it corresponds to the transpose $\mathfrak{M}_{m,n}(F) \to \mathfrak{M}_{n,m}(F)$, which is bijective. □

Exercise 13.6.11

Translate Exercises 13.6.9 and 13.6.10 into matrix language.

Solution. $I^t = I$, $(AB)^t = B^t A^t$, $(A^{-1})^t = (A^t)^{-1}$, $(A + B)^t = A^t + B^t$, $(cA)^t = cA^t$, and $A \mapsto A^t$ is an isomorphism $\mathfrak{M}_{m,n}(F) \to \mathfrak{M}_{n,m}(F)$. □

Exercise 13.6.12

For $L \in \mathfrak{L}(V, W)$, the square

$$\begin{array}{ccc} V & \xrightarrow{L} & W \\ {\scriptstyle \psi^V}\downarrow & & \downarrow{\scriptstyle \psi^W} \\ V^{**} & \xrightarrow[L^{**}]{} & W^{**} \end{array}$$

commutes; so with $V = V^{**}$, $W = W^{**}$, $L = L^{**}$ (in matrices, $A = (A^t)^t$).

Solution. For $v \in V$, $g \in W^*$: $L^{**}(\psi^V v)(g) = \psi^V v(L^* g) = (g \circ L)(v) = g(Lv) = \psi^W(Lv)(g)$. □

13.7 Duality

Motivation. The natural pairing $\varepsilon(v, f) = f(v)$ between V and V^* behaves like a non-degenerate bilinear form. Annihilators satisfy $\dim W^{\text{perp}} = \dim V - \dim W$, and the dual statements follow by symmetry through $V = V^{**}$.

V comes with the bilinear form $\varepsilon : V \times V^* \to F$, $\varepsilon(v, f) = f(v)$ (a **bilinear pairing**); we write $v \perp f$ if $\varepsilon(v, f) = 0$.

Notation 13.7.1

For $S \subseteq V$, $T \subseteq V^*$: $S^{\text{perp}} = \{f \in V^* : \varepsilon(v, f) = 0 \ \forall v \in S\}$ and $T^{\text{perp}} = \{v \in V : \varepsilon(v, f) = 0 \ \forall f \in T\}$ (the **annihilators**).

Exercise 13.7.2

(a) $S^{\text{perp}} = \langle S \rangle^{\text{perp}} \leq V^*$.
(b) $T^{\text{perp}} = \langle T \rangle^{\text{perp}} \leq V$.

Solution. Linearity of ε in each variable. □

Theorem 13.7.3 (Duality)

For $W \leq V$, $Y \leq V^*$: (a) $\dim W^{\text{perp}} = \dim V - \dim W$;
(b) $\dim Y^{\text{perp}} = \dim V - \dim Y$.

Proof. (a) Extend a basis $\{v_1, \dots, v_r\}$ of W to a basis $\{v_1, \dots, v_n\}$ of V. Then $\{v_{r+1}^*, \dots, v_n^*\}$ is a basis of W^{perp}: these vanish on W, and if $f = \sum a_i v_i^* \in W^{\text{perp}}$, then $a_i = f(v_i) = 0$ for $i \leq r$.

(b) (First proof.) Extend a basis $\{f_1, \dots, f_s\}$ of Y to a basis $\{f_1, \dots, f_n\}$ of V^* with dual basis $\{f_1^*, \dots, f_n^*\}$ of V^{**}. Since ψ^V is an isomorphism, it suffices that $\psi^V(Y^{\text{perp}}) = \langle f_{s+1}^*, \dots, f_n^* \rangle$. If $v \in Y^{\text{perp}}$ and $\psi^V(v) = \sum a_i f_i^*$, then $a_i = \psi^V(v)(f_i) = f_i(v) = 0$ for $i \leq s$. Conversely, $f_j^* = \psi^V(w_j)$ for a unique $w_j \in V$, and for $i \leq s < j$, $0 = f_j^*(f_i) = f_i(w_j)$, so $w_j \in Y^{\text{perp}}$. ■

(b) Second proof. V^* comes with $\delta : V^{**} \times V^* \to F$, $\delta(\alpha, f) = \alpha(f)$. Applying (a) to V^* and $Y \leq V^*$, the annihilator $Y^{\text{d-perp}} = \{\alpha \in V^{**} : \alpha(f) = 0 \ \forall f \in Y\}$ has dimension $\dim V^* - \dim Y = \dim V - \dim Y$. Under ψ^V, $\delta(\psi^V v, f) = f(v) = \varepsilon(v, f)$ (the square below commutes), so $\psi^V(Y^{\text{perp}}) = Y^{\text{d-perp}}$.

$$\begin{array}{ccc} V \times V^* & \xrightarrow{\varepsilon} & F \\ {\scriptstyle \psi^V \times I}\downarrow & & \downarrow{\scriptstyle I} \\ V^{**} \times V^* & \xrightarrow[\delta]{} & F \end{array}$$

■

That a proper subspace $W \neq V$ has a nonzero functional vanishing on it is easy; its "dual statement" needs the theorem:

Corollary 13.7.4

If $Y \leq V^*$, $Y \neq V^*$, then some $0 \neq v \in V$ has $f(v) = 0$ for all $f \in Y$.

Proof. By duality, $\dim Y^{\text{perp}} = \dim V - \dim Y > 0$. ■

Exercise 13.7.5

For $W \leq V$: (a) $(W^{\text{perp}})^{\text{perp}} = W$;
(b) with $V = V^{**}$, $(W^{\text{perp}})^{\text{d-perp}} = W$.

Solution. $W \leq (W^{\text{perp}})^{\text{perp}}$ is clear, and the dimensions agree by Theorem 13.7.3 applied twice.

(b) is (a) transported by ψ^V. □

Exercise 13.7.6

For $U, W \leq V$: (a) $(U + W)^{\text{perp}} = U^{\text{perp}} \cap W^{\text{perp}}$;
(b) $(U \cap W)^{\text{perp}} = U^{\text{perp}} + W^{\text{perp}}$.

Solution. (a) is immediate.

(b) $\supseteq$ is clear; the dimensions agree by Theorem 13.7.3, (a) and Corollary 12.4.22: $\dim(U^{\text{perp}} + W^{\text{perp}}) = (n - u) + (n - w) - (n - \dim(U + W)) = n - \dim(U \cap W)$. □

Exercise 13.7.7

For $L \in \mathfrak{L}(V,W)$: (a) $\ker(L^*) = (\operatorname{im} L)^{\text{perp}}$;
(b) $\operatorname{im}(L^*) = (\ker L)^{\text{perp}}$.

Solution. (a) $L^*g = 0 \iff g(Lv) = 0\ \forall v \iff g \in (\operatorname{im} L)^{\text{perp}}$. (b) $\operatorname{im} L^* \subseteq (\ker L)^{\text{perp}}$: $(g \circ L)(v) = 0$ for $v \in \ker L$. The dimensions agree: $\dim \operatorname{im} L^* = \dim W - \dim \ker L^* = \dim W - (\dim W - \operatorname{rk} L) = \operatorname{rk} L = \dim V - \dim \ker L$. □

(Compare Exercise 10.3.6 and Observation 10.7.9.)

13.8 B-identification

Motivation. A bilinear form B is the same as a linear map $\varphi_B : V \to V^*$, $v \mapsto B(\cdot, v)$, with matrix $[B]_{\mathfrak{B}}$. B is non-degenerate iff φ_B is an isomorphism; identifying $V = V^*$ through it turns B into the canonical pairing ε, which proves the perp theorem in one line.

Definition 13.8.1 (φ_B)

For a bilinear form B on V, $\varphi_B^V : V \to V^*$, $\varphi_B^V(v)(w) = B(w,v)$, is a linear map (well defined because B is linear in the first variable, linear because it is linear in the second).

Theorem 13.8.2

The following are equivalent: (1) B is non-degenerate; (2) φ_B^V is a monomorphism; (3) φ_B^V is an isomorphism; (4) every element of V^* is $\varphi_B^V(v)$ for some v.

Proof. $\ker \varphi_B^V = \{v : B(w,v) = 0\ \forall w\} = V^\perp$. So (1)⇔(2), and (2)⇔(3)⇔(4) by the dimension theorem ($\dim V = \dim V^*$). ■

The surjectivity of φ_B^V cannot be shown without the dimension theorem (and fails in infinite dimensions).

Observation 13.8.3

For a basis $\mathfrak{B}$ of V with dual basis $\mathfrak{B}^*$: $[\varphi_B^V]_{\mathfrak{B}}^{\mathfrak{B}^*} = [B]_{\mathfrak{B}}$.

Proof. With $[\varphi_B^V]_{\mathfrak{B}}^{\mathfrak{B}^*} = (b_{ij})$, $\varphi_B^V(v_j) = \sum_i b_{ij} v_i^*$, so $B(v_k, v_j) = \varphi_B^V(v_j)(v_k) = b_{kj}$. ■

So a bilinear form, an $n \times n$ matrix, and a linear map $V \to V^*$ are the same data once $\mathfrak{B}$ is fixed: dual spaces are unavoidable in the study of bilinear forms.

Second proof of Proposition 13.5.10. B is non-degenerate iff φ_B^V is an isomorphism, iff its matrix $[B]_{\mathfrak{B}}$ is invertible. ■

Exercise 13.8.4

Identifying $V = V^{**}$, the dual map $(\varphi_B^V)^* : V^{**} \to V^*$ becomes a map $V \to V^*$. Show that B is symmetric iff $(\varphi_B^V)^* = \varphi_B^V$.

Solution. $(\varphi_B^V)^*(\psi^V v)(w) = \psi^V(v)(\varphi_B^V w) = \varphi_B^V(w)(v) = B(v,w)$, while $\varphi_B^V(v)(w) = B(w,v)$. These agree for all v, w iff B is symmetric. (In matrices: $[B]^t = [B]$.) □

Notation 13.8.5 (B-identification)

If B is non-degenerate, φ_B^V is a natural isomorphism, and we identify $V = V^*$ through it, writing $v = \varphi_v = \varphi_B^V(v)$.

Remark 13.8.6

Transport B to V^*: $B^*(\varphi_v, \varphi_w) = B(w, v)$ is a non-degenerate bilinear form on V^*, giving $V^* = V^{**}$. For the identifications to be consistent we need $\varphi_{B^*}^{V^*} \circ \varphi_B^V = \psi^V$. Indeed $\varphi_{B^*}^{V^*}(\varphi_B^V(v))(\varphi_w) = B^*(\varphi_w, \varphi_v) = B(v, w) = \varphi_w(v) = \psi^V(v)(\varphi_w)$.

Exercise 13.8.7

For $f \in (\mathbb{R}^2)^*$, $f((x, y)^t) = x + y$, find the vector of $\mathbb{R}^2$ corresponding to f under $B_{\mathcal{E}}^J$-identification, for (a) $J = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$,
(b) $J = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$.

Solution. We need v with $X^t J v = x + y$, i.e. $Jv = (1, 1)^t$. (a) $v = (1, -1)^t$.
(b) $v = J^{-1}(1, 1)^t = \frac{1}{3}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)(1, 1)^t = (1, 1)^t$. □

Exercise 13.8.8

If B is non-degenerate and $\mathfrak{B} = \{v_i\}$ is a basis, there is a basis $\mathfrak{C} = \{w_j\}$ with $B(v_i, w_j) = \delta_{ij}$, and $[I]_{\mathfrak{C}}^{\mathfrak{B}} = [B]_{\mathfrak{B}}^{-1}$.

Solution. Let $w_j = (\varphi_B^V)^{-1}(v_j^*)$; then $B(v_i, w_j) = v_j^*(v_i) = \delta_{ij}$ (the basis "dual" to $\mathfrak{B}$ under the identification). Write $w_j = \sum_k c_{kj} v_k$, $C = [I]_{\mathfrak{C}}^{\mathfrak{B}} = (c_{kj})$. Then $\delta_{ij} = \sum_k B(v_i, v_k) c_{kj} = ([B]_{\mathfrak{B}} C)_{ij}$, so $C = [B]_{\mathfrak{B}}^{-1}$. □

Exercise 13.8.9

If V has an orthonormal basis $\{v_i\}$ (so (V, B) is a non-degenerate quadratic space), then $\varphi_B^V(v_i) = v_i^*$.

Solution. $\varphi_B^V(v_i)(v_k) = B(v_k, v_i) = \delta_{ki} = v_i^*(v_k)$. (Compare Observation 13.6.3.) □

Proposition 13.8.10

If B is non-degenerate, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{B} & F \\ {\scriptstyle I \times \varphi_B^V}\downarrow & & \downarrow{\scriptstyle I} \\ V \times V^* & \xrightarrow[\varepsilon]{} & F \end{array}$$

commutes: under B-identification, $B = \varepsilon$. In particular "$\perp$= perp".

Proof. $\varepsilon(v, \varphi_B^V w) = \varphi_B^V(w)(v) = B(v, w)$. ■

Theorem 13.8.11 (Perp theorem, non-degenerate case)

If B is non-degenerate and $W \leq V$, then under B-identification $W^\perp = W^{\mathrm{perp}}$; in particular $\dim W^\perp = \dim V - \dim W$.

Proof. By Proposition 13.8.10, $v \in W^\perp$ iff $\varphi_B^V(v) \in W^{\mathrm{perp}}$, i.e. $\varphi_B^V(W^\perp) = W^{\mathrm{perp}}$; apply Theorem 13.7.3(a). ■

Exercise 13.8.13

Let (V, B) be a non-degenerate quadratic space and $U \leq W \leq V$. Define $\vartheta : W \to (U^\perp/W^\perp)^*$, $\vartheta(w)(\bar{v}) = B(v, w)$ $(w \in W,\ v \in U^\perp)$. (a) ϑ is a well-defined linear map with $\ker \vartheta = U$. (b) There is a natural isomorphism $W/U \approx (U^\perp/W^\perp)^*$; in particular $W \approx (V/W^\perp)^*$. (c) There is a natural isomorphism $W/(W \cap W^\perp) \approx (W/(W \cap W^\perp))^*$.

Solution. (a) Note $W^\perp \leq U^\perp$. If $\bar{v} = \bar{v}'$, then $v - v' \in W^\perp$, so $B(v - v', w) = 0$: well defined. Linear in w. $\vartheta(w) = 0$ iff $w \perp U^\perp$ iff $w \in (U^\perp)^\perp = U$ (Exercise 13.5.13). (b) By (a) and the First Isomorphism Theorem, $W/U \approx \operatorname{im} \vartheta$. The dimensions agree: $\dim(U^\perp/W^\perp) = (n - u) - (n - w) = w - u$ (Theorem 13.5.11), so ϑ is onto. Take $U = 0$ for the special case. (c) The restriction of B to W induces $\bar{B}(\bar{w}, \bar{w}') = B(w, w')$ on $W/(W \cap W^\perp)$; it is well defined (the radical of $B|_W$ is $W \cap W^\perp$) and non-degenerate, so $\varphi_{\bar{B}}$ is a natural isomorphism (Theorem 13.8.2). □

13.9 Transpose operators

Motivation. Combining dual maps with B-identification defines the transpose L^t of a linear map between quadratic spaces, characterised by $C(Lv, w) = B(v, L^t w)$. Its matrix is $J_{\mathfrak{B}}^{-1}[L]^t J_{\mathfrak{C}}$, and orthogonal operators are exactly those with $L^t L = I$. This is the long-promised meaning of the transpose.

Throughout, (V, B) and (W, C) are non-degenerate quadratic spaces.

Definition 13.9.1 (Transpose map)

(a) For $L \in \mathfrak{L}(V, W)$, the **transpose** $L^t \in \mathfrak{L}(W, V)$ (with respect to B, C) is L^* under B- and C-identification, i.e. the unique linear map making

$$\begin{array}{ccc} W & \xrightarrow{L^t} & V \\ {\scriptstyle \varphi_C^W}\downarrow & & \downarrow{\scriptstyle \varphi_B^V} \\ W^* & \xrightarrow[L^*]{} & V^* \end{array}$$

commute: $L^t = (\varphi_B^V)^{-1} \circ L^* \circ \varphi_C^W$.
(b) For $L \in \mathfrak{L}(V, V)$, the **transpose operator** is $L^t = (\varphi_B^V)^{-1} \circ L^* \circ \varphi_B^V$.

(The dual map does not depend on B, C; the transpose does.)

Observation 13.9.2

If $L, M \in \mathfrak{L}(V, V)$ and $B(Lv, w) = B(Mv, w)$ for all v, w, then $L = M$ (likewise in the second variable).

Proof. $B(Lv - Mv, w) = 0$ for all w, so $Lv - Mv \in V^\perp = 0$ (by symmetry of B). ■

Proposition 13.9.3

L^t is the unique linear map with $C(Lv, w) = B(v, L^t w)$ for all $v \in V$, $w \in W$ (equivalently $C(w, Lv) = B(L^t w, v)$).

Proof. The definition says $L^*(\varphi_C^W(w)) = \varphi_B^V(L^t w)$. Evaluating at v: $C(Lv, w) = \varphi_w(Lv) = (L^*\varphi_w)(v) = \varphi_{L^t w}(v) = B(v, L^t w)$. Uniqueness follows from Observation 13.9.2 (non-degeneracy). ■

Proposition 13.9.4

For bases $\mathfrak{B}, \mathfrak{C}$ of V, W with $J_{\mathfrak{B}} = [B]_{\mathfrak{B}}$, $J_{\mathfrak{C}} = [C]_{\mathfrak{C}}$: (a) $[L^t]_{\mathfrak{C}}^{\mathfrak{B}} = (J_{\mathfrak{B}})^{-1}([L]_{\mathfrak{B}}^{\mathfrak{C}})^t J_{\mathfrak{C}}$; (b) for $L \in \mathfrak{L}(V, V)$ and $J = [B]_{\mathfrak{B}}$, $[L^t]_{\mathfrak{B}}^{\mathfrak{B}} = J^{-1}([L]_{\mathfrak{B}}^{\mathfrak{B}})^t J$.

Proof. From $L^t = (\varphi_B^V)^{-1} L^* \varphi_C^W$: $[L^t]_{\mathfrak{C}}^{\mathfrak{B}} = [(\varphi_B^V)^{-1}]_{\mathfrak{B}^*}^{\mathfrak{B}} [L^*]_{\mathfrak{C}^*}^{\mathfrak{B}^*} [\varphi_C^W]_{\mathfrak{C}}^{\mathfrak{C}^*}$, with $[L^*] = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^t$ (Observation 13.6.8) and $[\varphi_B^V] = J_{\mathfrak{B}}$, $[\varphi_C^W] = J_{\mathfrak{C}}$ (Observation 13.8.3).

(b) is a special case. ■

(This is the "secret" behind the transpose; compare Observation 13.1.8.)

Exercise 13.9.5

If $\mathfrak{B}, \mathfrak{C}$ are orthonormal bases, $[L^t]_{\mathfrak{C}}^{\mathfrak{B}} = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^t$.

Solution. $J_{\mathfrak{B}} = J_{\mathfrak{C}} = I$ in Proposition 13.9.4. □

Exercise 13.9.6

(a) For $L(x_1, x_2, x_3, x_4)^t = (x_4, x_1, x_2, x_3)^t$ on $\mathbb{R}^4$ with the dot product, find L^t. (b) For $J = \operatorname{diag}(1, -1)$ and $L(x, y)^t = (x + y, -x + 3y)^t$ on $(\mathbb{R}^2, B_{\mathcal{E}}^J)$, find L^t.

Solution. (a) $[L]$ is a permutation matrix; $L^t = L^{-1}$: $L^t(x_1, x_2, x_3, x_4)^t = (x_2, x_3, x_4, x_1)^t$. (b) $[L] = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$ and $J^{-1}[L]^t J = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$: $L^t = L$ ("self-transpose" for this form, though $[L]$ is not a symmetric matrix). □

Proposition 13.9.7

$\mathbf{O}(V, B) = \{L \in \mathrm{GL}(V) : L^t \circ L = I\}$.

Proof. For $L \in \mathrm{GL}(V)$: $B(v, w) = B(Lv, Lw) = B(v, (L^t \circ L)w)$ for all v, w iff $L^t L = I$ (Observation 13.9.2). ■

Remark 13.9.8

From $A^t J A = J$ one might look for an operator M with $L^t M L = M$; this is a dead end, since J is the matrix of a bilinear form, not of an operator.

Observation 13.9.9

$\mathbf{t} : \mathfrak{L}(V, W) \to \mathfrak{L}(W, V)$, $L \mapsto L^t$, is an isomorphism; in particular $(L + M)^t = L^t + M^t$ and $(cL)^t = cL^t$.

Proof. It is the composite of the isomorphism $*$ (Exercise 13.6.10) with the identifications. ■

Observation 13.9.10

For $L \in \mathfrak{L}(V,W)$, $M \in \mathfrak{L}(W,U)$: (a) $I^t = I$;
(b) $(M \circ L)^t = L^t \circ M^t$;
(c) if L is invertible, so is L^t, with $(L^t)^{-1} = (L^{-1})^t$;
(d) $(L^t)^t = L$.

Proof. (a), (b) from Exercise 13.6.9. (c) from (a), (b).

(d) By Proposition 13.9.3 twice: $C(Lv,w) = B(v,L^tw) = C((L^t)^tv,w)$; apply Observation 13.9.2. ■

Exercise 13.9.11

Prove Observations 13.9.9 and 13.9.10(d) directly from $L^t = (\varphi_B^V)^{-1}L^*\varphi_C^W$.

Solution. **t** is linear because $*$ and composition with fixed isomorphisms are. For (d), $(L^t)^t = (\varphi_C^W)^{-1}(L^t)^*\varphi_B^V$ and $(L^t)^* = (\varphi_C^W)^*L^{**}((\varphi_B^V)^{-1})^*$. Using $(\varphi_B^V)^* = \varphi_B^V$ (Exercise 13.8.4, B symmetric) and $L^{**} = L$ (Exercise 13.6.12), this gives L. □

Exercise 13.9.12

For $L \in \mathfrak{L}(V,V)$: (a) $\mathrm{tr}L^t = \mathrm{tr}L$;
(b) $\det L^t = \det L$;
(c) $\phi_{L^t} = \phi_L$;
(d) $m_{L^t} = m_L$.

Solution. By Proposition 13.9.4(b), $[L^t] = J^{-1}[L]^tJ$ is similar to $[L]^t$, which has the same trace, determinant and characteristic polynomial as $[L]$. For (d), $f(L^t) = f(L)^t$ for polynomials f (Observations 13.9.9, 13.9.10), so $f(L^t) = 0 \iff f(L) = 0$. □

Exercise 13.9.13

For $L \in \mathfrak{L}(V,W)$: (a) $\ker(L^t) = (\mathrm{im}\,L)^\perp$;
(b) $\mathrm{im}(L^t) = (\ker L)^\perp$.

Solution. (a) $L^tw = 0 \iff B(v,L^tw) = 0\ \forall v \iff C(Lv,w) = 0\ \forall v \iff w \in (\mathrm{im}\,L)^\perp$. (b) Apply (a) to L^t (using $(L^t)^t = L$) and take $\perp$: $\mathrm{im}\,L^t = ((\mathrm{im}\,L^t)^\perp)^\perp = (\ker L)^\perp$. □

Exercise 13.9.14

Explain the relation between Exercises 13.7.7 and 13.9.13 from the point of view of Proposition 13.8.10.

Solution. Under B- and C-identification $L^t = L^*$, $\perp$= perp (Proposition 13.8.10, Theorem 13.8.11); so Exercise 13.9.13 is Exercise 13.7.7 transported along these identifications. □

Chapter 14. Hermitian Forms

Hermitian forms on $\mathbb{C}$-vector spaces are Chapter 13 with a few bars (complex conjugations) added; the subtle differences justify a separate treatment. Proofs identical to those of Chapter 13 up to conjugation are only indicated. Throughout $F = \mathbb{C}$ and V is n-dimensional.

14.1 Hermitian forms

Motivation. Sesquilinear forms are linear in the first and conjugate linear in the second variable. They correspond to matrices via $H(v,w) = [v]^t J\overline{[w]}$, Hermitian ones to self-adjoint matrices, and their symmetry groups are the unitary groups $\{A : A^t J\bar{A} = J\}$.

Definition 14.1.1 (Sesquilinear and Hermitian forms)

$H : V \times V \to \mathbb{C}$ is **sesquilinear** if for all $u, v, w \in V$, $c \in \mathbb{C}$: (H1) $H(u+v,w) = H(u,w) + H(v,w)$, $H(cv,w) = cH(v,w)$; (H2) $H(u,v+w) = H(u,v) + H(u,w)$, $H(v,cw) = \bar{c}H(v,w)$. If moreover (H3) $H(v,w) = \overline{H(w,v)}$, H is a **Hermitian form** and (V,H) a **Hermitian space**.

Example 14.1.2

For a basis $\mathfrak{B} = \{v_i\}$, $[H]_\mathfrak{B} = (H(v_i,v_j))$ is the **matrix of** H, and for $v = \sum a_i v_i$, $w = \sum b_i v_i$:

$$H(v,w) = \sum_{i,j} a_i H(v_i,v_j)\bar{b}_j = [v]_\mathfrak{B}^t [H]_\mathfrak{B} \overline{[w]_\mathfrak{B}}.$$

Exercise 14.1.3

For $J \in \mathfrak{M}_{n,n}(\mathbb{C})$, $H_\mathfrak{B}^J(v,w) = [v]_\mathfrak{B}^t J\overline{[w]_\mathfrak{B}}$ is sesquilinear with $[H_\mathfrak{B}^J]_\mathfrak{B} = J$. (The dot product of $\mathbb{C}^n$ is $H_\mathcal{E}^I$.)

Solution. Linear in $[v]$, conjugate linear in $[w]$; $H_\mathfrak{B}^J(v_i,v_j) = e_i^t J e_j = J_{ij}$. □

Exercise 14.1.4

The sesquilinear forms $\mathfrak{Ses}(V)$ form a $\mathbb{C}$-vector space under pointwise operations.

Solution. As in Exercise 13.1.5. □

Theorem 14.1.5

$\Omega_\mathfrak{B} : \mathfrak{Ses}(V) \to \mathfrak{M}_{n,n}(\mathbb{C})$, $H \mapsto [H]_\mathfrak{B}$, is a vector space isomorphism.

Proof. Injective by Example 14.1.2, surjective by Exercise 14.1.3, linear entrywise. ■

Remark 14.1.6

With $\alpha_\mathfrak{B} : V \to \mathbb{C}^n$ and $J = [H]_\mathfrak{B}$, (V,H) and $(\mathbb{C}^n, H_\mathcal{E}^J)$ are essentially the same (the analogue of Remark 13.1.7).

Exercise 14.1.7

If $J \in \mathrm{GL}_n(\mathbb{C})$ and $A \in \mathfrak{M}_{n,n}(\mathbb{C})$, then $H^J_{\mathcal{E}}(AX, Y) = H^J_{\mathcal{E}}(X, (\bar{J}^{-1}A^*\bar{J})Y)$.

Solution. $(AX)^t J\bar{Y} = X^t A^t J\bar{Y}$, and $X^t J\,\overline{\bar{J}^{-1}A^*\bar{J}Y} = X^t JJ^{-1}\overline{A^*}J\bar{Y} = X^t A^t J\bar{Y}$, since $\overline{A^*} = A^t$. □

Exercise 14.1.8

A sesquilinear H is Hermitian iff $J = [H]_{\mathfrak{B}}$ satisfies $J^t = \bar{J}$, i.e. $J^* = J$. Such J are **self-adjoint** (**Hermitian**) matrices.

Solution. (H3) on basis vectors is $J_{ij} = \overline{J_{ji}}$. Conversely, by Example 14.1.2, $\overline{H(w,v)} = \overline{[w]^t J\overline{[v]}} = \overline{[w]}^t \bar{J}[v] = [v]^t \bar{J}^t[w] = [v]^t J[w]$ when $\bar{J}^t = J$. □

Remark 14.1.9

The Hermitian forms $\mathfrak{Her}(V)$ are not closed under multiplication by non-real scalars (iH is not Hermitian if $H \neq 0$), so $\mathfrak{Her}(V)$ is only a real vector space. Such subtle differences persist throughout Chapters 13 and 14.

Definition 14.1.10 (Unitary group)

$\mathbf{U}(V,H) = \{L \in \mathrm{GL}(V) : H(Lv, Lw) = H(v,w)\ \forall v, w\}$, the **unitary group**; its elements are **unitary operators**; this is **unitary geometry**.

Exercise 14.1.11

For a basis $\{v_i\}$, $\mathbf{U}(V,H) = \{L \in \mathrm{GL}(V) : H(Lv_i, Lv_j) = H(v_i, v_j)\ \forall i, j\}$.

Solution. By sesquilinearity, both sides of $H(Lv, Lw) = H(v,w)$ expand into the same combination of these values. □

Proposition 14.1.12

With $J = [H]_{\mathfrak{B}}$ (self-adjoint), $\{[L]^{\mathfrak{B}}_{\mathfrak{B}} : L \in \mathbf{U}(V,H)\} = \{A \in \mathrm{GL}_n(\mathbb{C}) : A^t J\bar{A} = J\}$.

Proof. As in Proposition 13.3.5: $H(Lv, Lw) = [v]^t([L]^t J\overline{[L]})\overline{[w]}$, which equals $[v]^t J\overline{[w]}$ for all v, w iff $[L]^t J\overline{[L]} = J$ (Exercise 9.3.7, applied with $\overline{[w]}$ ranging over $\mathbb{C}^n$). ■

This settles Remark 10.6.4 in the complex case ($J = I$).

Definition 14.1.13 (Matrix unitary groups)

For self-adjoint J, $\mathbf{U}^J_n(\mathbb{C}) = \{A \in \mathrm{GL}_n(\mathbb{C}) : A^t J\bar{A} = J\}$. (So $\mathbf{U}(n) = \mathbf{U}^I_n(\mathbb{C})$.)

Exercise 14.1.14

If J is invertible self-adjoint and $A \in \mathbf{U}^J_n(\mathbb{C})$, then $|\det A| = 1$.

Solution. $\det A \cdot \det J \cdot \overline{\det A} = \det J \neq 0$, so $|\det A|^2 = 1$. □

14.2 Non-degenerate Hermitian forms

Motivation. Orthogonality, non-degeneracy, the perp theorem and orthogonal bases transfer from quadratic spaces to Hermitian spaces, using the real and imaginary parts of H where Chapter 13 used polarization.

Definition 14.2.1 (Orthogonality)

$v \perp w$ if $H(v,w) = 0$; $S \perp T$ similarly; $S^\perp = \{v : H(w,v) = 0 \ \forall w \in S\}$. For Hermitian H, $H(v,w) = 0 \iff H(w,v) = \overline{H(v,w)} = 0$: orthogonality is symmetric.

Exercise 14.2.2

$S^\perp = \langle S \rangle^\perp \leq V$.

Solution. Conjugate linearity in the second variable still gives a subspace, and linearity in the first gives $S^\perp = \langle S \rangle^\perp$. □

Definition 14.2.3 (Orthogonal bases)

$\mathfrak{B}$ is an **orthogonal basis** if $[H]_\mathfrak{B}$ is diagonal and an **orthonormal basis** if $[H]_\mathfrak{B} = I$.

Definition 14.2.4 (Non-degenerate)

H is **non-degenerate** if $V^\perp = 0$. (E.g. the dot product $H_\mathcal{E}^I$ of $\mathbb{C}^n$.)

Proposition 14.2.5

H is non-degenerate iff $\det[H]_\mathfrak{B} \neq 0$.

Proof. As in the first proof of Proposition 13.5.10, replacing Y by $\bar{Y}$: H is degenerate iff some $\bar{Y} \neq 0$ has $X^t J \bar{Y} = 0$ for all X, iff $J\bar{Y} = 0$ for some $\bar{Y} \neq 0$. ■

Theorem 14.2.6 (Perp theorem, non-degenerate case)

If H is non-degenerate and $W \leq V$, then $\dim W^\perp = \dim V - \dim W$.

Proof. As in Theorem 13.5.11: with K the first m rows of J, $v \in W^\perp$ iff $K\overline{[v]} = 0$, i.e. $\overline{W^\perp}$ (in coordinates) is $\ker L_K$. By Exercise 10.3.10, $\dim W^\perp = \dim \ker L_K = n - m$ (rank and dimension theorems). ■

Exercise 14.2.7

For a non-degenerate Hermitian space and $U, W \leq V$: $(W^\perp)^\perp = W$, $(U+W)^\perp = U^\perp \cap W^\perp$, $U^\perp + W^\perp = (U \cap W)^\perp$.

Solution. As in Exercise 13.5.13, using Theorem 14.2.6. □

Exercise 14.2.8

If H is Hermitian and $H(w,w) \neq 0$, then $V = \langle w \rangle \oplus \langle w \rangle^{\perp}$.

Solution. $v = \frac{H(v,w)}{H(w,w)} w + \left(v - \frac{H(v,w)}{H(w,w)} w \right)$, and the second term lies in $\langle w \rangle^{\perp}$ by Hermitian symmetry ($H(w,\cdot) = \overline{H(\cdot,w)}$); $\langle w \rangle \cap \langle w \rangle^{\perp} = 0$. □

Observation 14.2.9

If H is Hermitian and $H(v,v) = 0$ for all v, then $H = 0$.

Proof. $H(v+w,v+w) - H(v,v) - H(w,w) = 2 \operatorname{Re} H(v,w)$, and as in the proof of Observation 10.6.5, $H(v+iw,v+iw) - H(v,v) - H(w,w) = 2 \operatorname{Im} H(v,w)$. Both vanish, so $H(v,w) = 0$. ■

Theorem 14.2.10

Every nonzero Hermitian space (V,H) has an orthogonal basis (H need not be non-degenerate).

Proof [Supplement]. As in Theorem 13.5.16: if $H = 0$ any basis works; otherwise pick w with $H(w,w) \neq 0$ (Observation 14.2.9), split $V = \langle w \rangle \oplus \langle w \rangle^{\perp}$ (Exercise 14.2.8) and induct on dimension. ■

14.3 H-identification and adjoint operators

Motivation. $v \mapsto H(\cdot, v)$ is now *conjugate* linear, but still a natural bijection $V \to V^*$ for non-degenerate H. Through it the dual map becomes the adjoint $L^\star$, with matrix $\bar{J}^{-1}[L]^* \bar{J}$, and unitary operators are those with $L^\star L = I$.

Remark 14.3.1

$\varphi_H^V : V \to V^*$, $\varphi_H^V(v)(w) = H(w,v)$, is well defined (linear in w), but it is *not* linear: H is conjugate linear in its second variable.

Exercise 14.3.2

$\varphi_H^V(u+v) = \varphi_H^V(u) + \varphi_H^V(v)$ and $\varphi_H^V(cv) = \bar{c}\varphi_H^V(v)$: φ_H^V is conjugate linear.

Solution. $H(w,u+v) = H(w,u) + H(w,v)$ and $H(w,cv) = \bar{c}H(w,v)$. □

Observation 14.3.3

For $W \leq V$: (a) $\varphi_H^V(W) \leq V^*$;
(b) if H is non-degenerate, $\dim W = \dim \varphi_H^V(W)$.

Proof. (a) Conjugate linear images of subspaces are subspaces ($c\varphi(w) = \varphi(\bar{c}w)$).

(b) If $\{w_j\}$ is a basis of W, then $\{\varphi(w_j)\}$ spans $\varphi(W)$. It is independent: $\sum c_j \varphi(w_j) = \varphi(\sum \bar{c}_j w_j) = 0$ forces $\sum \bar{c}_j w_j \in V^{\perp} = 0$ (as in Exercise 10.3.10). ■

Theorem 14.3.4

If H is non-degenerate, φ_H^V is a bijection (an isomorphism of abelian groups $(V,+)\to(V^*,+)$); every element of V^* is $\varphi_H^V(v)$ for a unique v.

Proof. Injective: $\varphi(u)=\varphi(v)$ gives $H(w,u-v)=0$ for all w, so $u=v$. Surjective: Observation 14.3.3(b) with $W=V$. ■

Since φ_H^V is not linear, it has no matrix, and there is no second proof of Proposition 14.2.5 via φ_H.

Notation 14.3.5 (H-identification)

For non-degenerate H, the basis-free bijection φ_H^V is used to identify $V=V^*$, $v=\varphi_v=\varphi_H^V(v)$. (This identification is only conjugate linear, and Remark 13.8.6 cannot be carried over.)

Exercise 14.3.6

If H is non-degenerate and $\{v_i\}$ is a basis, there is a basis $\{w_j\}$ with $H(v_i,w_j)=\delta_{ij}$.

Solution. $w_j=(\varphi_H^V)^{-1}(v_j^*)$; these form a basis by Observation 14.3.3(b). □

Exercise 14.3.7

If V has an orthonormal basis $\{v_i\}$, then $\varphi_H^V(v_i)=v_i^*$.

Solution. $\varphi_H^V(v_i)(v_k)=H(v_k,v_i)=\delta_{ki}$. □

Exercise 14.3.8

If H is non-degenerate, the square

$$\begin{array}{ccc} V\times V & \xrightarrow{H} & \mathbb{C} \\ {\scriptstyle I\times\varphi_H^V}\downarrow & & \downarrow{\scriptstyle I} \\ V\times V^* & \xrightarrow[\varepsilon]{} & \mathbb{C} \end{array}$$

commutes: under H-identification, $H=\varepsilon$.

Solution. $\varepsilon(v,\varphi_H^V w)=\varphi_H^V(w)(v)=H(v,w)$. □

Theorem 14.3.9 (Perp theorem, non-degenerate case)

If H is non-degenerate and $W\le V$, then under H-identification $W^\perp=W^{\mathrm{perp}}$; in particular $\dim W^\perp=\dim V-\dim W$.

Proof. By Exercise 14.3.8, $v\in W^\perp \iff \varphi_v\in W^{\mathrm{perp}}$, i.e. $\varphi_H^V(W^\perp)=W^{\mathrm{perp}}$. By Observation 14.3.3(b) and Theorem 13.7.3, $\dim W^\perp=\dim W^{\mathrm{perp}}=\dim V-\dim W$. ■

From now on (V,H) and (W,K) are non-degenerate Hermitian spaces with bases $\mathfrak{B},\mathfrak{C}$.

Definition 14.3.10 (Adjoint map)

(a) For $L \in \mathfrak{L}(V,W)$, the **adjoint map** $L^\star \in \mathfrak{L}(W,V)$ (with respect to H, K) is L^* under H- and K-identification:

$$\begin{array}{ccc} W & \xrightarrow{L^\star} & V \\ {\scriptstyle \varphi_K^W}\downarrow & & \downarrow{\scriptstyle \varphi_H^V} \\ W^* & \xrightarrow[L^*]{} & V^* \end{array}$$

i.e. $L^\star = (\varphi_H^V)^{-1} \circ L^* \circ \varphi_K^W$, which is linear (a composite of two conjugate linear maps and a linear one).
(b) For $L \in \mathfrak{L}(V,V)$, the **adjoint operator** is $L^\star = (\varphi_H^V)^{-1} \circ L^* \circ \varphi_H^V$.

Proposition 14.3.11

$L^\star$ is the unique linear map with $K(Lv,w) = H(v,L^\star w)$ for all $v \in V$, $w \in W$ (equivalently $K(w,Lv) = H(L^\star w, v)$).

Proof. As in Proposition 13.9.3: $K(Lv,w) = \varphi_w(Lv) = (L^*\varphi_w)(v) = \varphi_{L^\star w}(v) = H(v,L^\star w)$; uniqueness by non-degeneracy. The second form follows by conjugating. ■

Proposition 14.3.12

(a) With $J_{\mathfrak{B}} = [H]_{\mathfrak{B}}$, $J_{\mathfrak{C}} = [K]_{\mathfrak{C}}$: $[L^\star]_{\mathfrak{C}}^{\mathfrak{B}} = \overline{J_{\mathfrak{B}}}^{-1}([L]_{\mathfrak{B}}^{\mathfrak{C}})^* \overline{J_{\mathfrak{C}}}$. (b) For $L \in \mathfrak{L}(V,V)$ and $J = [H]_{\mathfrak{B}}$: $[L^\star]_{\mathfrak{B}}^{\mathfrak{B}} = \bar{J}^{-1}([L]_{\mathfrak{B}}^{\mathfrak{B}})^* \bar{J}$.

Proof. φ_H, φ_K are not linear, so we use Proposition 14.3.11. It reads $([Lv]_{\mathfrak{C}})^t J_{\mathfrak{C}} \overline{[w]_{\mathfrak{C}}} = [v]_{\mathfrak{B}}^t J_{\mathfrak{B}} \overline{[L^\star w]_{\mathfrak{B}}}$, i.e. $[v]^t([L]_{\mathfrak{B}}^{\mathfrak{C}})^t J_{\mathfrak{C}} \overline{[w]} = [v]^t J_{\mathfrak{B}} \overline{[L^\star]_{\mathfrak{C}}^{\mathfrak{B}}}\, \overline{[w]}$ for all v, w. By Exercise 9.3.7, $([L]_{\mathfrak{B}}^{\mathfrak{C}})^t J_{\mathfrak{C}} = J_{\mathfrak{B}} \overline{[L^\star]_{\mathfrak{C}}^{\mathfrak{B}}}$; conjugate and solve. (Without the bars this is a second proof of Proposition 13.9.4.) ■

Exercise 14.3.13

If $\mathfrak{B}, \mathfrak{C}$ are orthonormal, $[L^\star]_{\mathfrak{C}}^{\mathfrak{B}} = ([L]_{\mathfrak{B}}^{\mathfrak{C}})^*$.

Solution. $J_{\mathfrak{B}} = J_{\mathfrak{C}} = I$ in Proposition 14.3.12. □

Proposition 14.3.14

$\mathbf{U}(V,H) = \{L \in \mathrm{GL}(V) : L^\star \circ L = I\}$.

Proof. $H(Lv,Lw) = H(v,L^\star Lw)$ equals $H(v,w)$ for all v, w iff $L^\star L = I$ (non-degeneracy, as in Observation 13.9.2). ■

Observation 14.3.15

For $L, M \in \mathfrak{L}(V,W)$, $c \in \mathbb{C}$: (a) $(L+M)^\star = L^\star + M^\star$;
(b) $(cL)^\star = \bar{c}L^\star$.

Proof. (b) $(cL)^\star = (\varphi_H^V)^{-1}(cL^*)\varphi_K^W = \bar{c}(\varphi_H^V)^{-1}L^*\varphi_K^W$, since $(\varphi_H^V)^{-1}$ is conjugate linear. (a) is similar (or use Proposition 14.3.11). Here "$\star = *$" would give the false $(cL)^\star = cL^\star$. ■

Exercise 14.3.16

For $L \in \mathfrak{L}(V,W)$, $M \in \mathfrak{L}(W,U)$: (a) $I^\star = I$;
(b) $(M \circ L)^\star = L^\star \circ M^\star$;
(c) if L is invertible, $(L^\star)^{-1} = (L^{-1})^\star$;
(d) $(L^\star)^\star = L$.

Solution. Use Proposition 14.3.11:

(b) $U(MLv,u) = K(Lv, M^\star u) = H(v, L^\star M^\star u)$.

(d) $K(Lv,w) = H(v,L^\star w) = \overline{H(L^\star w, v)} = \overline{K(w,(L^\star)^\star v)} = K((L^\star)^\star v, w)$. (a), (c) follow. □

Exercise 14.3.17

For $L \in \mathfrak{L}(V,V)$: (a) $\mathrm{tr} L^\star = \overline{\mathrm{tr} L}$;
(b) $\det L^\star = \overline{\det L}$;
(c) $\phi_{L^\star}(t) = \overline{\phi_L}(t)$ (conjugate coefficients);
(d) $m_{L^\star} = \overline{m_L}$.

Solution. $[L^\star] = \bar{J}^{-1}[L]^* \bar{J}$ is similar to $[L]^* = \overline{[L]}^t$, whose trace, determinant and characteristic polynomial are the conjugates of those of $[L]$.

(d) $f(L^\star) = (\bar{f}(L))^\star$ by Observation 14.3.15 and Exercise 14.3.16. □

Exercise 14.3.18

For $L \in \mathfrak{L}(V,W)$: (a) $\ker L^\star = (\mathrm{im}\, L)^\perp$;
(b) $\mathrm{im}\, L^\star = (\ker L)^\perp$.

Solution. As in Exercise 13.9.13, using Proposition 14.3.11, Exercise 14.2.7 and $(L^\star)^\star = L$. □

14.4 Selected exercises

Motivation. A further fact every Hermitian space satisfies: a sesquilinear form is determined by its values $H(v,v)$.

Exercise 14.4.1 (Polarization)

Let H be a sesquilinear form on the $\mathbb{C}$-vector space V.
(a) Show that $H(v,w) = \frac{1}{4}\sum_{k=0}^{3} i^k\, H(v + i^k w,\, v + i^k w)$ for all $v, w \in V$.
(b) Show that H is Hermitian iff $H(v,v) \in \mathbb{R}$ for all v.
(c) Show that $L \in \mathrm{GL}(V)$ lies in $\mathbf{U}(V,H)$ iff $H(Lv,Lv) = H(v,v)$ for all v.

Solution [Supplement].

(a) For $c \in \mathbb{C}$ with $|c| = 1$, (H1) and (H2) give

$$H(v + cw, v + cw) = H(v,v) + \bar{c}\, H(v,w) + c\, H(w,v) + |c|^2 H(w,w).$$

Take $c = i^k$, multiply by i^k and sum over $k = 0, \ldots, 3$. Since $i^k \overline{i^k} = 1$, $\sum_k i^k = 0$ and $\sum_k i^{2k} = 0$, only the middle term with $H(v,w)$ survives, with coefficient 4.

(b) If H is Hermitian, $H(v,v) = \overline{H(v,v)}$ by (H3). Conversely let $H(v,v) \in \mathbb{R}$ for all v, and set $H'(v,w) = \overline{H(w,v)}$. Then $H'(cv,w) = \overline{\bar{c} H(w,v)} = c H'(v,w)$ and $H'(v,cw) = \bar{c} H'(v,w)$, so H' is

sesquilinear, and $H'(v,v) = \overline{H(v,v)} = H(v,v)$. Applying (a) to H and to H' gives $H' = H$, which is (H3).

(c) If L is unitary, take $w = v$. Conversely, $H_L(v,w) = H(Lv, Lw)$ is sesquilinear and agrees with H on all pairs (u,u); by (a), $H_L = H$. (Compare Observation 14.2.9, the special case $H(v,v) = 0$.) □

Chapter 15. The Spectral Theorem

Traditionally the set of eigenvalues of an operator is its *spectrum*, and theorems describing operators by (orthonormal) eigen-decompositions are called spectral theorems. With the tools of Chapters 13–14 we can now go much further than the bare hands of Chapter 9.

15.1 Notation and terminology

Motivation. Transpose and adjoint operators give a single notation T^* for quadratic and Hermitian spaces. Subspaces invariant under both T and T^* make induction on dimension possible, and the conditions $T^* = T$, $T^*T = I$, $TT^* = T^*T$ single out the symmetric, orthogonal and normal operators.

Notation 15.1.1

"$T \in \mathfrak{LM}$" means $T \in \mathfrak{L}(V,V)$ or $T \in \mathfrak{M}_{n,n}(F)$, with $\dim V = n$; a matrix T is regarded as L_T on $V = F^n$. We assume $V \neq 0$.

Notation 15.1.2

(a) For $F = \mathbb{C}$ and a non-degenerate Hermitian space (V,H), T^* means the adjoint $T^\star$; for a matrix A, $T^* = (L_A)^\star$. (b) For a non-degenerate quadratic space (V,B), T^* means the transpose T^t; we usually write T^t to stress the quadratic case.

Remark 15.1.3

(a) Dual maps have done their job of introducing transposes and adjoints and do not appear again. (b) For $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ in $(\mathbb{C}^n, H)$, $[(L_A)^\star]_\mathcal{E} = \bar{J}^{-1}A^*\bar{J}$ ($J = [H]_\mathcal{E}$), which must be distinguished from the adjoint matrix A^*; they agree when $\mathcal{E}$ is orthonormal (e.g. the Hermitian dot product). (c) Likewise $[(L_A)^t]_\mathcal{E} = J^{-1}A^tJ$ in (F^n, B), which equals the transpose matrix A^t when $\mathcal{E}$ is orthonormal (e.g. the ordinary inner product).

Observation 15.1.4

If $W \leq V$ is both T-invariant and T^*-invariant, then $(T|_W)^* = T^*|_W$.

Proof. Writing $\langle\,,\rangle$ for H or B: $\langle T|_W w, w'\rangle = \langle Tw, w'\rangle = \langle w, T^*w'\rangle = \langle w, T^*|_W w'\rangle$ for $w, w' \in W$; apply Propositions 14.3.11, 13.9.3 (on W, when $\langle\,,\rangle|_W$ is non-degenerate, e.g. for inner products). ■

Definition 15.1.5 (Special operators)

For $T \in \mathfrak{LM}$: (a) if $T^* = T$, T is **self-adjoint** (**Hermitian**) in the case (V,H), and **symmetric** in the case (V,B); (b) if $T^*T = I$, T is **unitary** (case (V,H)) or **orthogonal** (case (V,B)); (c) if $TT^* = T^*T$, T is **normal**. ($A \in \mathfrak{M}_{n,n}(\mathbb{C})$ is a **normal matrix** if $AA^* = A^*A$.)

Exercise 15.1.6

(a) Self-adjoint (symmetric) operators are normal.
(b) Unitary (orthogonal) operators are normal.

Solution. (a) $TT^* = T^2 = T^*T$.

(b) $T^*T = I$ forces T invertible with $T^{-1} = T^*$, so $TT^* = I = T^*T$. □

15.2 Normal operators

Motivation. On a Hermitian inner product space the operators that are diagonal in some orthonormal basis are exactly the normal ones. As a consequence every unitary matrix is conjugate in $\mathbf{U}(n)$ to a diagonal matrix of numbers of modulus 1.

Notation 15.2.1

In this section $F = \mathbb{C}$ and $(V, \langle\,,\rangle)$ is an n-dimensional space with a Hermitian inner product.

Observation 15.2.2

If V has an orthonormal basis of eigenvectors of T, then T is normal.

Proof. With such $\mathfrak{B}$, $[H]_{\mathfrak{B}} = I$ and $[T]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, so by Proposition 14.3.12 $[T^*]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(\bar\lambda_1, \dots, \bar\lambda_n)$. Diagonal matrices commute. ■

Observation 15.2.3

T is normal iff $\langle Tv, Tw\rangle = \langle T^*v, T^*w\rangle$ for all v, w.

Proof. If T is normal, $\langle Tv, Tw\rangle = \langle v, T^*Tw\rangle = \langle v, TT^*w\rangle = \langle T^*v, T^*w\rangle$. Conversely, the same chain read backwards gives $\langle v, T^*Tw\rangle = \langle v, TT^*w\rangle$ for all v, w, so $T^*T = TT^*$. ■

Observation 15.2.4

If T is normal and $\lambda \in \mathbb{C}$, then $T - \lambda I$ is normal.

Proof. $(T - \lambda I)^* = T^* - \bar\lambda I$ (Observation 14.3.15, Exercise 14.3.16), and $(T - \lambda I)(T^* - \bar\lambda I) = TT^* - \bar\lambda T - \lambda T^* + |\lambda|^2 I = (T^* - \bar\lambda I)(T - \lambda I)$. ■

Observation 15.2.5

If T is normal and $Tv = \lambda v$, then $T^*v = \bar\lambda v$.

Proof. By Observations 15.2.3, 15.2.4: $0 = \langle (T - \lambda I)v, (T - \lambda I)v\rangle = \langle (T^* - \bar\lambda I)v, (T^* - \bar\lambda I)v\rangle$, so $(T^* - \bar\lambda I)v = 0$ by positive definiteness. ■

Theorem 15.2.6 (Spectral theorem, positive definite Hermitian case)

For $T \in \mathfrak{LM}$ (Notation 15.2.1) the following are equivalent: (1) T is normal; (2) V has an orthonormal basis of eigenvectors of T.

Proof. (2)⇒(1) is Observation 15.2.2. Let T be normal; induct on n. Over $\mathbb{C}$, T has a unit eigenvector v, $Tv = \lambda v$. By Gram–Schmidt $V = \langle v\rangle \oplus \langle v\rangle^{\perp}$; let $W = \langle v\rangle^{\perp}$.

(i) W is T-invariant: for $u \in W$, $\langle v, Tu\rangle = \langle T^*v, u\rangle = \langle \bar\lambda v, u\rangle = \bar\lambda\langle v, u\rangle = 0$ (Observation 15.2.5).

(iii) W is T^*-invariant: $\langle v, T^*u\rangle = \langle Tv, u\rangle = \lambda\langle v, u\rangle = 0$.

(ii) By Observation 15.1.4, $(T|_W)^* = T^*|_W$, so $T|_W$ is normal. By induction W has an orthonormal basis of eigenvectors of $T|_W$; add v. ■

So normal operators act, in a suitable orthonormal frame, by scalar multiplication on each coordinate axis.

Definition 15.2.7 (Maximal tori)

With $\mathbf{D}_n(\mathbb{C})$ the invertible diagonal matrices, $\mathbf{T}_{\mathbf{U}(n)} = \mathbf{U}(n) \cap \mathbf{D}_n(\mathbb{C})$ and $\mathbf{T}_{\mathbf{SU}(n)} = \mathbf{SU}(n) \cap \mathbf{D}_n(\mathbb{C}) = \mathbf{T}_{\mathbf{U}(n)} \cap \mathrm{SL}_n(\mathbb{C})$.

Exercise 15.2.8

(a) $\mathbf{T}_{\mathbf{U}(n)} = \{\mathrm{diag}(\lambda_1, \dots, \lambda_n) : |\lambda_i| = 1\}$. (b) $\mathbf{T}_{\mathbf{U}(n)} \approx S^1 \times \cdots \times S^1 \approx \mathbf{U}(1) \times \cdots \times \mathbf{U}(1)$ (n factors). (c) $\mathbf{T}_{\mathbf{SU}(n)} \approx S^1 \times \cdots \times S^1$ ($n-1$ factors).

Solution. (a) A diagonal matrix has orthonormal columns iff $|\lambda_i| = 1$.

(b) $\mathrm{diag}(\lambda_i) \mapsto (\lambda_i)$.

(c) The last entry is determined by $\lambda_n = (\lambda_1 \cdots \lambda_{n-1})^{-1}$ (Exercise 11.9.13). □

Corollary 15.2.9 (Spectral theorem for $\mathbf{U}(n)$)

If $A \in \mathbf{U}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{U}(n)}$ for some $U \in \mathbf{U}(n)$. Hence $\mathbf{U}(n) = \bigcup_{U \in \mathbf{U}(n)} U\, \mathbf{T}_{\mathbf{U}(n)}\, U^{-1}$.

Proof. In $\mathbb{C}^n$ with the Hermitian dot product, $(L_A)^\star = L_{A^*}$ (Remark 15.1.3(b)), so L_A is unitary, hence normal (Exercise 15.1.6). By Theorem 15.2.6 there is an orthonormal eigenbasis $\mathfrak{B}$. Then $[L_A]^{\mathfrak{B}}_{\mathfrak{B}} \in \mathbf{U}(n)$ is diagonal (Remark 10.6.4, Proposition 14.1.12), so $[I]^{\mathfrak{B}}_{\mathcal{E}} A [I]^{\mathcal{E}}_{\mathfrak{B}} \in \mathbf{T}_{\mathbf{U}(n)}$, and $U = [I]^{\mathcal{E}}_{\mathfrak{B}} \in \mathbf{U}(n)$ (Exercise 10.6.9(5)). ■

(So every element of $\mathbf{U}(n)$ is conjugate to an element of $\mathbf{T}_{\mathbf{U}(n)}$, and $\mathbf{U}(n)$ is the union of the conjugates of $\mathbf{T}_{\mathbf{U}(n)}$.)

Exercise 15.2.10

If $U \in \mathbf{U}(n)$, then $\alpha U \in \mathbf{SU}(n)$ for some $\alpha \in \mathbb{C}$.

Solution. $|\det U| = 1$, so $\det U = e^{i\theta}$; take $\alpha = e^{-i\theta/n}$, so that $\det(\alpha U) = \alpha^n \det U = 1$ and $\alpha U \in \mathbf{U}(n)$ ($|\alpha| = 1$). □

Corollary 15.2.11 (Spectral theorem for $\mathbf{SU}(n)$)

If $A \in \mathbf{SU}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{SU}(n)}$ for some $U \in \mathbf{SU}(n)$; so $\mathbf{SU}(n) = \bigcup_{U \in \mathbf{SU}(n)} U\, \mathbf{T}_{\mathbf{SU}(n)}\, U^{-1}$.

Proof. By Corollary 15.2.9, $U^{-1}AU$ is diagonal unitary with determinant $\det A = 1$, i.e. in $\mathbf{T}_{\mathbf{SU}(n)}$, for some $U \in \mathbf{U}(n)$. Replace U by $\alpha U \in \mathbf{SU}(n)$ (Exercise 15.2.10); $(\alpha U)^{-1} A (\alpha U) = U^{-1}AU$. ■

Exercise 15.2.12

In Corollary 15.2.9, can U always be chosen in $\mathbf{SU}(n)$, even when $A \notin \mathbf{SU}(n)$?

Solution. Yes: replace U by αU as in the proof of Corollary 15.2.11; the conjugate is unchanged. □

Exercise 15.2.13

If $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ is normal ($AA^* = A^*A$), then $U^{-1}AU \in \mathbf{D}_n(\mathbb{C}) \cup \{\text{singular diagonal}\}$, i.e. $U^{-1}AU$ is diagonal, for some $U \in \mathbf{SU}(n)$.

Solution. L_A is normal for the dot product, so Theorem 15.2.6 gives an orthonormal eigenbasis; proceed as in Corollary 15.2.9 and Exercise 15.2.12. □

Exercise 15.2.14

For $A = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right) \in \mathbf{SU}(2)$, find $U \in \mathbf{SU}(2)$ with $U^{-1}AU \in \mathbf{T}_{\mathbf{SU}(2)}$.

Solution. The eigenvalues are $\frac{1 \pm i}{\sqrt{2}} = e^{\pm i\pi/4}$, with unit eigenvectors $\frac{1}{\sqrt{2}}(1,1)^t$ and $\frac{1}{\sqrt{2}}(-1,1)^t$. $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ has $\det U = 1$, and $U^{-1}AU = \operatorname{diag}(e^{i\pi/4}, e^{-i\pi/4})$. □

15.3 Symmetric operators

Motivation. Over $\mathbb{R}$ eigenvalues may fail to exist, but symmetric operators always have real eigenvalues and are diagonal in an orthonormal basis. Applied to the matrix of a quadratic form, this diagonalizes quadratic forms by a rotation of axes and characterises positive definiteness by eigenvalues.

Notation 15.3.1

In this section (through Exercise 15.3.11) $F = \mathbb{R}$, $(V, \langle\,,\rangle)$ is an n-dimensional real inner product space, and we write T^t for T^*.

Proposition 15.3.2

(a) A symmetric operator T has a real eigenvalue.
(b) A real symmetric matrix has a real eigenvalue.

Proof. $(V, \langle\,,\rangle)$ is isomorphic to $(\mathbb{R}^n, \text{dot product})$ (§10.5), so we may take $T = L_A$, $A \in \mathfrak{M}_{n,n}(\mathbb{R})$; then $(L_A)^t = L_{A^t}$ and $A = A^t$. View $L_A : \mathbb{C}^n \to \mathbb{C}^n$ with the Hermitian dot product, which extends the dot product of $\mathbb{R}^n$. Take an eigenvector $X \in \mathbb{C}^n$, $AX = \lambda X$, $\lambda \in \mathbb{C}$. Since $A^* = A^t = A$, Observation 10.7.1 gives $\lambda\langle X, X\rangle = \langle AX, X\rangle = \langle X, A^*X\rangle = \langle X, \lambda X\rangle = \bar{\lambda}\langle X, X\rangle$, so $\lambda = \bar{\lambda} \in \mathbb{R}$. ■

So the characteristic polynomial of every symmetric operator splits over $\mathbb{R}$.

Observation 15.3.3

If V has an orthonormal basis of eigenvectors of T, then T is symmetric.

Proof. With such $\mathfrak{B}$, $[B]_{\mathfrak{B}} = I$ and $[T^t]^{\mathfrak{B}}_{\mathfrak{B}} = I^{-1}([T]^{\mathfrak{B}}_{\mathfrak{B}})^t I = [T]^{\mathfrak{B}}_{\mathfrak{B}}$ (Proposition 13.9.4), since diagonal matrices equal their transposes. ■

Theorem 15.3.4 (Spectral theorem, positive definite real case)

For $T \in \mathfrak{LM}$ (Notation 15.3.1) the following are equivalent: (1) T is symmetric; (2) V has an orthonormal basis of eigenvectors of T.

Proof. (2)⇒(1) is Observation 15.3.3. For (1)⇒(2), copy the proof of Theorem 15.2.6 with $*$ replaced by ^{t} and no conjugations: Proposition 15.3.2 provides a real unit eigenvector v; $W = \langle v \rangle^\perp$ is T-invariant ($\langle v, Tu \rangle = \langle Tv, u \rangle = \lambda \langle v, u \rangle = 0$); $T|_W$ is symmetric; induct. ■

Corollary 15.3.5 (Spectral theorem for real symmetric matrices)

For $A \in \mathfrak{M}_{n,n}(\mathbb{R})$ (dot product on $\mathbb{R}^n$) the following are equivalent: (1) A is symmetric; (2) $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of A; (3) $U^{-1}AU$ is diagonal for some $U \in \mathbf{O}(n)$.

Proof. Theorem 15.3.4 for L_A, with U the orthogonal matrix whose columns are the orthonormal eigenbasis. ■

Diagonalizing quadratic forms. For $2Q(x_1, \dots, x_n) = \sum_i a_{ii} x_i^2 + 2 \sum_{i<j} a_{ij} x_i x_j$ and symmetric $J = (a_{ij})$, $2Q(X) = X^t J X$. Choose $U \in \mathbf{O}(n)$ with $U^{-1} J U = D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Since $U^t = U^{-1}$, with $X = UX'$,

$$2Q(X) = X^t U D U^{-1} X = (U^{-1}X)^t D (U^{-1}X) = \lambda_1 (x_1')^2 + \cdots + \lambda_n (x_n')^2 :$$

the quadratic form is **diagonalized**. (The point is $U^t = U^{-1}$.)

Example 15.3.6

For $J = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $2Q(x, y) = 2x^2 + 2xy + 2y^2$, and $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right) J \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right)$. Substituting $x = \frac{1}{\sqrt{2}}x' + \frac{1}{\sqrt{2}}y'$, $y = -\frac{1}{\sqrt{2}}x' + \frac{1}{\sqrt{2}}y'$ gives $2x^2 + 2xy + 2y^2 = (x')^2 + 3(y')^2$.

Example 15.3.7

The curve $2x^2 + 2xy + 2y^2 = 4$: in the orthonormal eigenbasis $\mathfrak{B} = \{v, w\} = \{\frac{1}{\sqrt{2}}(1, -1)^t, \frac{1}{\sqrt{2}}(1, 1)^t\}$ with $U = [I]^{\mathcal{E}}_{\mathfrak{B}}$, $X = UX'$ means $X = x'v + y'w$. The equation becomes $(x')^2 + 3(y')^2 = 4$: an ellipse with axes along v and w, i.e. a standard ellipse rotated clockwise by 45°.

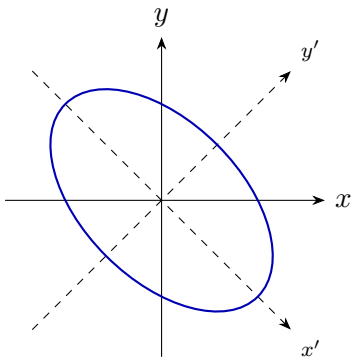


Figure. $2x^2 + 2xy + 2y^2 = 4$ *is the ellipse* $(x')^2 + 3(y')^2 = 4$ *in the rotated frame.*

Exercise 15.3.8

$x^2+xy+y^2-x-y+\frac{1}{3}\geq 0$ for all real x,y.

Solution. Substituting $x=u+\frac{1}{3}$, $y=v+\frac{1}{3}$ kills the linear terms and the constant: the expression becomes u^2+uv+v^2. Its matrix $\begin{pmatrix}1 & \frac{1}{2}\\ \frac{1}{2} & 1\end{pmatrix}$ has eigenvalues $\frac{3}{2},\frac{1}{2}>0$, so $u^2+uv+v^2=\frac{3}{2}(u')^2+\frac{1}{2}(v')^2\geq 0$. □

Exercise 15.3.9

Sketch (a) $2x^2-2\sqrt{2}xy+3y^2=4$;
(b) $x^2-6xy+y^2=2$;
(c) $11x^2+24xy+y^2=15$.

Solution. (a) $J=\begin{pmatrix}2 & -\sqrt{2}\\ -\sqrt{2} & 3\end{pmatrix}$ has eigenvalues $1,4$, with eigenvectors $(\sqrt{2},1)$, $(-1,\sqrt{2})$: the ellipse $(x')^2+4(y')^2=4$. (b) $J=\begin{pmatrix}1 & -3\\ -3 & 1\end{pmatrix}$ has eigenvalues -2 (eigenvector $(1,1)$) and 4 (eigenvector $(-1,1)$): the hyperbola $4(y')^2-2(x')^2=2$. (c) $J=\left(\begin{smallmatrix}11 & 12\\ 12 & 1\end{smallmatrix}\right)$ has eigenvalues 19 (eigenvector $(3,2)$) and -7 (eigenvector $(-2,3)$): the hyperbola $19(x')^2-7(y')^2=15$. □

Exercise 15.3.10

Why are the high-school conics (with centre at the origin) exactly ellipses (including circles) and hyperbolas, plus parabolas?

Solution. After rotating axes (Corollary 15.3.5) a quadratic equation $X^tJX+\ell(X)+c=0$ has quadratic part $\lambda_1(x')^2+\lambda_2(y')^2$. Completing squares in the variables with $\lambda_i\neq 0$: if $\lambda_1\lambda_2>0$ one gets an ellipse (or a point or nothing), if $\lambda_1\lambda_2<0$ a hyperbola (or a pair of lines), and if exactly one $\lambda_i=0$ a parabola (or degenerate lines). □

Exercise 15.3.11

For a finite-dimensional real quadratic space (V,B) with basis $\mathfrak{B}$, the following are equivalent: (1) B is positive definite; (2) all eigenvalues of $J=[B]_{\mathfrak{B}}$ are positive.

Solution. $B(v,v)=X^tJX$ with $X=[v]_{\mathfrak{B}}$. Diagonalize $J=UDU^t$ with $U\in\mathbf{O}(n)$ (Corollary 15.3.5): $X^tJX=\sum\lambda_i(x_i')^2$ with $X'=U^tX$ ranging over all of $\mathbb{R}^n$. This is positive for all $X'\neq 0$ iff all $\lambda_i>0$. (This answers Question 13.1.14.) □

Exercise 15.3.12

For a finite-dimensional Hermitian space (V,H) with basis $\mathfrak{B}$, the following are equivalent: (1) H is positive definite; (2) all eigenvalues of $J=[H]_{\mathfrak{B}}$ are positive reals.

Solution. $H(v,v)=X^tJ\bar{X}=\overline{\bar{X}^t\bar{J}X}$ with $X=[v]_{\mathfrak{B}}$. $\bar{J}$ is self-adjoint, hence unitarily diagonalizable with real eigenvalues (normal, Theorem 15.2.6, and Proposition 15.3.2's argument), which are those of J (conjugates of real numbers). Then $H(v,v)=\sum\lambda_i|x_i'|^2$, positive for all $X\neq 0$ iff all $\lambda_i>0$. □

15.4 Orthogonal operators

Motivation. A real orthogonal operator need not be diagonalizable (rotations), but it splits V into mutually orthogonal invariant lines and planes. Hence every element of $\mathbf{SO}(n)$ is conjugate in

$\mathbf{SO}(n)$ to a block-diagonal matrix of plane rotations: $\mathbf{SO}(n)$ is a mixture of copies of $\mathbf{SO}(2)$.

Here $F = \mathbb{R}$.

Proposition 15.4.1

Every $T \in \mathfrak{LM}$ over $\mathbb{R}$ has an invariant subspace W with $\dim W = 1$ or 2.

Proof. Factor the minimal polynomial $m_T = p_1 \cdots p_s$ into monic irreducibles in $\mathbb{R}[t]$. Some $0 \neq v$ has $p_1(T)v = 0$: if $s = 1$ take any $v \neq 0$; if $s \geq 2$, minimality gives w with $v = p_2(T) \cdots p_s(T)w \neq 0$. Irreducible real polynomials have degree 1 or 2. If $p_1 = t + c$, then $Tv = -cv$; if $p_1 = t^2 + at + b$, then $T^2v = -aTv - bv$. In either case $W = \langle v, Tv \rangle$ is T-invariant, of dimension 1 or 2. ■

Theorem 15.4.2 (Spectral theorem for real orthogonal operators)

If V is a finite-dimensional real inner product space and T is orthogonal, there are subspaces $W_1, \ldots, W_s$ of V with (1) $V = W_1 \oplus \cdots \oplus W_s$; (2) each W_i is T-invariant; (3) $W_i \perp W_j$ for $i \neq j$; (4) $\dim W_i \in \{1, 2\}$.

Proof. Induction on $\dim V$. Take W as in Proposition 15.4.1; $V = W \oplus W^\perp$ (Gram–Schmidt). $W^\perp$ is T-invariant: T is injective, so $TW = W$, and every $w' \in W$ is Tw for some $w \in W$; for $v \in W^\perp$, $\langle w', Tv \rangle = \langle Tw, Tv \rangle = \langle w, T^tTv \rangle = \langle w, v \rangle = 0$. $T|_{W^\perp}$ is orthogonal on the inner product space $W^\perp$; apply induction. ■

Corollary 15.4.3 (Spectral theorem for $\mathbf{O}(n)$)

For $A \in \mathbf{O}(n)$ there are $A_i \in \mathbf{SO}(2)$ and $U \in \mathbf{O}(n)$ with
$A \in \mathbf{SO}(n)$, $n = 2m$: $U^{-1}AU = \operatorname{diag}(A_1, \ldots, A_m)$;
$A \in \mathbf{SO}(n)$, $n = 2m + 1$: $U^{-1}AU = \operatorname{diag}(1, A_1, \ldots, A_m)$;
$A \notin \mathbf{SO}(n)$, $n = 2m$: $U^{-1}AU = \operatorname{diag}(-1, 1, A_1, \ldots, A_{m-1})$;
$A \notin \mathbf{SO}(n)$, $n = 2m + 1$: $U^{-1}AU = \operatorname{diag}(-1, A_1, \ldots, A_m)$.

Proof. (i) Apply Theorem 15.4.2 to L_A on $\mathbb{R}^n$ with the dot product; collecting orthonormal bases of the W_i gives an orthonormal basis $\mathfrak{B}$, $[I]^{\mathfrak{B}}_{\mathcal{E}} \in \mathbf{O}(n)$ (Observation 9.3.6(5)), and $[I]^{\mathfrak{B}}_{\mathcal{E}}\, A\, [I]^{\mathcal{E}}_{\mathfrak{B}} = [L_A]^{\mathfrak{B}}_{\mathfrak{B}} = \operatorname{diag}(B_1, \ldots, B_s) \in \mathbf{O}(n)$ with B_i of size 1 or 2.

(ii) $\operatorname{diag}(B_1, \ldots, B_s) \in \mathbf{O}(n)$ iff each $B_i \in \mathbf{O}(1)$ or $\mathbf{O}(2)$. So (a) $B_i = \pm 1$ if $\dim W_i = 1$;

(b) if $B_i \in \mathbf{O}(2) - \mathbf{SO}(2)$, B_i is a reflection, so $U_i^{-1}B_iU_i = \operatorname{diag}(-1, 1)$ for some $U_i \in \mathbf{O}(2)$ (§9.4); inserting U_i we may assume $B_i = \operatorname{diag}(-1, 1)$.

(iii) Reordering $\mathfrak{B}$ (a permutation matrix, which lies in $\mathbf{O}(n)$) gives $\operatorname{diag}(\pm 1, \ldots, \pm 1, C_1, \ldots, C_r)$ with $C_j \in \mathbf{SO}(2)$. Now $I_2 = R_0$ and $-I_2 = R_\pi$ lie in $\mathbf{SO}(2)$, so pairs of 1's or of -1's can be absorbed into rotations. At most one 1 and at most one -1 remain, and since $\det A = (-1)^{\#(-1)}$, reordering once more yields the four stated forms according to the parity of n and whether $\det A = \pm 1$. ■

Definition 15.4.4 (Maximal torus of $\mathbf{SO}(n)$)

(a) For $n = 2m$: $\mathbf{T}_{\mathbf{SO}(n)} = \{\operatorname{diag}(A_1, \ldots, A_m) : A_i \in \mathbf{SO}(2)\}$. (b) For $n = 2m + 1$: $\mathbf{T}_{\mathbf{SO}(n)} = \{\operatorname{diag}(1, A_1, \ldots, A_m) : A_i \in \mathbf{SO}(2)\}$.

Exercise 15.4.5

(a) $\mathbf{T}_{\mathbf{SO}(n)} \leq \mathbf{SO}(n)$.
(b) $\mathbf{T}_{\mathbf{SO}(n)} \approx \mathbf{SO}(2) \times \cdots \times \mathbf{SO}(2) \approx S^1 \times \cdots \times S^1$ ($m = \lfloor n/2 \rfloor$ factors).

Solution. Block-diagonal products and inverses are computed blockwise, and $\det = \prod \det A_i = 1$; $\operatorname{diag}(\dots, A_i, \dots) \mapsto (A_i)$ is an isomorphism, and $\mathbf{SO}(2) \approx S^1$ (Example 11.9.19). □

Corollary 15.4.6 (Spectral theorem for $\mathbf{SO}(n)$)

If $A \in \mathbf{SO}(n)$, then $U^{-1}AU \in \mathbf{T}_{\mathbf{SO}(n)}$ for some $U \in \mathbf{SO}(n)$. Hence $\mathbf{SO}(n) = \bigcup_{U \in \mathbf{SO}(n)} U\, \mathbf{T}_{\mathbf{SO}(n)}\, U^{-1}$.

Proof. By Corollary 15.4.3, $U^{-1}AU = \operatorname{diag}(1, R_{\theta_1}, \dots, R_{\theta_m}) \in \mathbf{T}_{\mathbf{SO}(n)}$ for some $U \in \mathbf{O}(n)$ (for even n ignore the leading 1; assume $n \geq 2$). If $U \in \mathbf{SO}(n)$ we are done. Otherwise let $S = \operatorname{diag}(1, I_2, \dots, I_2, \operatorname{diag}(-1, 1)) \in \mathbf{O}(n) - \mathbf{SO}(n)$, which flips one coordinate of the last plane. By Observation 9.5.9, $\operatorname{diag}(-1, 1) R_\theta \operatorname{diag}(-1, 1) = R_{-\theta}$, so $(US)^{-1}A(US) = \operatorname{diag}(1, R_{\theta_1}, \dots, R_{\theta_{m-1}}, R_{-\theta_m}) \in \mathbf{T}_{\mathbf{SO}(n)}$, and $US \in \mathbf{SO}(n)$. ■

Exercise 15.4.7

In Corollary 15.4.3, can U be chosen in $\mathbf{SO}(n)$ even when $A \notin \mathbf{SO}(n)$?

Solution. Yes. In the normal forms for $A \notin \mathbf{SO}(n)$ there is a block -1; replacing the corresponding basis vector u by $-u$ multiplies $\det U$ by -1 and leaves $U^{-1}AU$ unchanged (u spans a 1-dimensional invariant subspace). □

So orthogonal operators of Euclidean space, in a suitable orthonormal frame, rotate planes and flip at most one axis: "$\mathbf{SO}(n)$ is a mixture of $\mathbf{SO}(2)$'s", and the elements $\operatorname{diag}(-1, R_\theta) \in \mathbf{O}(3)$ and $\operatorname{diag}(R_\theta, R_\eta) \in \mathbf{SO}(4)$ of Examples 9.6.7 and 9.6.10 are in fact the typical ones.

Exercise 15.4.8

For $A \in \mathbf{SO}(n)$ below, find $U \in \mathbf{SO}(n)$ with $U^{-1}AU \in \mathbf{T}_{\mathbf{SO}(n)}$: (a) $A = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{smallmatrix}\right)$;
(b) $A = \left(\begin{smallmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{smallmatrix}\right)$.

Solution. (a) $\det A = 1$. The eigenvalue 1 has unit eigenvector $u_1 = \frac{1}{\sqrt{2}}(1, 1, 0)^t$, and $A = -I$ on $u_1^\perp = \langle u_2, u_3 \rangle$ with $u_2 = \frac{1}{\sqrt{2}}(-1, 1, 0)^t$, $u_3 = e_3$. With $U = (u_1, u_2, u_3)$ ($\det U = 1$), $U^{-1}AU = \operatorname{diag}(1, R_\pi)$. (b) Eigenvalue 1: $u_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0)^t$, $u_2 = \frac{1}{\sqrt{2}}(0, 0, 1, -1)^t$; eigenvalue -1: $u_3 = \frac{1}{\sqrt{2}}(1, -1, 0, 0)^t$, $u_4 = \frac{1}{\sqrt{2}}(0, 0, 1, 1)^t$. With $U = (u_1, u_2, u_3, u_4)$, $U^{-1}AU = \operatorname{diag}(R_0, R_\pi)$; $\det U = 1$ (if it were -1, replace u_4 by $-u_4$, which keeps $R_\pi = -I_2$). □

15.5 Epilogue

Motivation. Without positive definiteness the spectral theorems fail: a vector may be orthogonal to itself, and symmetric or self-adjoint operators need not be diagonalizable. Over $\mathbb{C}$, Schur's triangularization survives. And the perp theorem, proved without Gram–Schmidt, applies verbatim to binary codes.

In §§15.2–15.4 Gram–Schmidt was essential. In a non-degenerate quadratic space, if $Tv = \lambda v$ with $B(v,v) = 0$, the decomposition $\langle v\rangle \oplus \langle v\rangle^{\perp}$ makes no sense.

A false proof. "Let (V,H) be a non-degenerate finite-dimensional Hermitian space and T normal. Take an eigenvector v, put $\bar{V} = V/\langle v\rangle$, let $\bar{T}$ be induced, define $\bar{H}(\bar{u},\bar{w}) = H(u,w)$, and induct." The flaw: $\bar{H}$ is not well defined unless $v \perp V$, which non-degeneracy forbids. The statement itself is false:

Example 15.5.1

For $J = \operatorname{diag}(1,-1)$ and $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right)$ in $(\mathbb{R}^2, B^J_{\mathcal{E}})$, L_A is symmetric (Exercise 13.9.6(b)), but $\phi_A(t) = (t-2)^2$ and $A \neq 2I$, so L_A is not diagonalizable.

Example 15.5.2

For $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ in $(\mathbb{C}^2, H^J_{\mathcal{E}})$: $H^J_{\mathcal{E}}(A(x,y)^t,(z,w)^t) = (x+y)\bar{w} + y\bar{z} = H^J_{\mathcal{E}}((x,y)^t, A(z,w)^t)$, so L_A is self-adjoint, yet not diagonalizable.

(Orthogonal and unitary geometry continue beyond the second-year course.)

Theorem 15.5.3 (Schur)

If $F = \mathbb{C}$ and $(V, \langle\,,\rangle)$ is an n-dimensional Hermitian inner product space, then for every $T \in \mathfrak{LM}$ there is an orthonormal basis $\mathfrak{B}$ with $[T]^{\mathfrak{B}}_{\mathfrak{B}}$ upper triangular.

Proof. Take a unit eigenvector v of T^*, $T^*v = \mu v$. $V = \langle v\rangle^{\perp} \oplus \langle v\rangle$, and $\langle v\rangle^{\perp}$ is T-invariant: for $w \perp v$, $\langle v, Tw\rangle = \langle T^*v, w\rangle = \mu\langle v,w\rangle = 0$. By induction $\langle v\rangle^{\perp}$ has an orthonormal basis $\mathfrak{C}$ with $[T|_{\langle v\rangle^{\perp}}]^{\mathfrak{C}}_{\mathfrak{C}}$ upper triangular; $\mathfrak{B} = \mathfrak{C} \cup \{v\}$, with v placed last, works. (It suffices to treat $T = L_A$, where Observation 10.7.1 provides T^*.) ∎

Over $\mathbb{C}$ only, this proof is simpler than those of §7.3 and §12.5, but those work over any algebraically closed field.

Field-independent proofs and coding theory. $\mathbb{Z}_2 = \{\bar{0},\bar{1}\}$ with $\bar{1}\cdot\bar{1} = \bar{1}$ (and all other products $\bar{0}$) is a field.

Definition 15.5.5 (Binary codes)

A $\mathbb{Z}_2$-subspace C of $(\mathbb{Z}_2)^n$ is a **linear binary code of length** n; with the ordinary inner product $B^I_{\mathcal{E}}$ on $(\mathbb{Z}_2)^n$, $C^{\perp}$ is the **dual code**.

Theorem 15.5.6

If C is a linear code in $(\mathbb{Z}_2)^n$, then $\dim C + \dim C^{\perp} = n$; so if $\dim C = m$, $|C| = 2^m$ and $|C^{\perp}| = 2^{n-m}$.

Proof. $B^I_{\mathcal{E}}$ is non-degenerate on $(\mathbb{Z}_2)^n$ ($\det I = 1 \neq 0$), so this is the perp theorem, Theorem 13.5.11 or 13.8.11, whose proofs work over any field. Note that Gram–Schmidt is unavailable: e.g. $B^I_{\mathcal{E}}((\bar{1},\bar{1})^t,(\bar{1},\bar{1})^t) = \bar{1}+\bar{1} = \bar{0}$. ∎

Chapter 16. A Taste of Topology

A first taste of topology for matrix groups, assuming only the analysis of $\mathbb{R}^n$. Identify $\mathfrak{M}_{n,n}(\mathbb{R}) = \mathbb{R}^{n^2}$ and $\mathfrak{M}_{n,n}(\mathbb{C}) = \mathbb{C}^{n^2} = \mathbb{R}^{2n^2}$ (coordinates in any fixed order), with the Euclidean norm, distance and continuity.

16.1 Matrix group isomorphisms

Motivation. Matrix groups live in Euclidean space, so we may ask which of their properties are topological. Homeomorphisms (here informally "same-shape maps") that are also group isomorphisms identify matrix groups both as groups and as shapes.

Exercise 16.1.1

Under $\mathbb{C}^n = \mathbb{R}^{2n}$, the Hermitian dot product of $\mathbb{C}^n$ and the dot product of $\mathbb{R}^{2n}$ give the same norm (and metric).

Solution. $|a+ib|^2 = a^2 + b^2$ coordinatewise (compare Remark 10.2.20). □

Example 16.1.2

$\det : \mathfrak{M}_{n,n}(\mathbb{R}) \to \mathbb{R}$ is a polynomial map, hence continuous, so $\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} - \{0\})$ is an open subset of $\mathbb{R}^{n^2}$; likewise $\mathrm{GL}_n(\mathbb{C})$ is open in $\mathbb{R}^{2n^2}$.

Example 16.1.3

$\mathrm{SL}_n(\mathbb{R}) = \det^{-1}(1)$ is a closed subset of $\mathbb{R}^{n^2}$; it is a **closed subgroup** of $\mathrm{GL}_n(\mathbb{R})$. Likewise $\mathrm{SL}_n(\mathbb{C})$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$.

Example 16.1.4

$\mathbf{O}(n) = \{A : A^tA - I = 0\}$ is the common zero set of n^2 polynomials in n^2 variables, hence closed; it is a closed subgroup of $\mathrm{GL}_n(\mathbb{R})$ (and a closed subset of $\mathfrak{M}_{n,n}(\mathbb{R})$).

Exercise 16.1.5

$\mathbf{U}(n)$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$.

Solution. $A^*A = I$ is a system of polynomial equations in the real and imaginary parts of the entries. □

Exercise 16.1.6

(a) $\mathbf{SO}(n)$ is a closed subgroup of $\mathbf{O}(n)$.
(b) $\mathbf{SU}(n)$ is a closed subgroup of $\mathbf{U}(n)$.

Solution. Each is cut out by an additional polynomial equation $\det A = 1$ (real and imaginary parts). □

Question 16.1.7

For $F = \mathbb{R}$ in Exercise 12.4.12, is $\operatorname{im}\psi$ a closed subgroup of $\mathrm{SL}_3(\mathbb{R})$?

Definition 16.1.8 ("Same shape")

For $X \subseteq \mathbb{R}^r$, $Y \subseteq \mathbb{R}^s$, a bijection $f : X \to Y$ with f and f^{-1} continuous is a **homeomorphism** ("same-shape map"); then X and Y have the **same shape** (are homeomorphic).

Exercise 16.1.9

"Same shape" is an equivalence relation.

Solution. Identities, inverses and composites of homeomorphisms are homeomorphisms. □

Exercise 16.1.10

If $f : X \to Y$ is a homeomorphism and $Z \subseteq X$: (a) Z open in $X \Rightarrow f(Z)$ open in Y; (b) Z closed in $X \Rightarrow f(Z)$ closed in Y.

Solution. $f(Z) = (f^{-1})^{-1}(Z)$, and preimages of (relatively) open/closed sets under the continuous map f^{-1} are open/closed. □

Example 16.1.11

$(a, b) \subseteq \mathbb{R}$ and $X = \{(x, 1)^t : a < x < b\} \subseteq \mathbb{R}^2$ have the same shape, although (a, b) is open in $\mathbb{R}$ and X is not open in $\mathbb{R}^2$. Any two open intervals have the same shape.

Example 16.1.12

$\exp : \mathbb{R} \to (0, \infty)$ is a homeomorphism (Example 11.5.7).

Example 16.1.13

$\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$ is a homeomorphism.

Exercise 16.1.14

$[a, b)$ and $[a, \infty)$ have the same shape.

Solution. $x \mapsto a + \tan\left(\frac{\pi}{2} \cdot \frac{x-a}{b-a}\right)$ is a continuous bijection $[a, b) \to [a, \infty)$ with continuous inverse. □

So openness, closedness and boundedness are not invariants of shape.

Exercise 16.1.15

A circle and an ellipse have the same shape. What about a circle and a square?

Solution. $(x, y) \mapsto (ax, by)$ maps the unit circle onto $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The square (its boundary) also has the same shape as the circle, via radial projection $v \mapsto v/\|v\|$ from the centre. □

Exercise 16.1.16

Explain how a topologist turns a soccer ball with a needle hole inside out.

Solution. A sphere with a (small open) hole has the same shape as a closed disk; stretch the hole until the punctured sphere is a disk, flip the disk over, and shrink the hole back. Every step is a homeomorphism. □

Exercise 16.1.17

(a) Do $\mathbb{R}$ and $\mathbb{R}^2$ have the same shape?
(b) $\mathbb{R}$ and S^1?

Solution. [Supplement] (a) No: removing one point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$, and a homeomorphism restricts to a homeomorphism of the complements.

(b) No: S^1 is compact and $\mathbb{R}$ is not (compactness is preserved, Proposition 16.2.1). □

Now return to $\mathfrak{M}_{n,n}(F) = \mathbb{R}^N$.

Exercise 16.1.18

For a matrix group $G \leq \mathrm{GL}_n(F)$: (a) for fixed $g \in G$, left multiplication $\lambda_g : G \to G$ is a continuous bijection, hence (its inverse is $\lambda_{g^{-1}}$) a homeomorphism; likewise right multiplication (compare Cayley's theorem); (b) inversion $x \mapsto x^{-1}$ is a homeomorphism of G; (c) for fixed g, $\mathrm{Int}(g) : G \to G$ is a homeomorphism.

Solution. Matrix multiplication is polynomial in the entries, hence continuous. Inversion is continuous by the adjugate formula $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$ and is its own inverse. $\mathrm{Int}(g) = \lambda_g \circ \rho_{g^{-1}}$. □

Observation 16.1.19

If H is a subgroup of a matrix group G and $g \in G$, then H and the coset gH have the same shape.

Proof. $\lambda_g(H) = gH$, and λ_g restricts to a homeomorphism $H \to gH$ (Exercise 16.1.18(a)). ■

Definition 16.1.20 (Matrix group isomorphism)

For matrix groups $G \leq \mathrm{GL}_n(F) \subseteq \mathbb{R}^N$ and $H \leq \mathrm{GL}_m(F') \subseteq \mathbb{R}^M$ ($F, F' \in \{\mathbb{R}, \mathbb{C}\}$, possibly different), $\varphi : G \to H$ is a **matrix group isomorphism** if it is a group isomorphism and a homeomorphism; then G and H are **isomorphic as matrix groups**, $G \approx_{\mathrm{mat}} H$.

Example 16.1.21

$\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a matrix group isomorphism.

Exercise 16.1.22

$\mathbf{SO}(2) \approx_{\mathrm{mat}} S^1 \approx_{\mathrm{mat}} \mathbf{U}(1)$.

Solution. $R_\theta \mapsto e^{i\theta}$, i.e. $\left(\begin{smallmatrix} x & -y \\ y & x \end{smallmatrix}\right) \mapsto x+iy$, is a group isomorphism (Exercise 11.9.20) and a restriction of a linear map with linear inverse, hence a homeomorphism; $S^1 = \mathbf{U}(1)$. □

Observation 16.1.23

If $H \leq G \leq \mathrm{GL}_n(F)$ and $g \in G$, then $H \approx_{\mathrm{mat}} gHg^{-1}$.

Proof. $\mathrm{Int}(g)$ restricts to a group isomorphism $H \to gHg^{-1}$, which is a homeomorphism by Exercise 16.1.18(c). ■

16.2 Compactness and connectedness

Motivation. Compactness and connectedness are invariants of shape. $\mathbf{O}(n)$, $\mathbf{U}(n)$ and their special subgroups are compact, $\mathrm{GL}_n(\mathbb{R})$ and $\mathbf{O}(1,1)$ are not. The spectral theorems, expressing $\mathbf{SO}(n)$ and $\mathbf{U}(n)$ as unions of conjugates of a connected torus, show that these groups are connected.

In $\mathbb{R}^N$, X is **compact** iff it is closed and bounded, and (for the sets considered here) **connected** means arcwise connected.

Proposition 16.2.1

If $f : X \to Y$ is continuous ($X \subseteq \mathbb{R}^N$, $Y \subseteq \mathbb{R}^M$): (a) X compact $\Rightarrow$ $f(X)$ compact; (b) X connected $\Rightarrow$ $f(X)$ connected.

Proof. Well known from analysis. ■

So compactness and connectedness are invariants of shape, independent of the ambient $\mathbb{R}^N$.

Example 16.2.2

$\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{SL}_n(\mathbb{R})$ ($n \geq 2$) are not bounded in $\mathbb{R}^{n^2}$ (e.g. $\mathrm{diag}(t, t^{-1}, 1, \dots)$), hence not compact; likewise $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{SL}_n(\mathbb{C})$.

Example 16.2.3

For $A \in \mathbf{O}(n) \subseteq \mathbb{R}^{n^2}$, $\|A\| = \sqrt{n}$ (its columns are unit vectors, Observation 9.3.6(5)), so $\mathbf{O}(n)$ is bounded; it is closed (Example 16.1.4), hence a compact group.

Example 16.2.4

$\mathbf{SO}(n)$, a closed subgroup of $\mathbf{O}(n)$, is compact.

Exercise 16.2.5

$\mathbf{U}(n)$ and $\mathbf{SU}(n)$ are compact.

Solution. As in Examples 16.2.3, 16.2.4: $\|A\| = \sqrt{n}$ for $A \in \mathbf{U}(n)$ (Exercise 10.6.9(5), Exercise 16.1.1), and they are closed (Exercises 16.1.5, 16.1.6). □

Example 16.2.6

(a) $\mathbf{SO}^\circ(1,1)$ is unbounded in $\mathbb{R}^4$ ($\cosh t \to \infty$), so $\mathbf{SO}^\circ(1,1)$ and $\mathbf{O}(1,1)$ are not compact. (b) Alternatively, $\mathbf{SO}^\circ(1,1) \approx_{\mathrm{mat}} (\mathbb{R}, +)$ (Exercise 13.4.5; the inverse $\left(\begin{smallmatrix} x & y \\ y & x \end{smallmatrix}\right) \mapsto \operatorname{arsinh} y$ is continuous), and $\mathbb{R}$ is not compact. (c) Since $\mathbf{SO}(2) \approx_{\mathrm{mat}} S^1$, the compact $\mathbf{SO}(2)$ and the

non-compact $\mathbf{SO}^\circ(1,1)$ are not isomorphic as matrix groups. They are not even isomorphic as abstract groups: $\mathbf{SO}(2)$ has elements of every finite order, $\mathbf{SO}^\circ(1,1) \approx \mathbb{R}$ has no nontrivial element of finite order (compare Exercise 13.4.6).

Example 16.2.7

$\det : \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times = \mathbb{R} - \{0\}$ is a continuous surjection and $\mathbb{R}^\times$ is not connected; so $\mathrm{GL}_n(\mathbb{R})$ is not connected.

Exercise 16.2.8

$\mathbf{O}(n)$ and $\mathbf{O}(1,1)$ are not connected.

Solution. det maps each continuously onto $\{\pm 1\}$, which is not connected. □

Proposition 16.2.9

For $X_i \subseteq \mathbb{R}^{N_i}$, identify $X_1 \times \cdots \times X_r \subseteq \mathbb{R}^{N_1+\cdots+N_r}$. (a) If all X_i are compact, so is the product.
(b) If all X_i are connected, so is the product.

Proof. Well known. ■

Definition 16.2.10 (Torus)

$\mathbf{T} \subseteq \mathbb{R}^N$ is a **torus** if it has the same shape as $S^1 \times \cdots \times S^1$. A torus is compact and connected; $S^1 \times S^1$ has the shape of a doughnut.

Exercise 16.2.11

$\mathbf{T}_{\mathbf{U}(n)}$, $\mathbf{T}_{\mathbf{SU}(n)}$ (Definition 15.2.7) and $\mathbf{T}_{\mathbf{SO}(n)}$ (Definition 15.4.4) are tori.

Solution. The isomorphisms of Exercises 15.2.8 and 15.4.5 with products of circles are coordinate maps (with continuous inverses), hence matrix group isomorphisms, in particular homeomorphisms. □

Corollary 16.2.12 (Corollary of the spectral theorems)

$\mathbf{U}(n)$, $\mathbf{SU}(n)$ and $\mathbf{SO}(n)$ are connected groups.

Proof. We treat $\mathbf{SO}(n)$; the others follow from Corollaries 15.2.9 and 15.2.11 in the same way. By Corollary 15.4.6, $\mathbf{SO}(n) = \bigcup_{U \in \mathbf{SO}(n)} U\,\mathbf{T}_{\mathbf{SO}(n)}\,U^{-1}$. Each $U\mathbf{T}_{\mathbf{SO}(n)}U^{-1} \approx_{\mathrm{mat}} \mathbf{T}_{\mathbf{SO}(n)}$ (Observation 16.1.23) is connected (Exercise 16.2.11), and all of them contain I. A union of connected sets with a common point is connected. ■

Since $\mathbf{O}(n)$ is not connected, we look at its **connected components** (maximal connected subsets); every $X \subseteq \mathbb{R}^N$ is uniquely the disjoint union of its components (**component decomposition**).

Observation 16.2.13

(a) $\mathbf{SO}(n)$ is a connected component of $\mathbf{O}(n)$.
(b) The coset decomposition $\mathbf{O}(n) = \mathbf{SO}(n) \amalg (\mathbf{O}(n) - \mathbf{SO}(n))$ is the component decomposition of $\mathbf{O}(n)$.

Proof. $\mathbf{SO}(n)$ is connected (Corollary 16.2.12), and so is $\mathbf{O}(n) - \mathbf{SO}(n) = S \cdot \mathbf{SO}(n)$ (Observation 16.1.19). No connected subset can meet both, since det would map it continuously onto $\{\pm 1\}$. ■

Observation 16.2.14

(a) $\mathbf{SO}^\circ(1,1)$ is a connected component of $\mathbf{O}(1,1)$.
(b) The coset decomposition of $\mathbf{O}(1,1)$ modulo $\mathbf{SO}^\circ(1,1)$ (§13.4) is its component decomposition.

Proof [Supplement]. $\mathbf{SO}^\circ(1,1) \approx \mathbb{R}$ is connected, and so are its cosets (Observation 16.1.19). The four cosets are separated by the continuous functions det and $\operatorname{sign}(A_{11})$ ($|A_{11}| \geq 1$ never vanishes), which are constant on connected sets and take the four combinations of values ± 1 on the four cosets. ■

That the component containing the identity is a subgroup is no accident (proof postponed). Thus $\mathbf{O}(2)$ has the shape of two disjoint circles and $\mathbf{O}(1,1)$ that of four disjoint lines.